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Masarik et al.

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(45) **Date of Patent:** **Aug. 12, 2025**

(54) **FIREARM AMMUNITION AVAILABILITY
DETECTION SYSTEM**

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(71) Applicant: **Maztech Industries, LLC**, Irvine, CA
(US)

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(US)

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(73) Assignee: **Maztech Industries, LLC**, Irvine, CA
(US)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/526,592**

(22) Filed: **Dec. 1, 2023**

Primary Examiner — John Cooper

(65) **Prior Publication Data**

US 2024/0344790 A1 Oct. 17, 2024

(74) *Attorney, Agent, or Firm* — Knobbe, Martens, Olson
& Bear, LLP

(57) **ABSTRACT**

Related U.S. Application Data

(63) Continuation of application No. 17/929,648, filed on
Sep. 2, 2022, now Pat. No. 11,859,935, which is a
(Continued)

A linear encoder may be used to accurately determine a
number of cartridges within a magazine. Further, using a set
of magnetic sensors within a buffer tube of a firearm, it is
possible to determine whether a cartridge is loaded within a
chamber of the weapon and/or whether the firearm is
jammed. The determination of cartridges within the maga-
zine in conjunction with the determination of whether a
cartridge is in a chamber of a firearm can give a user an
accurate ammunition count. Further, the use of the linear
encoder makes it possible to accurately determine the car-
tridge count when different cartridges are loaded into the
magazine or when the magazine degrades in quality over
time. Moreover, the ability for magazines to communicate
with a firearm enables a user to determine a total available
ammunition to the user in a single display without individu-
ally checking each magazine.

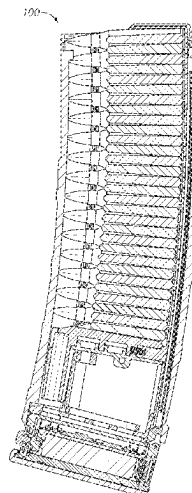
(51) **Int. Cl.**
F41A 9/62 (2006.01)
F41A 9/70 (2006.01)
G01D 5/12 (2006.01)

(52) **U.S. Cl.**
CPC **F41A 9/62** (2013.01); **F41A 9/70**
(2013.01); **G01D 5/12** (2013.01)

(58) **Field of Classification Search**
CPC F41A 9/70; F41A 9/61; F41A 9/62; F41A
5/12; G01D 5/12

(Continued)

21 Claims, 46 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/129,066, filed on Dec. 21, 2020, now Pat. No. 11,466,947, which is a continuation of application No. 16/746,770, filed on Jan. 17, 2020, now Pat. No. 10,900,726, which is a continuation of application No. 16/297,288, filed on Mar. 8, 2019, now Pat. No. 10,557,676.

- (60) Provisional application No. 62/640,451, filed on Mar. 8, 2018.

(58) **Field of Classification Search**

USPC 89/33.01, 33.02, 33.1; 42/1.02
See application file for complete search history.

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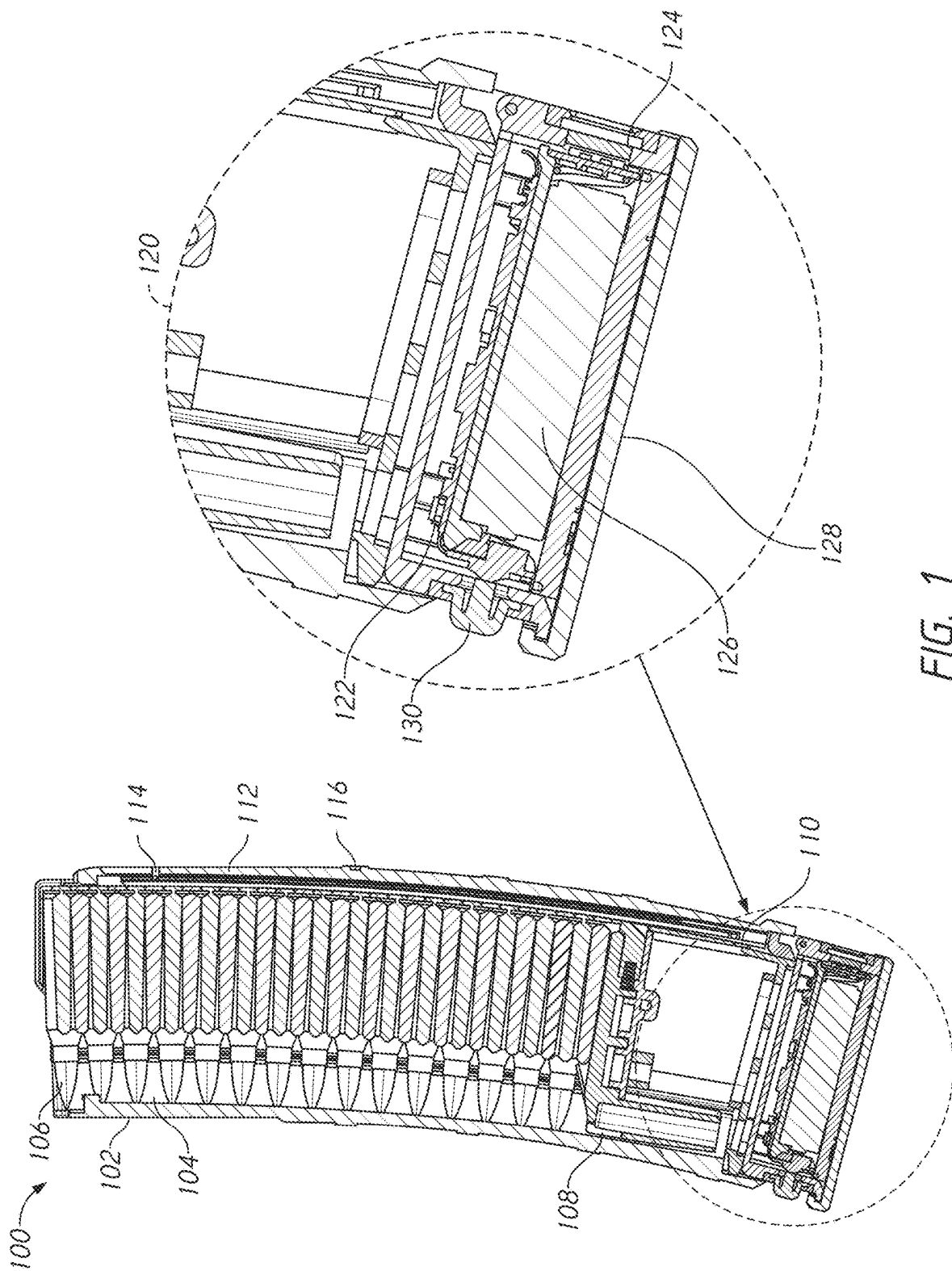
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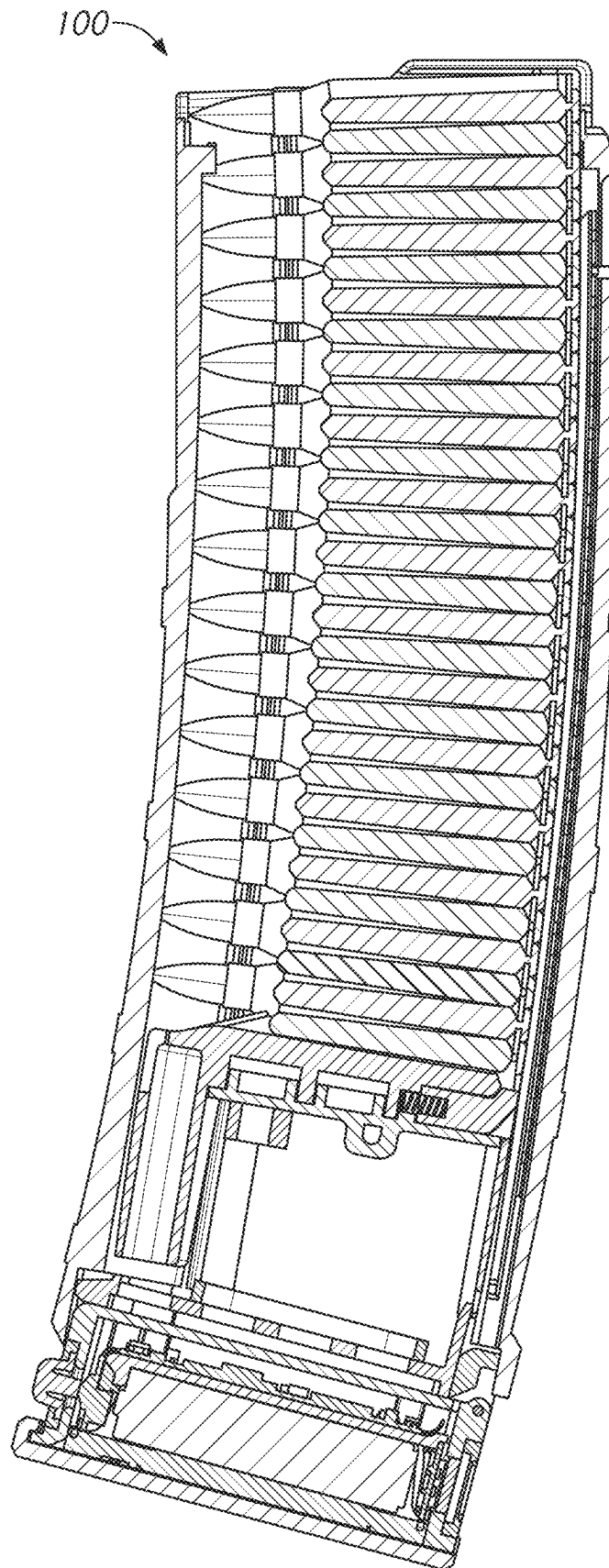


FIG. 2

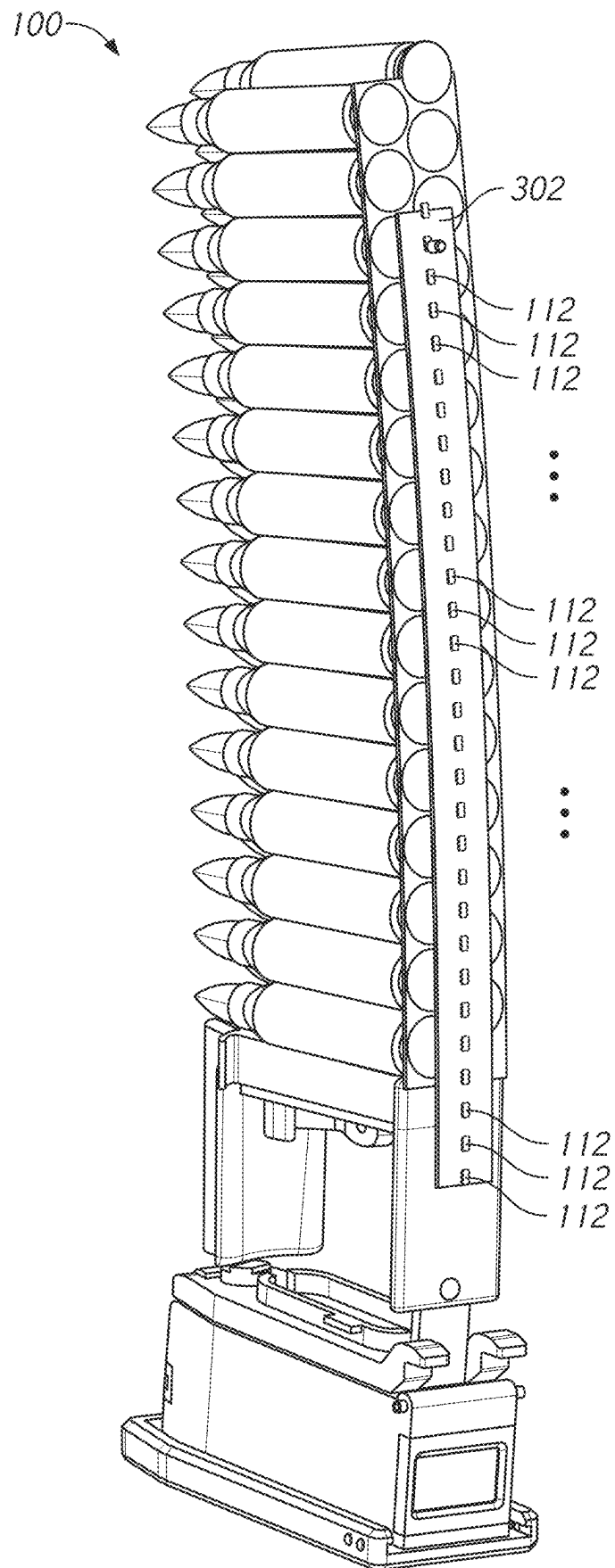


FIG. 3

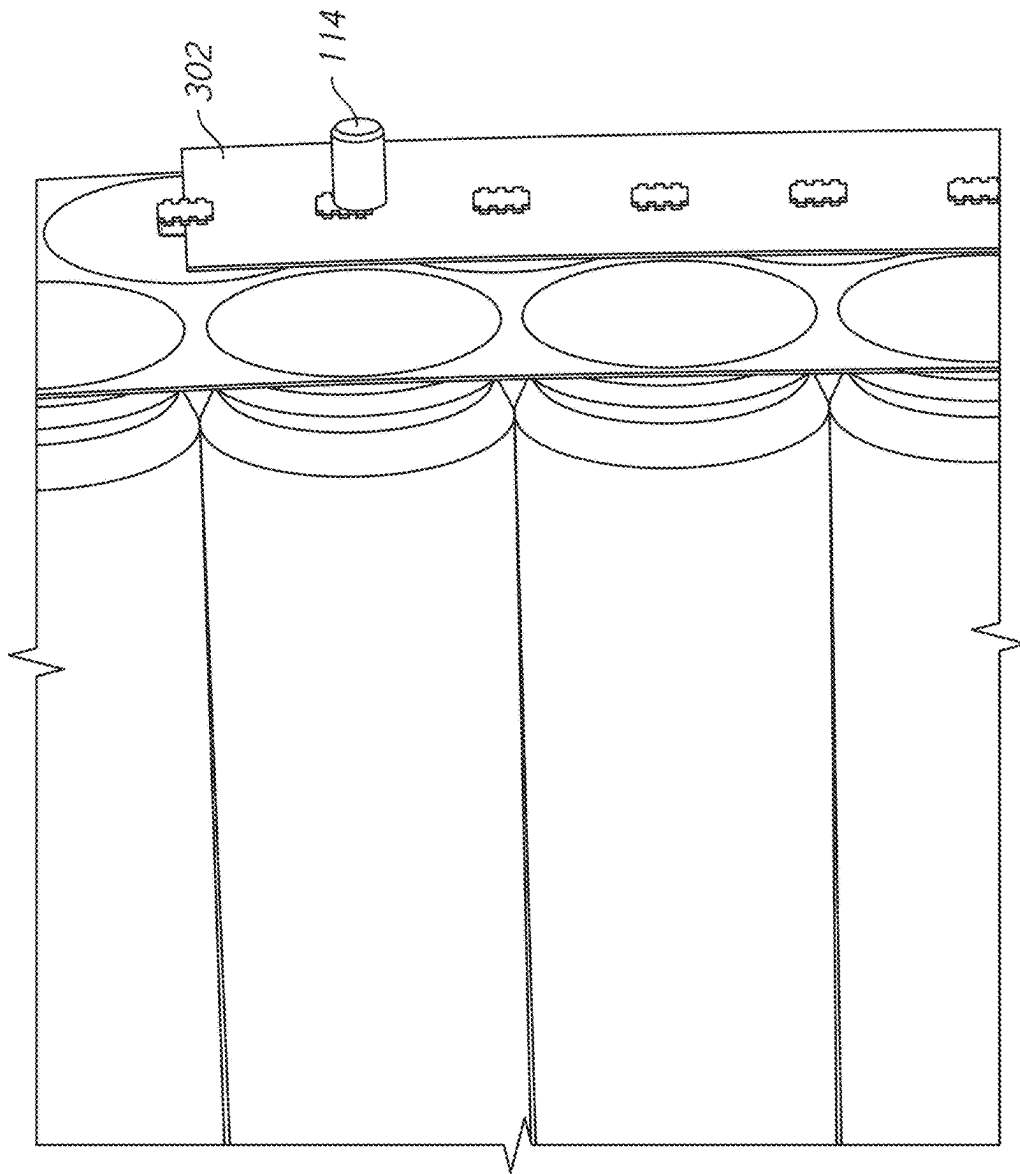


FIG. 4A

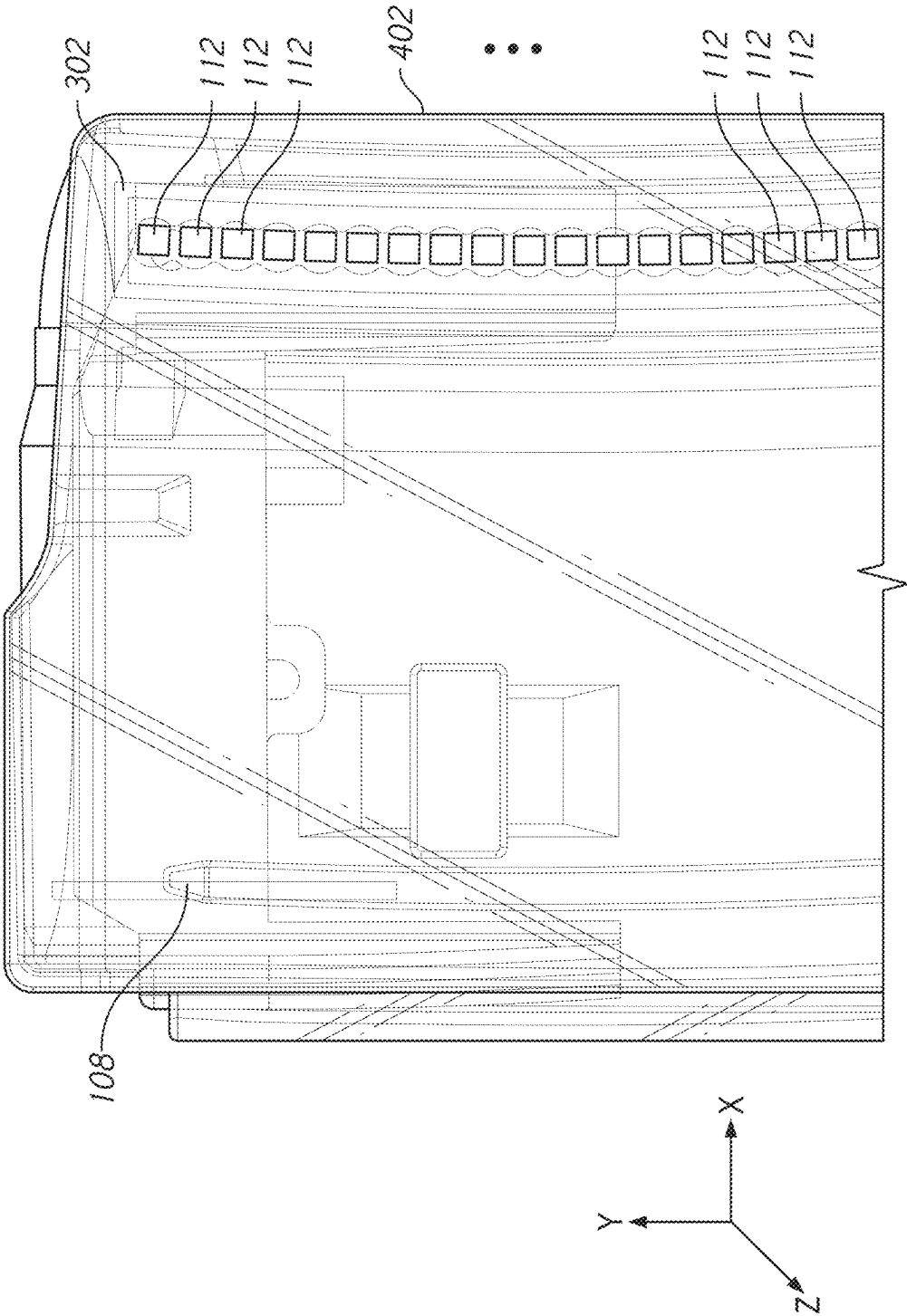


FIG. 4B

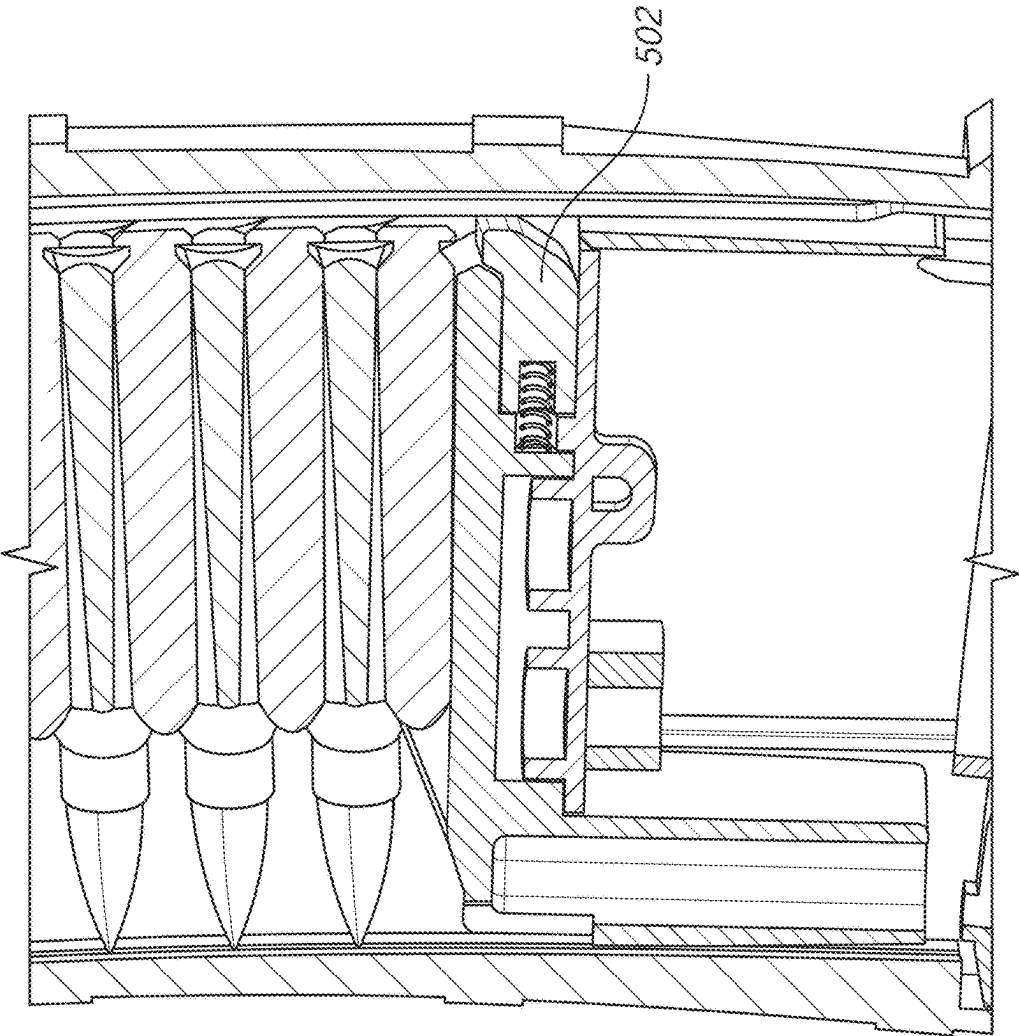


FIG. 5

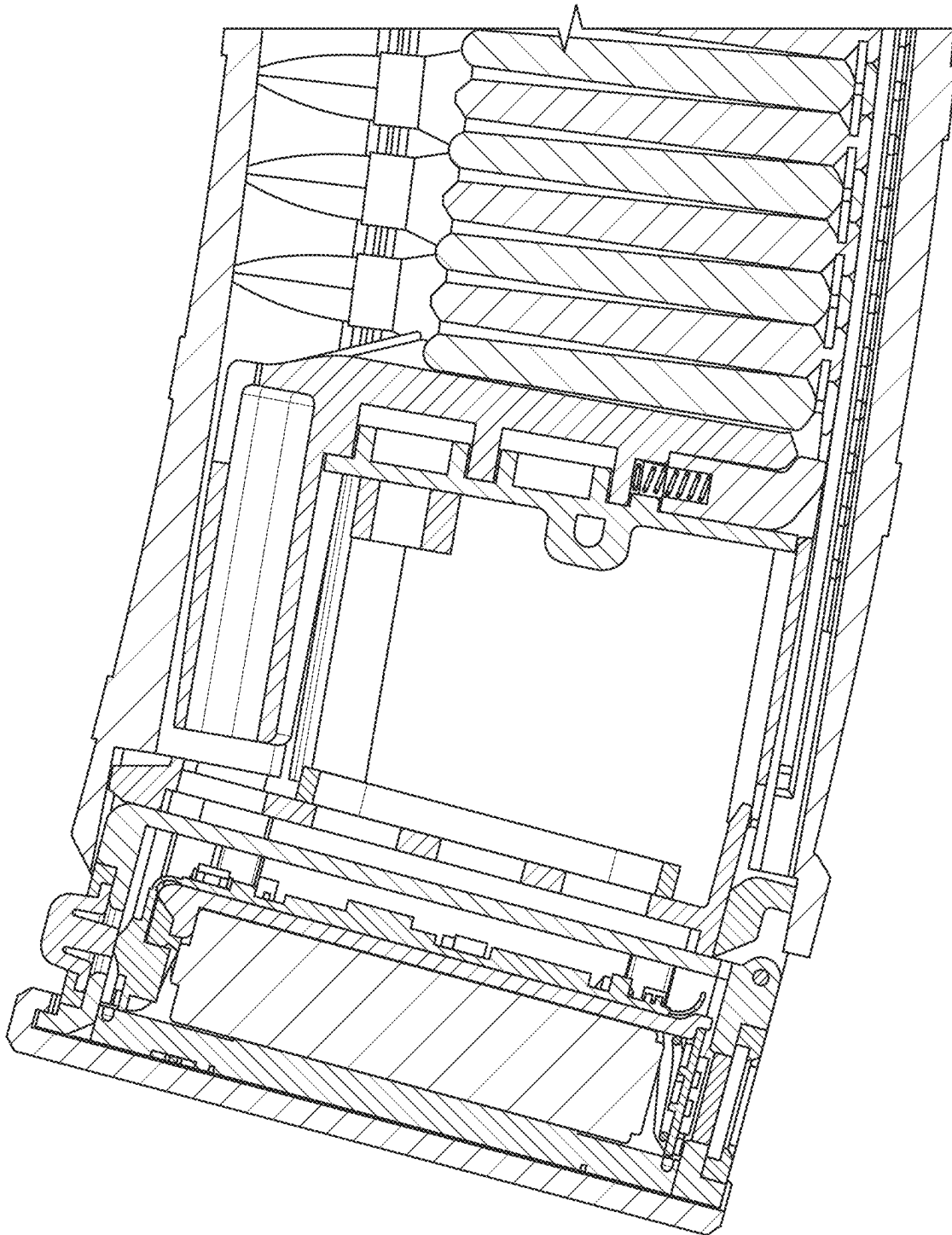
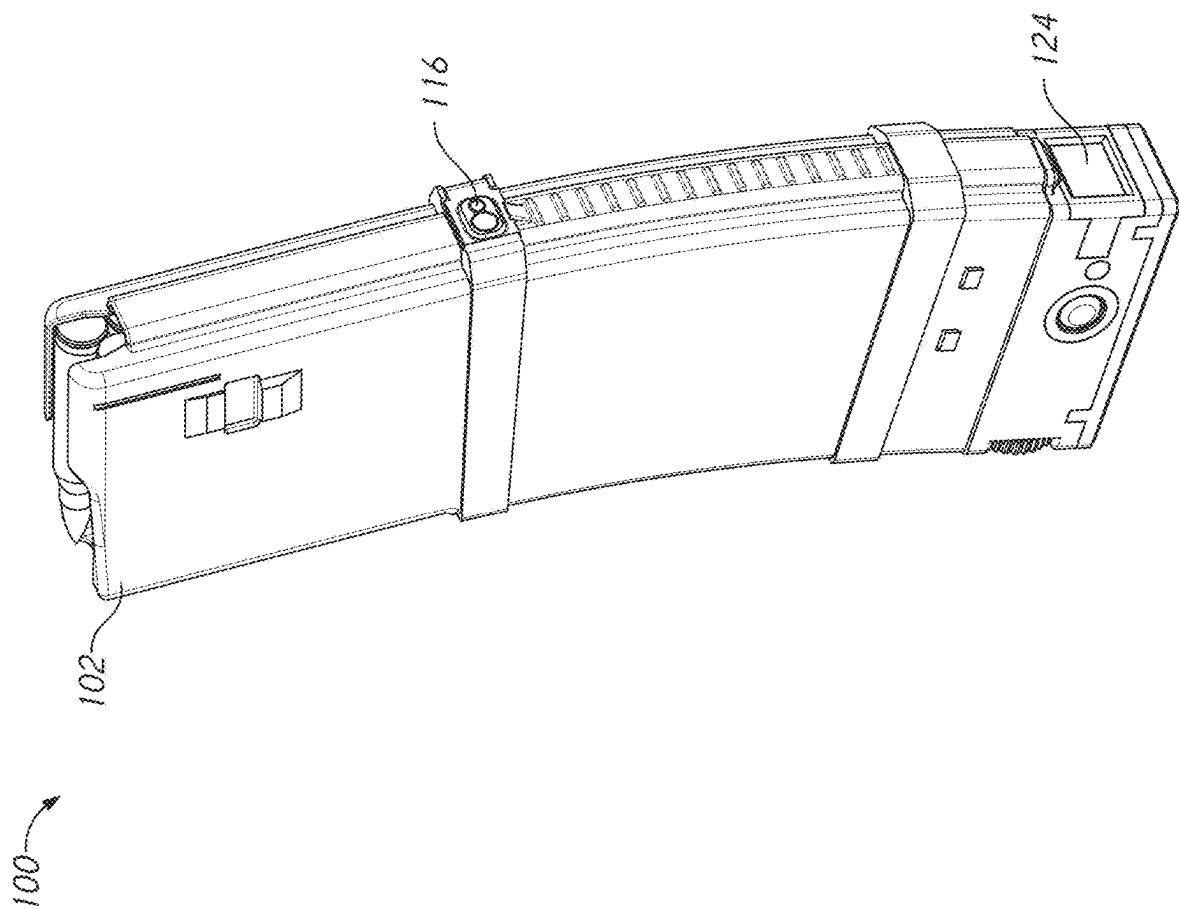


FIG. 6



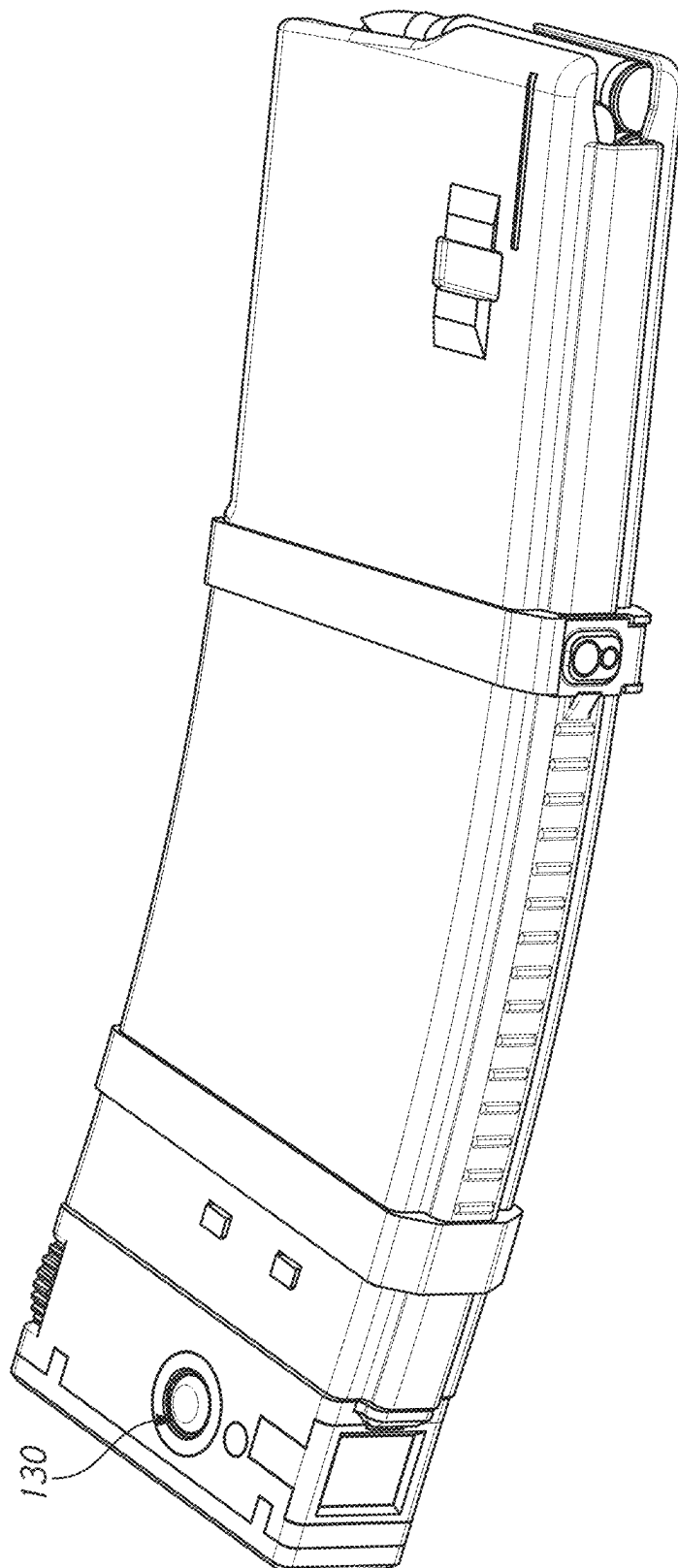


FIG. 8A

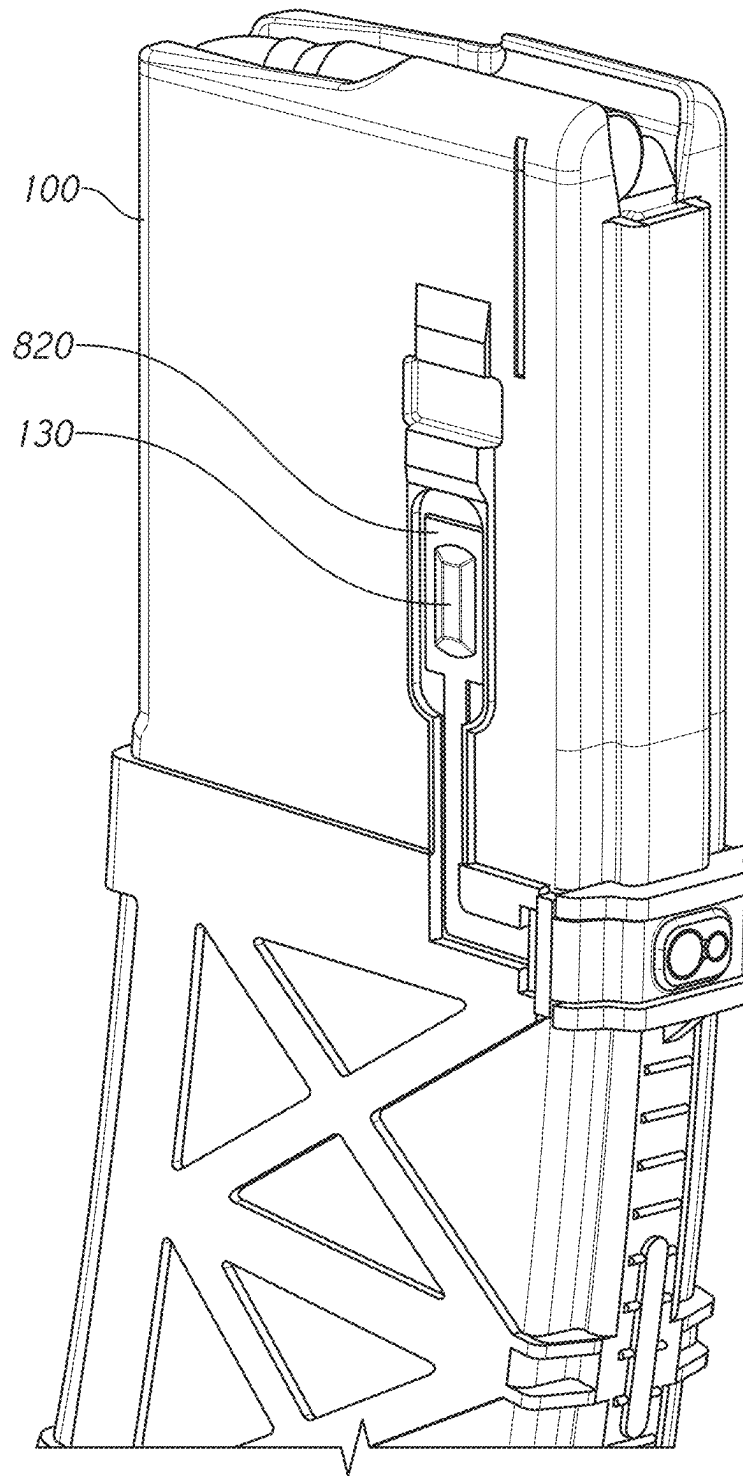


FIG. 8B

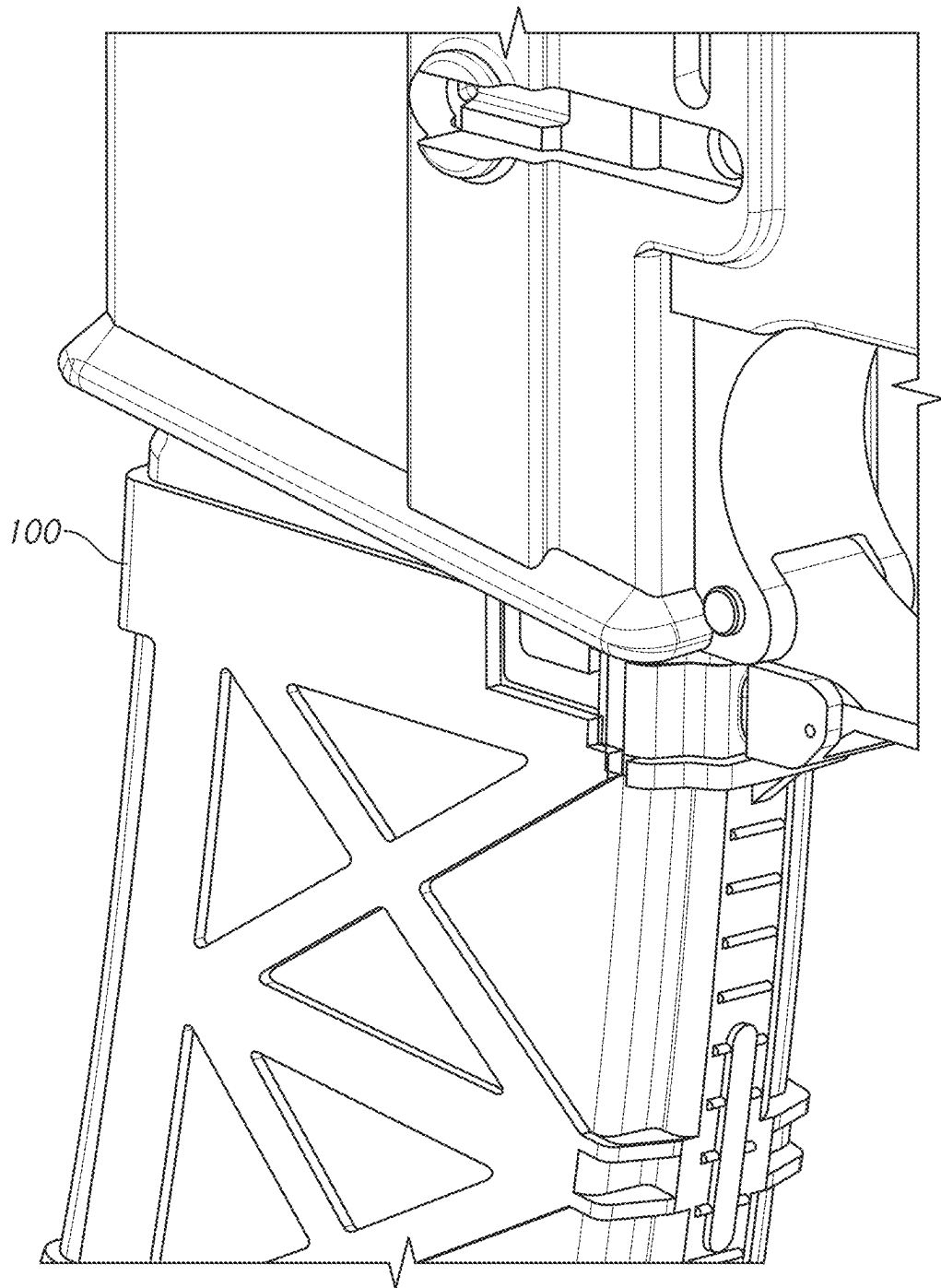


FIG. 8C

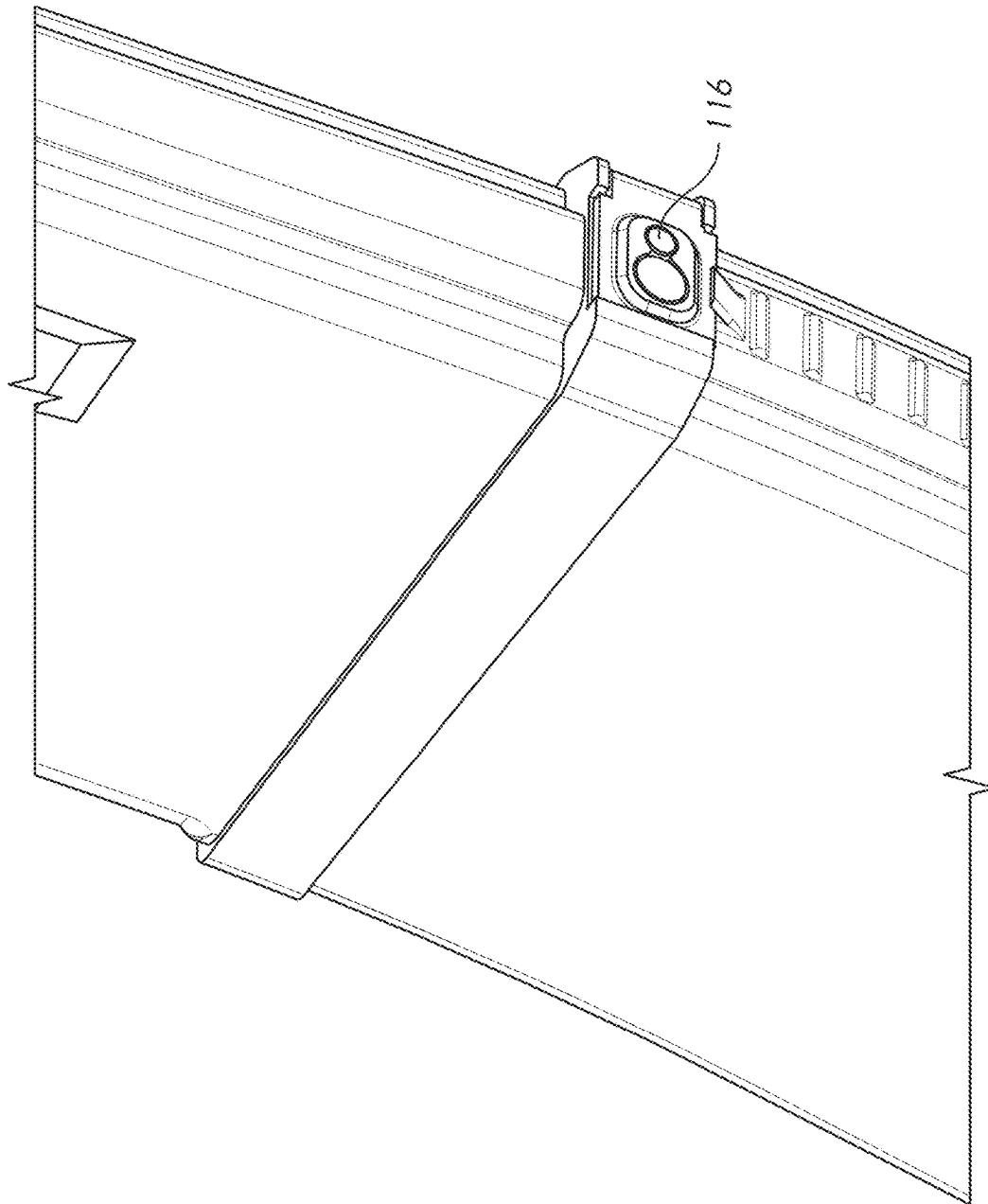


FIG. 9A

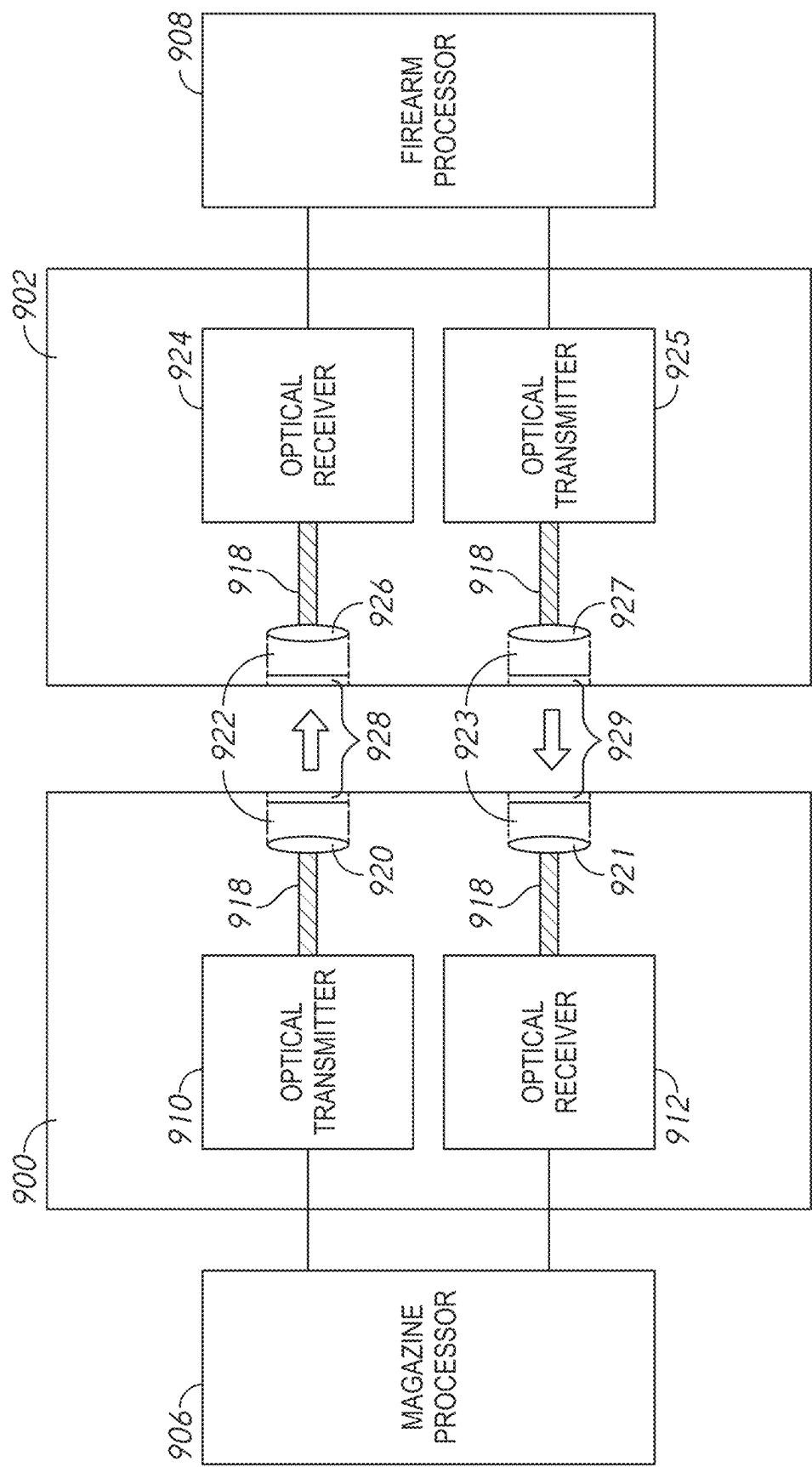


FIG. 9B

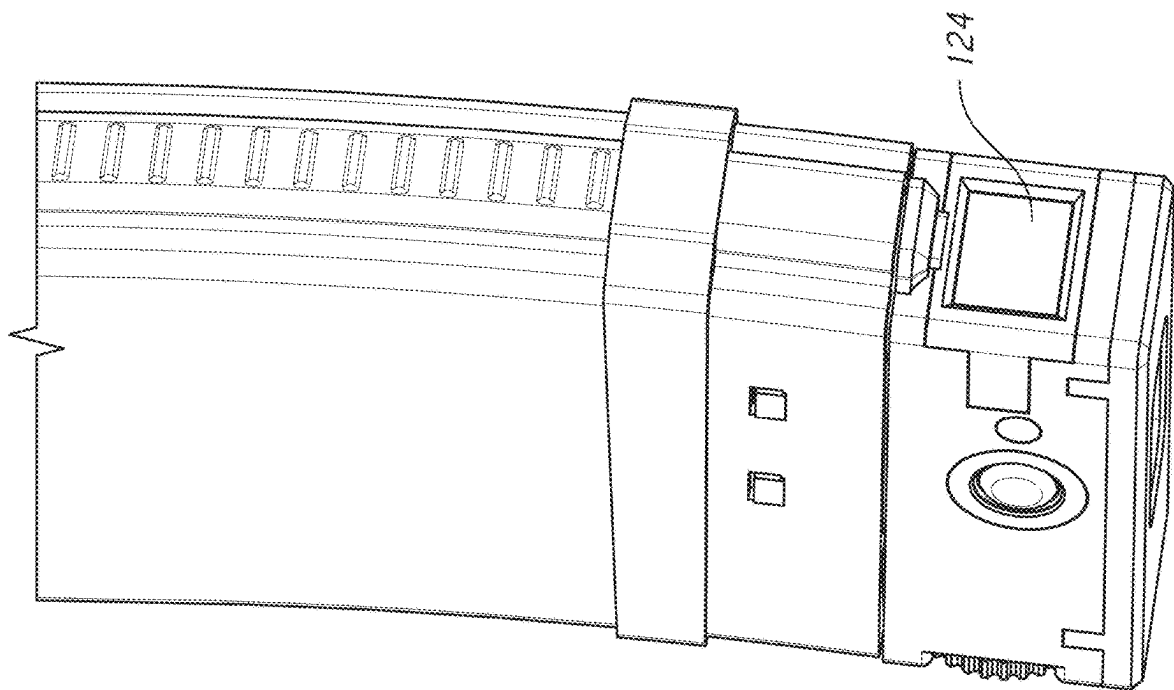


FIG. 10

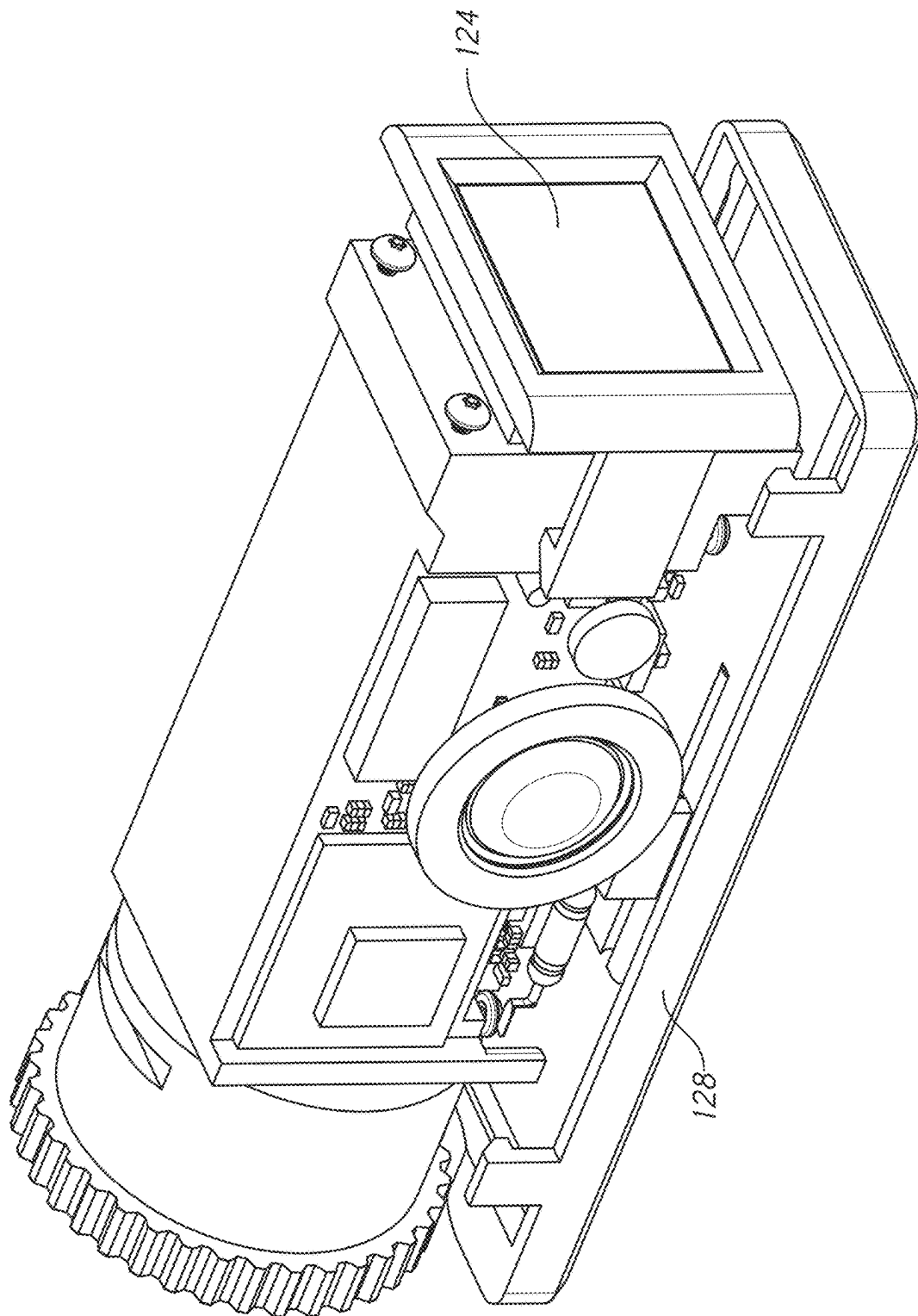


FIG. 11A

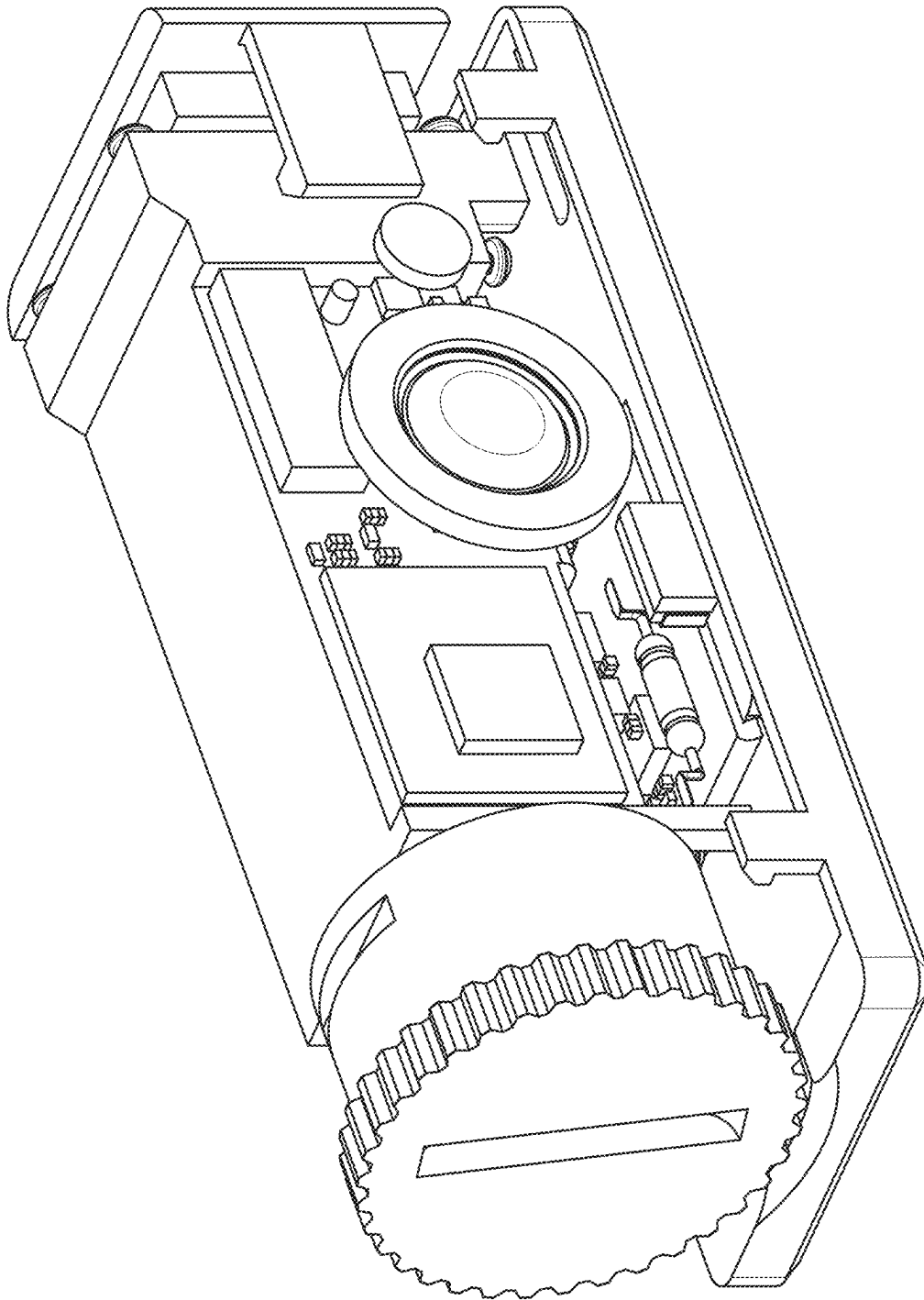


FIG. 11B

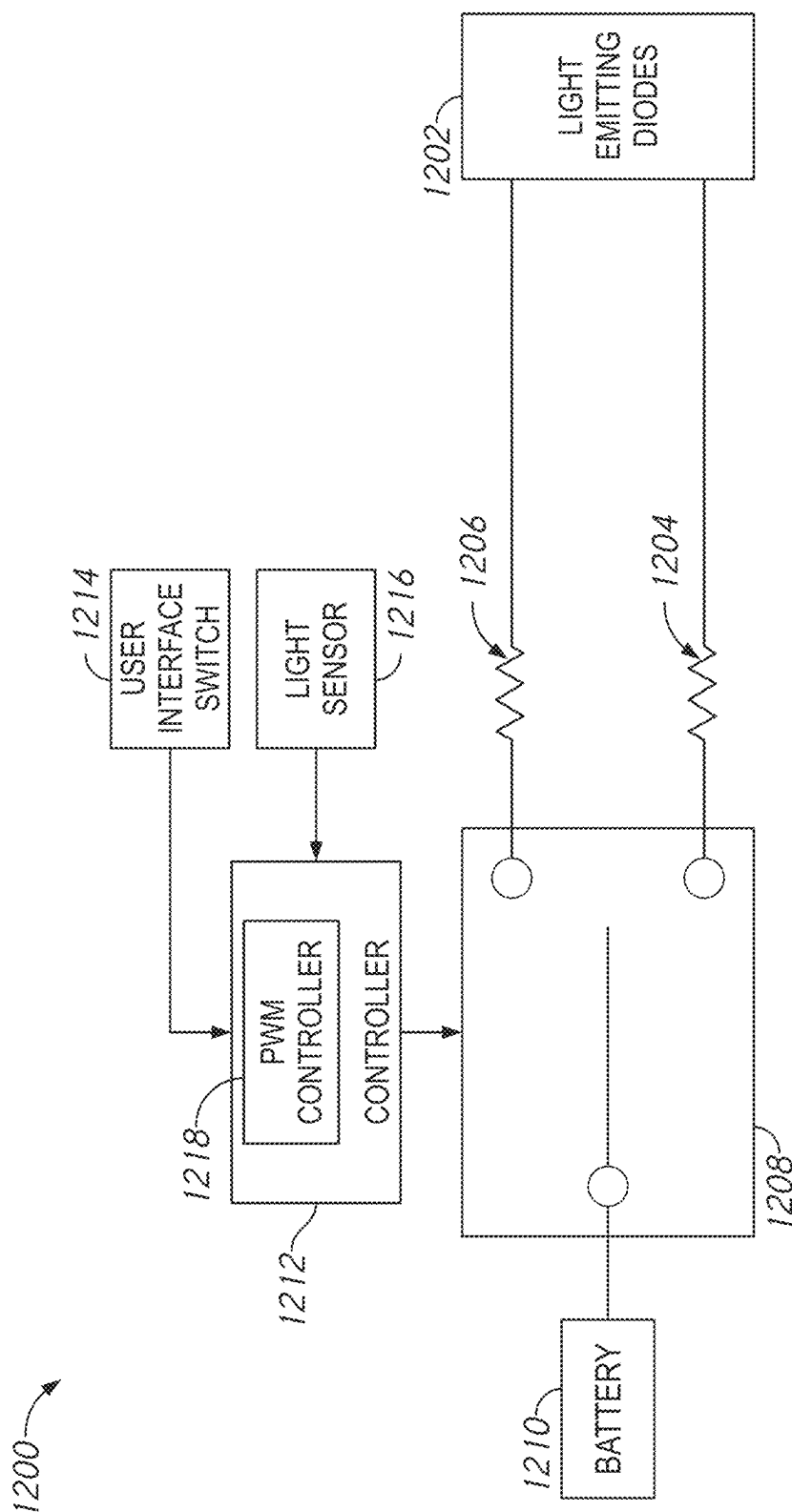


FIG. 12

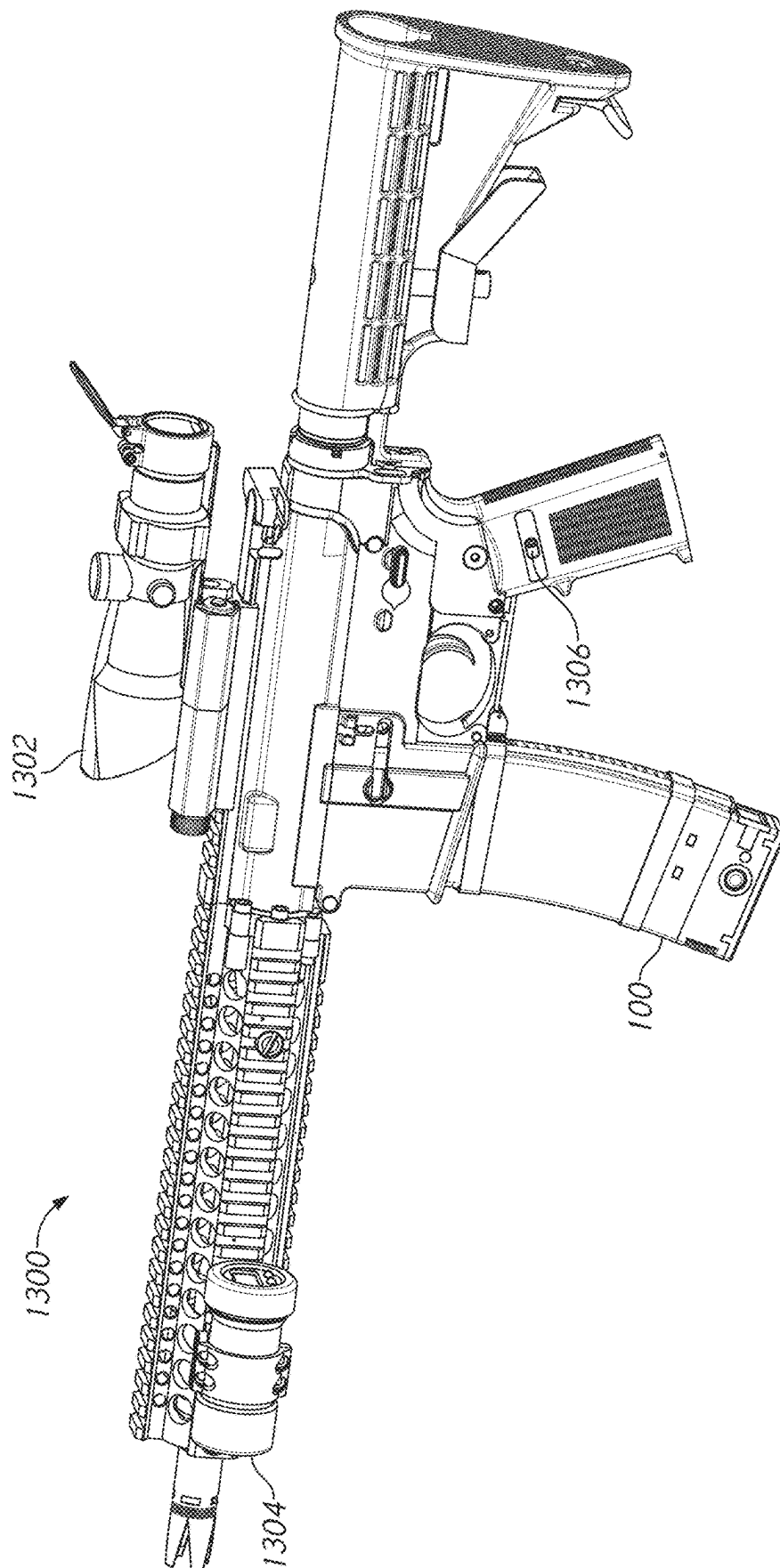


FIG. 13

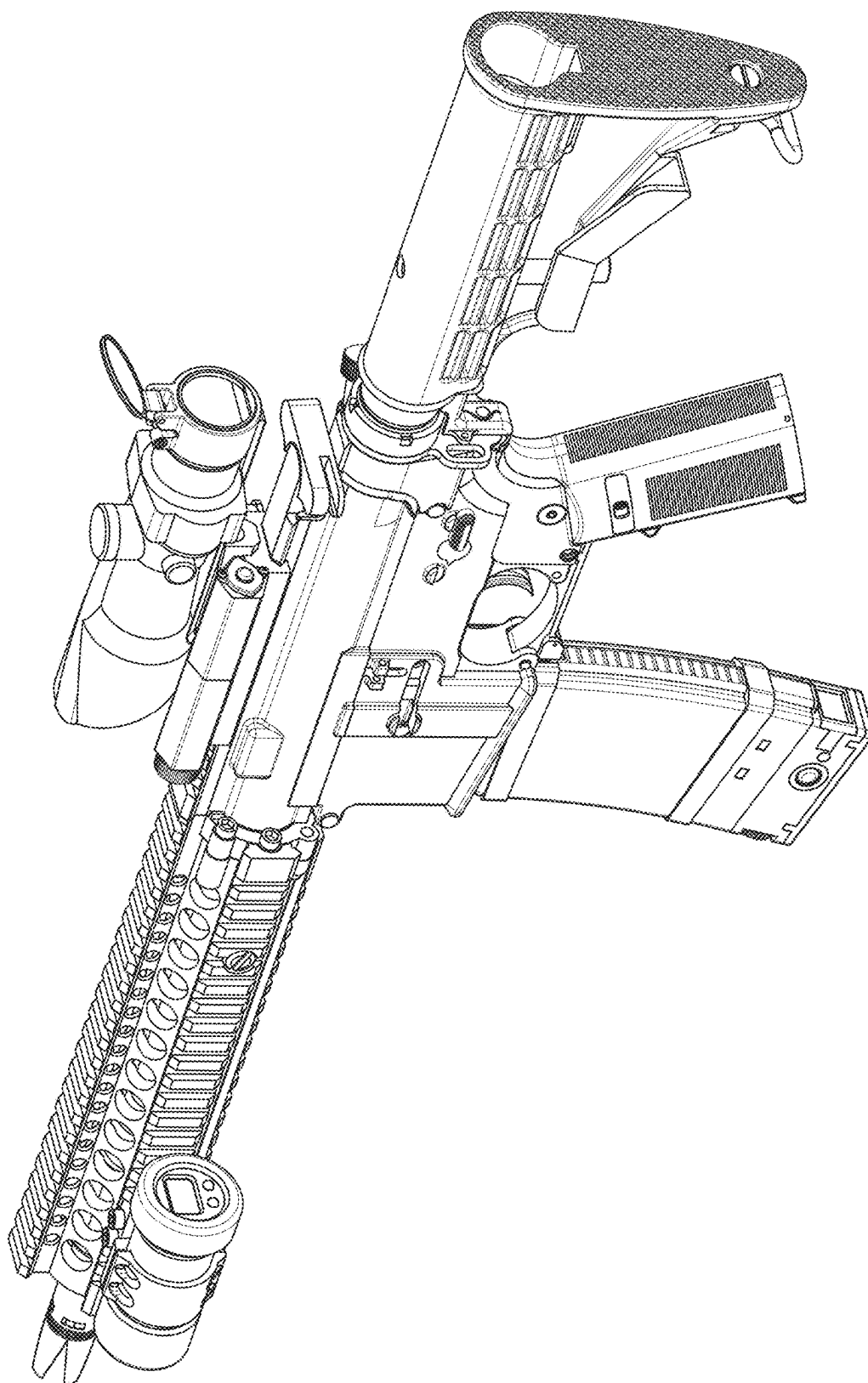


FIG. 14

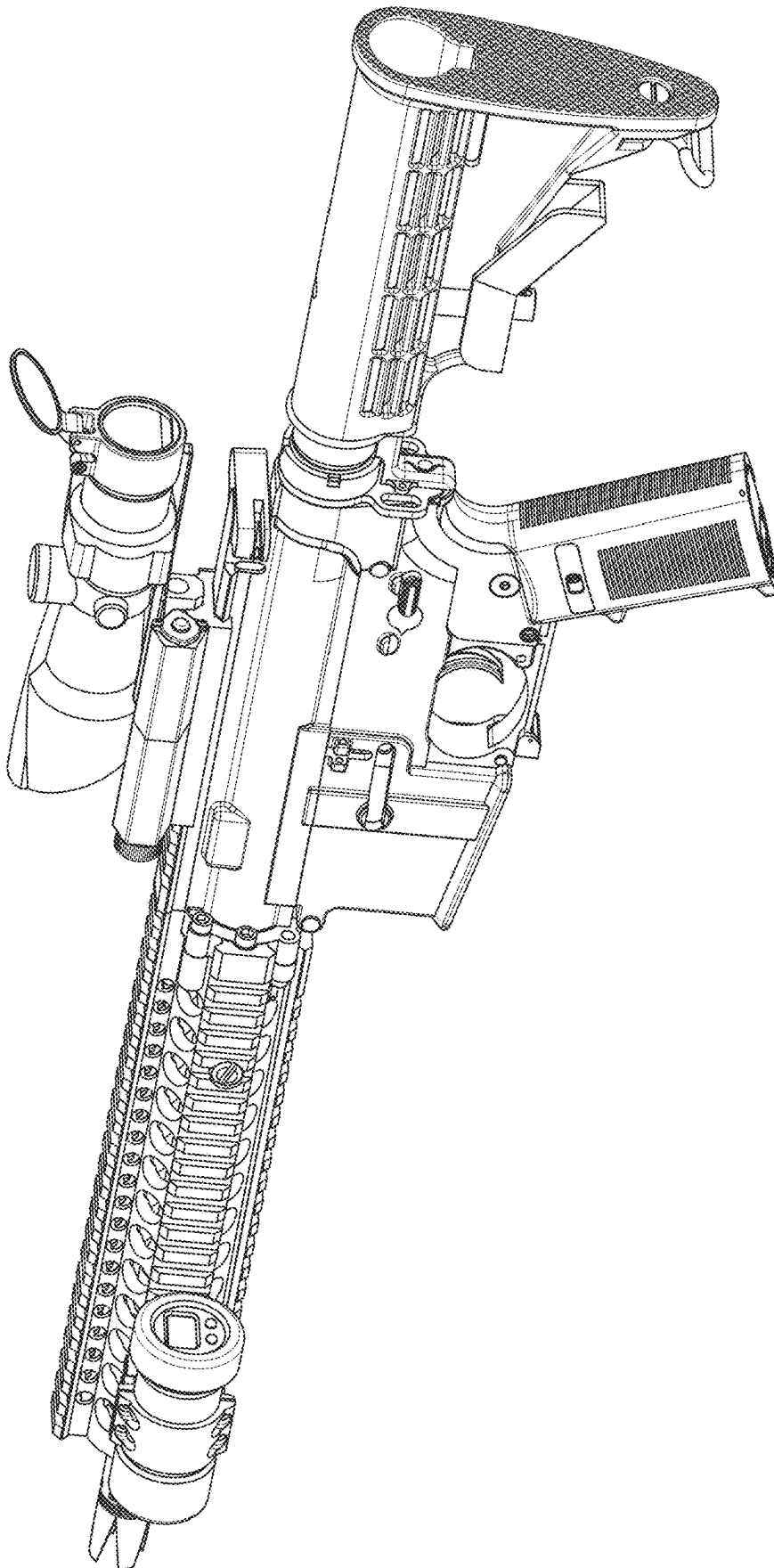


FIG. 15

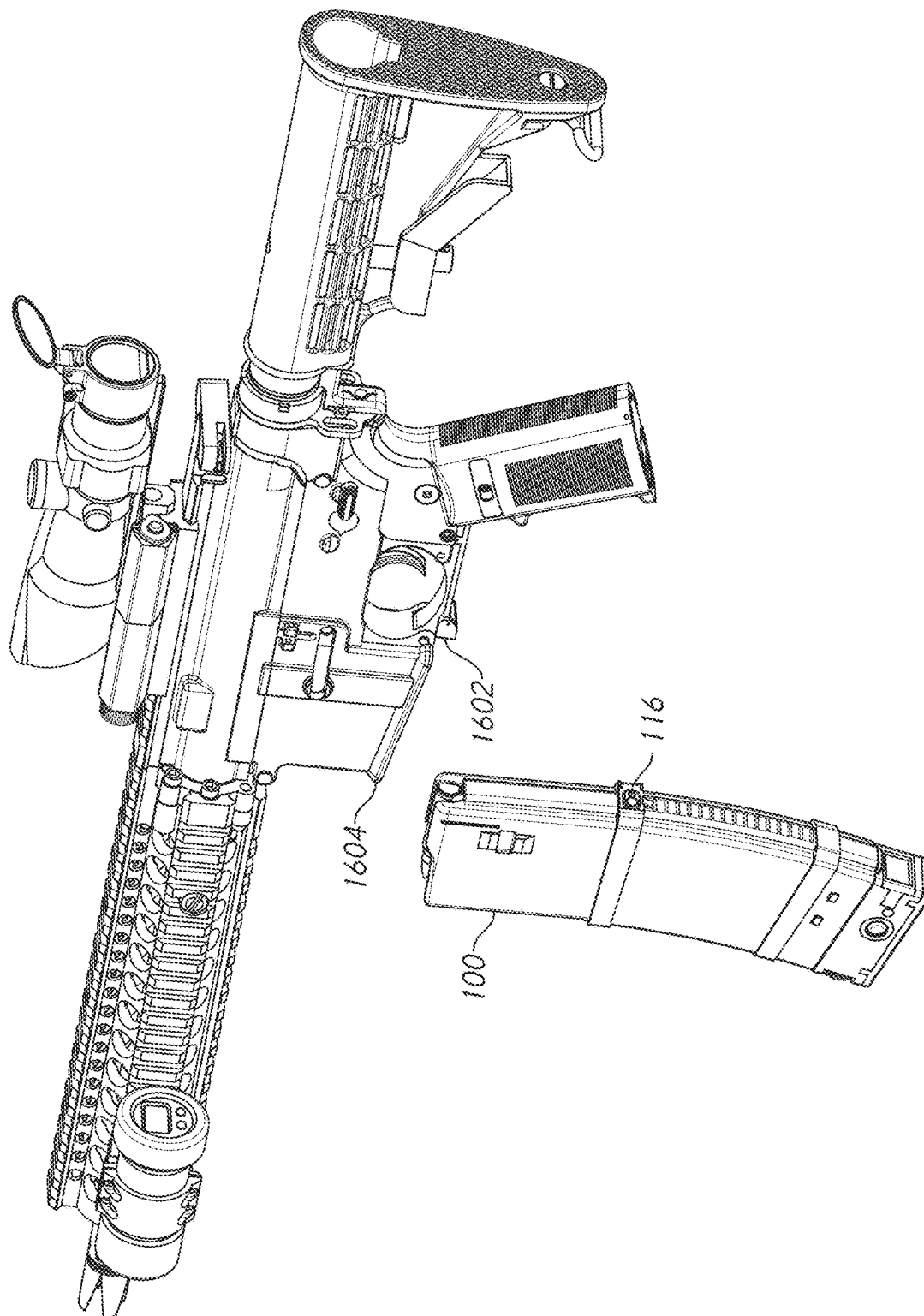


FIG. 16

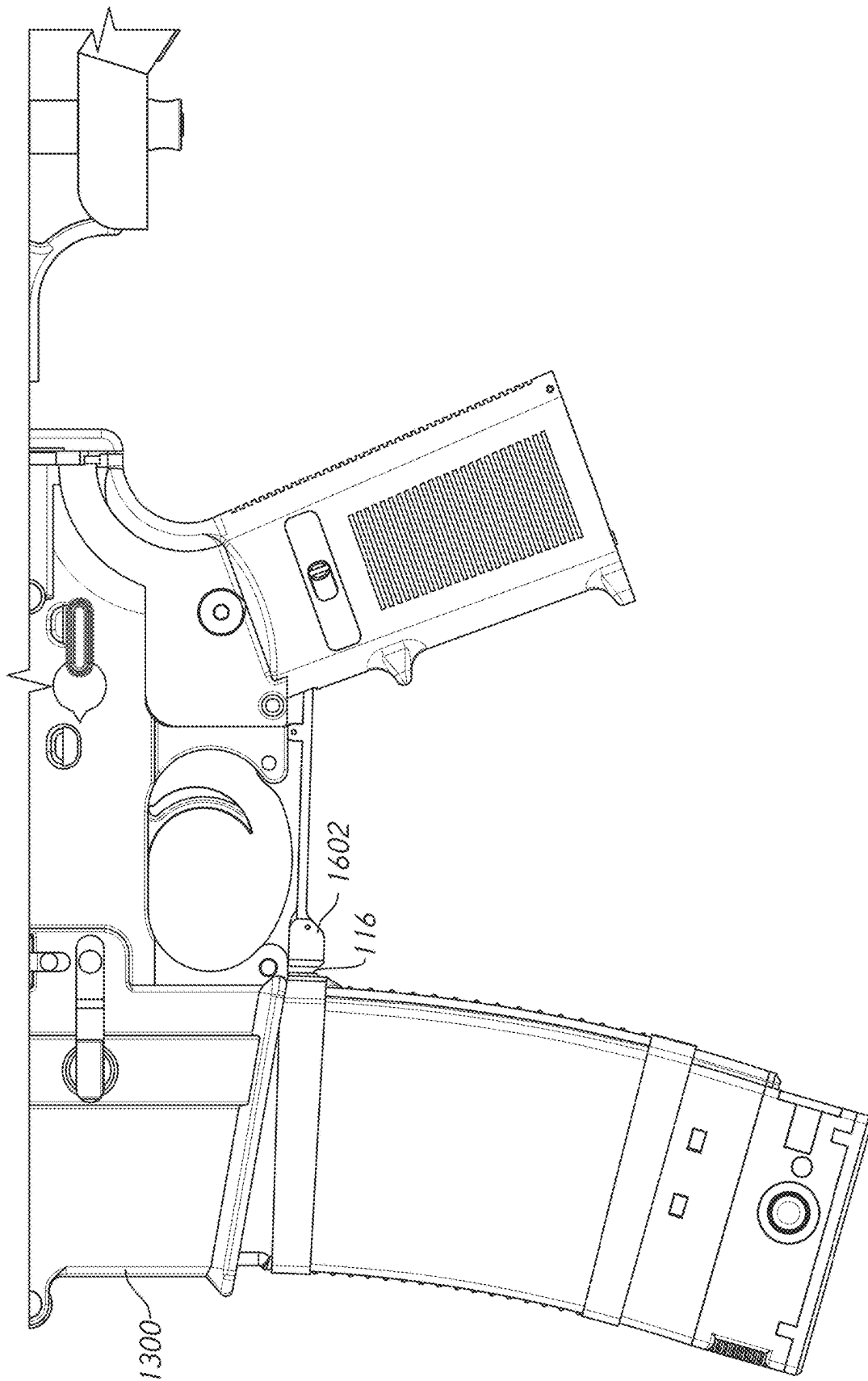


FIG. 17

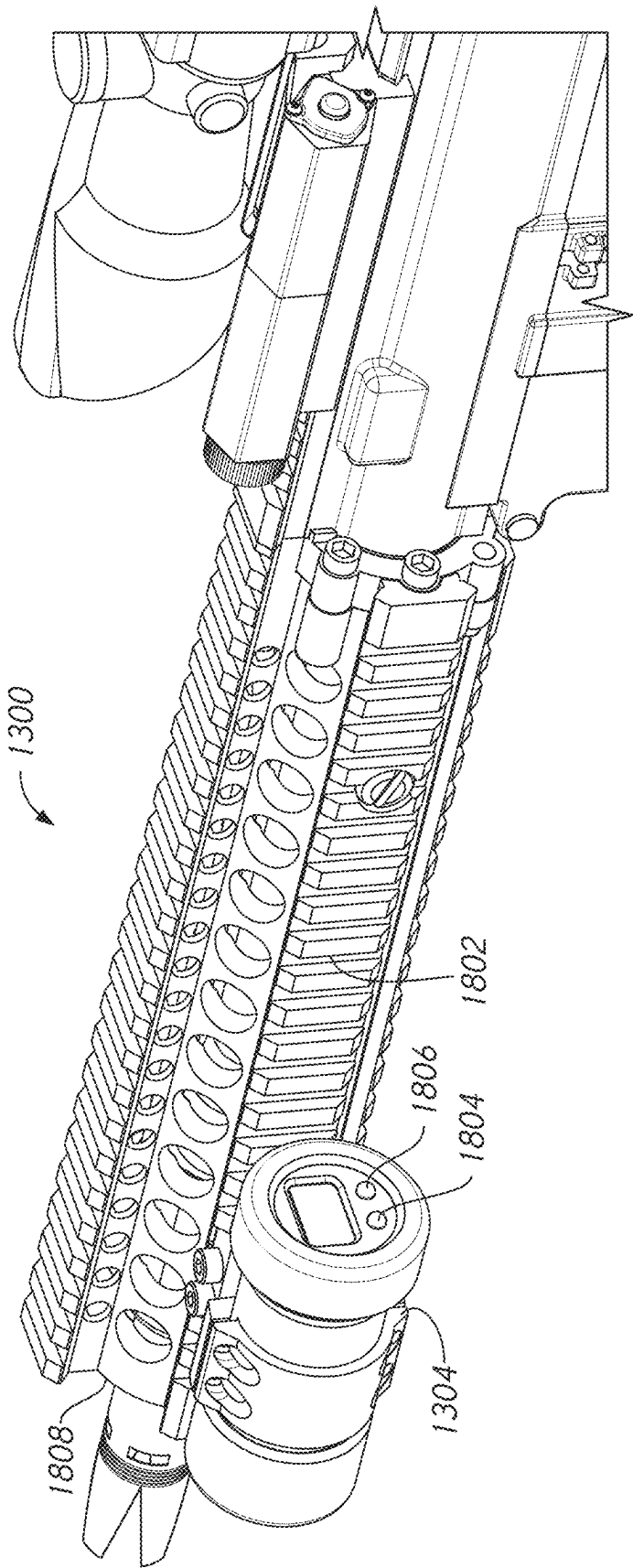


FIG. 18

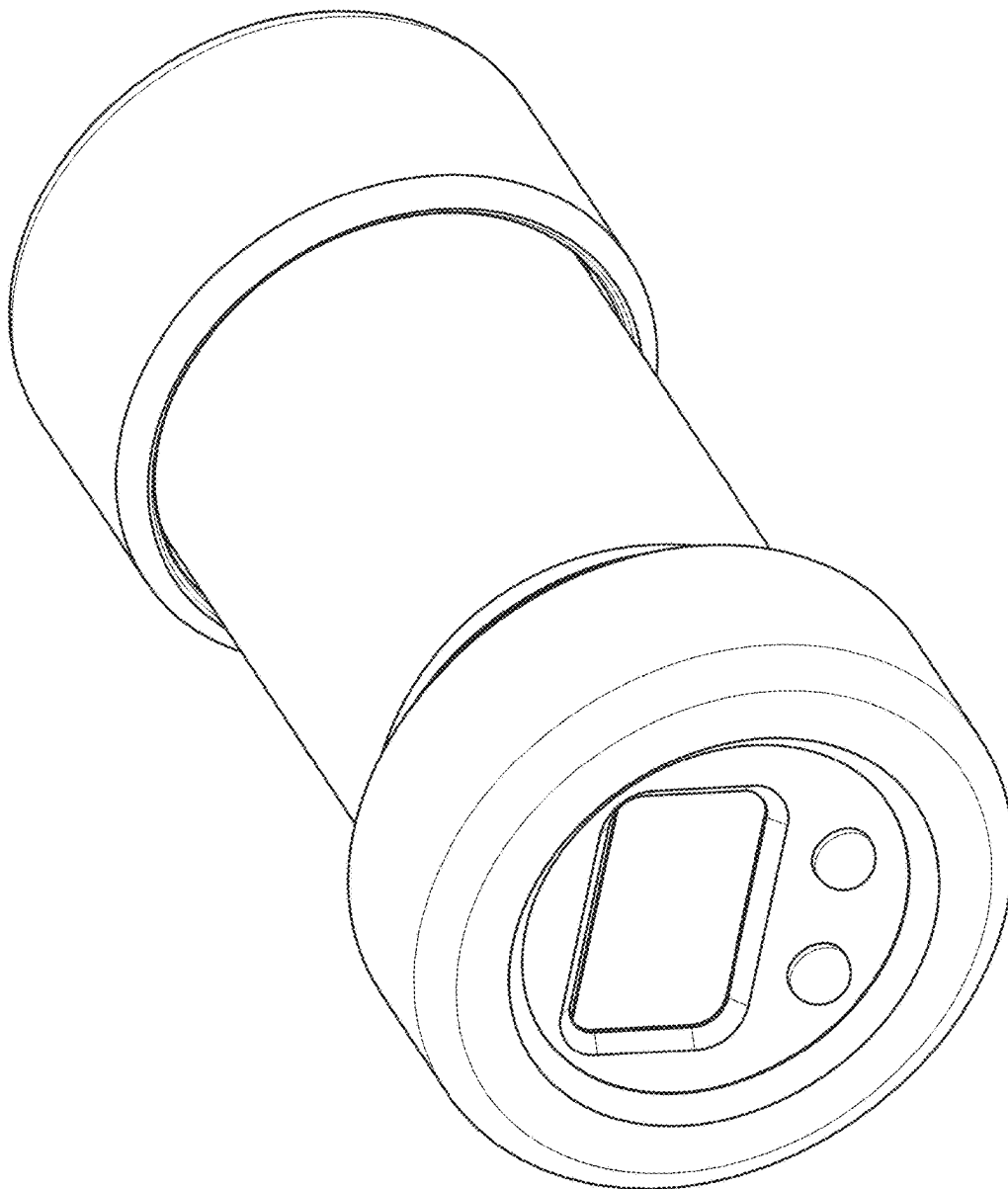


FIG. 19

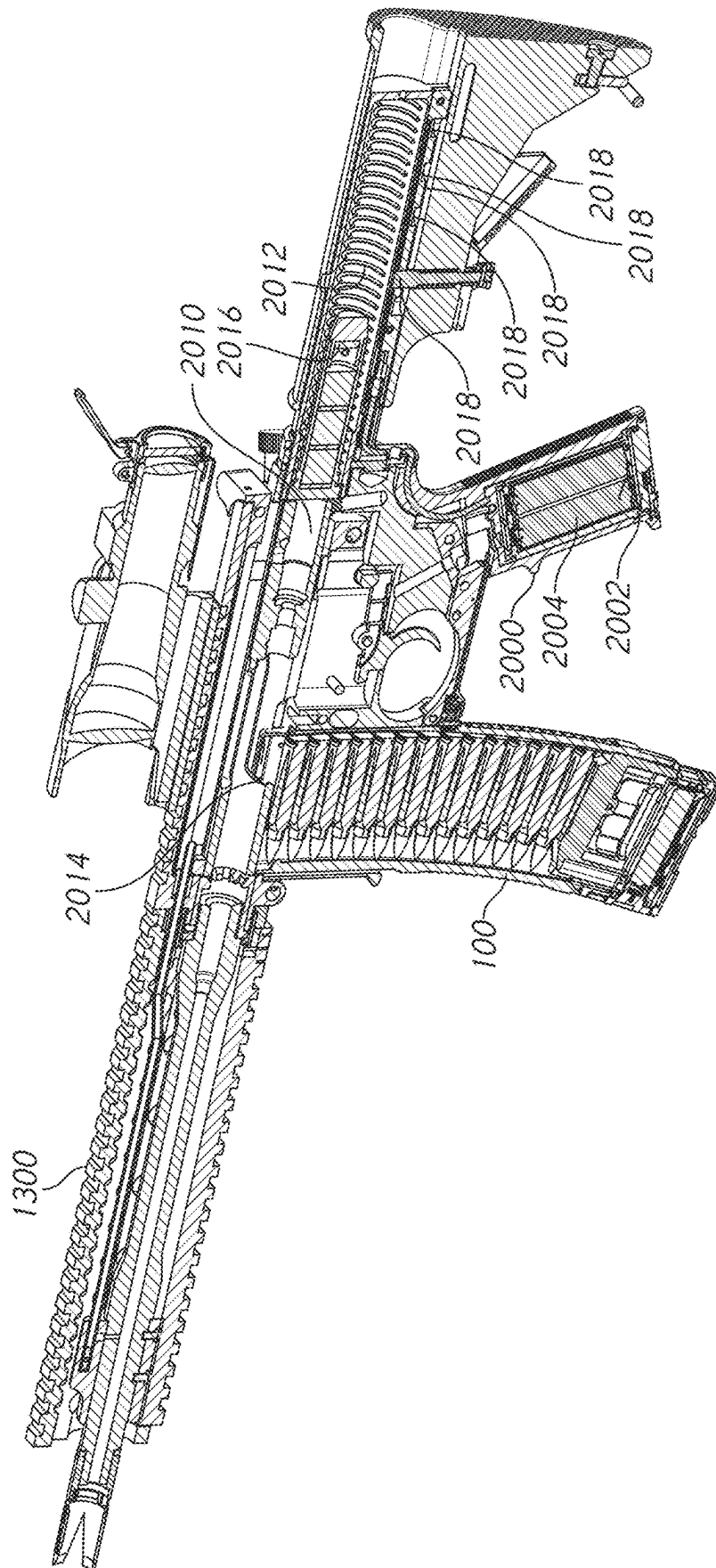


FIG. 20

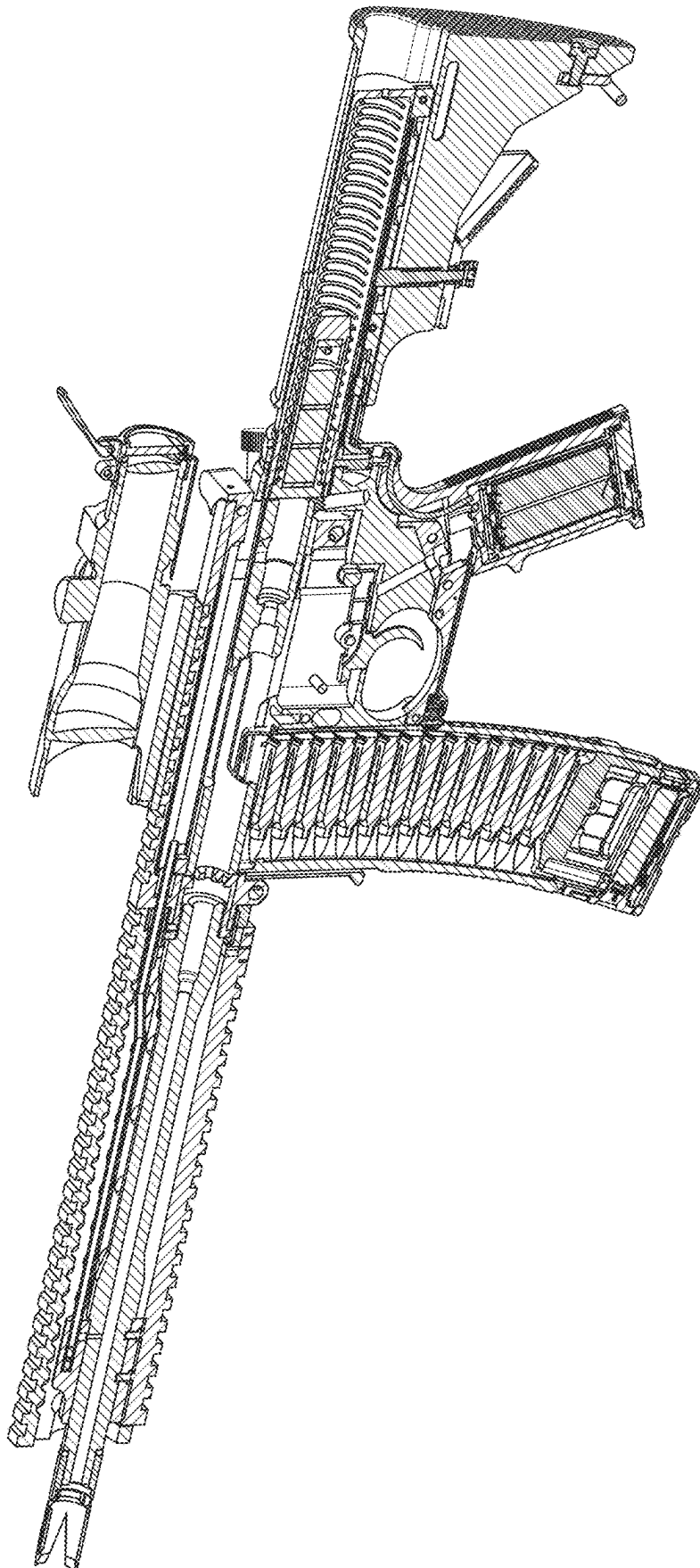


FIG. 21

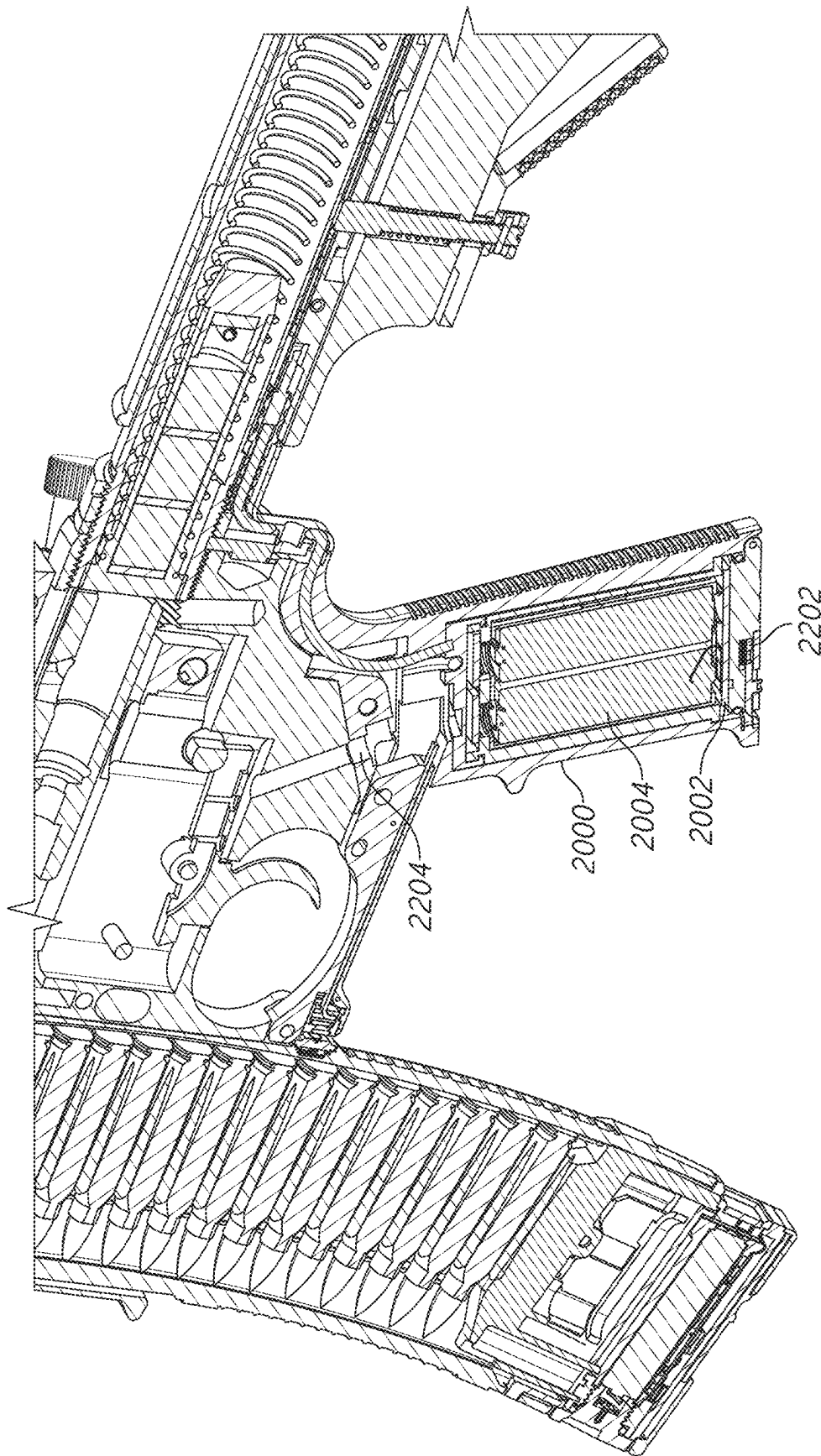


FIG. 22

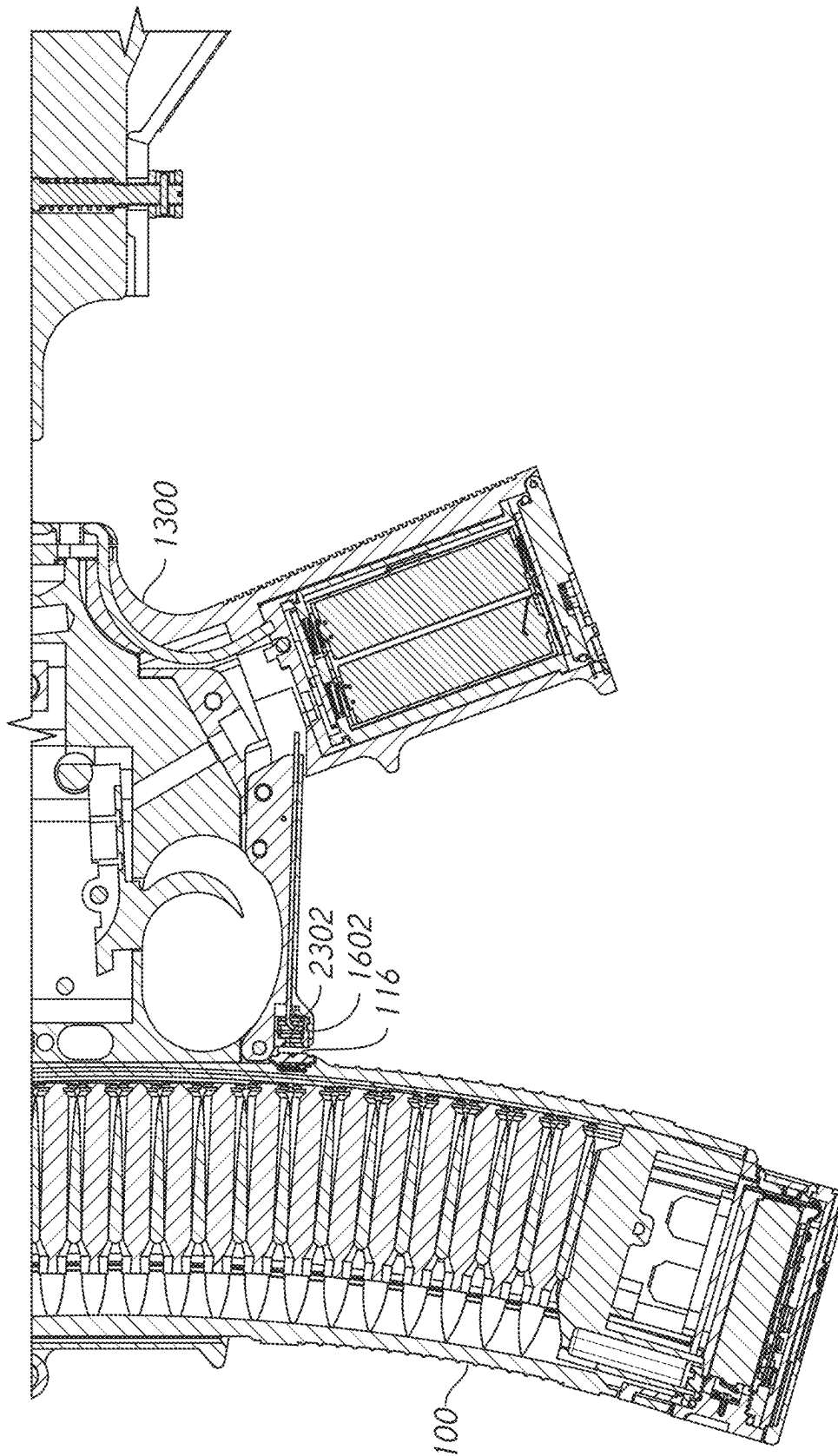
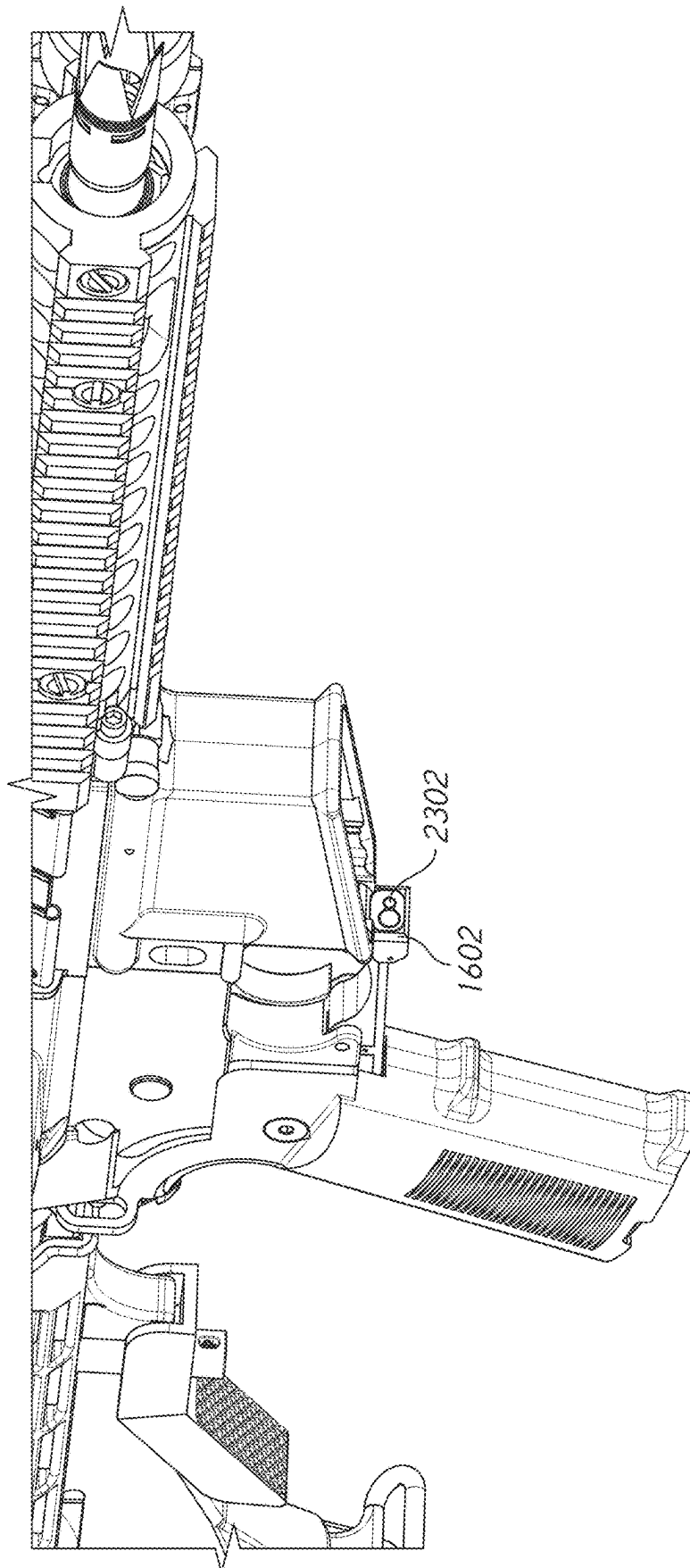


FIG. 23



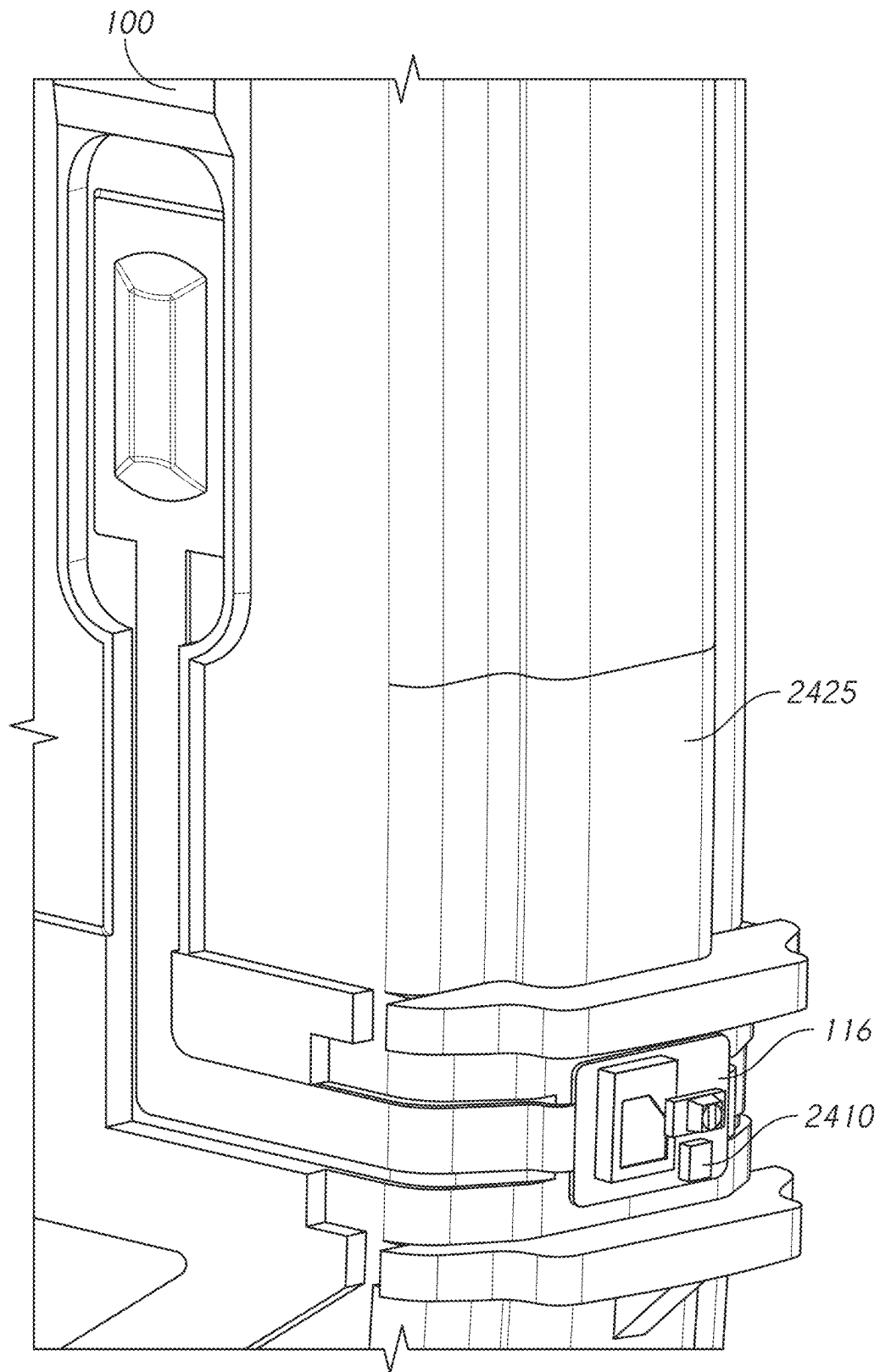


FIG. 24B

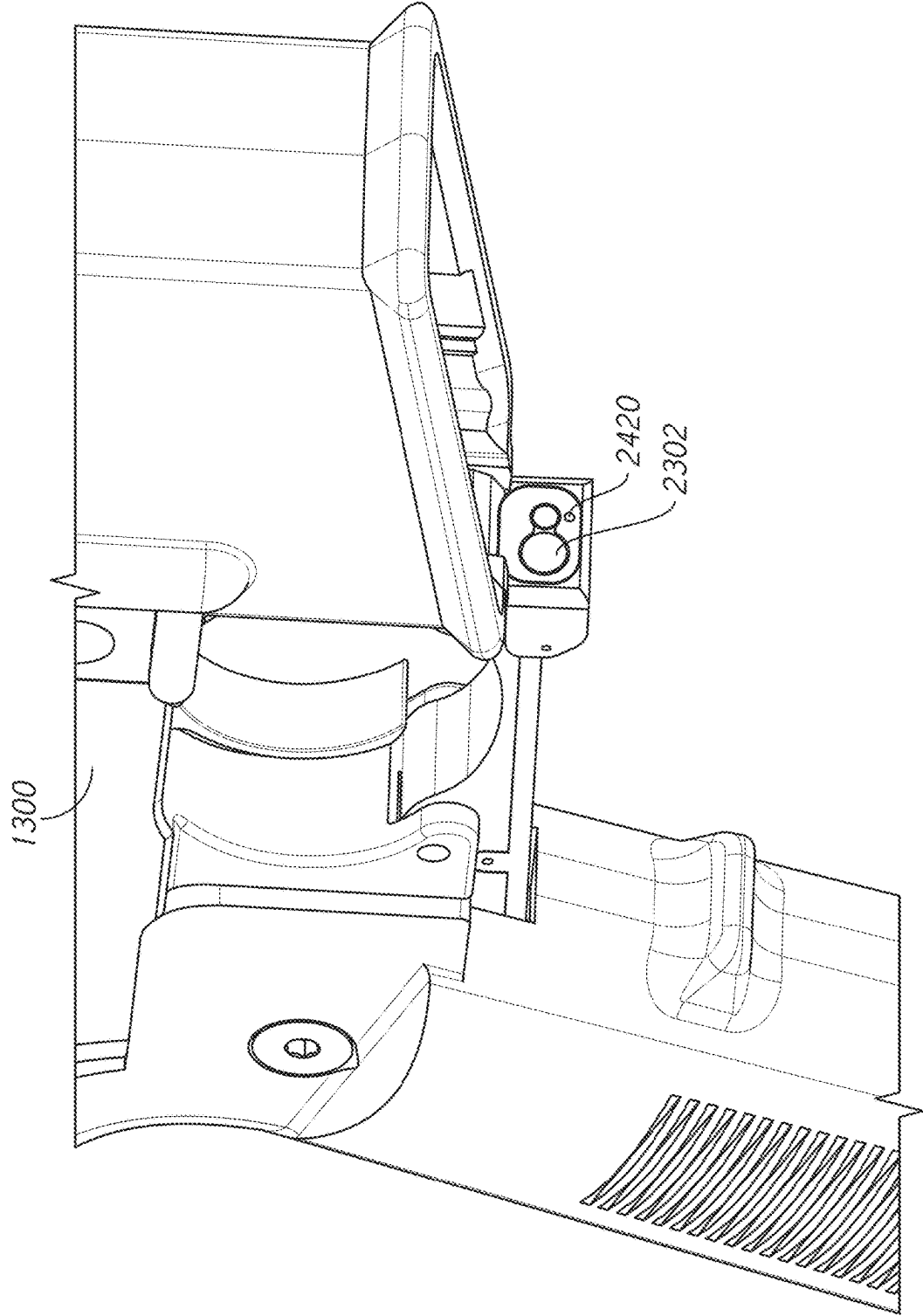


FIG. 24C

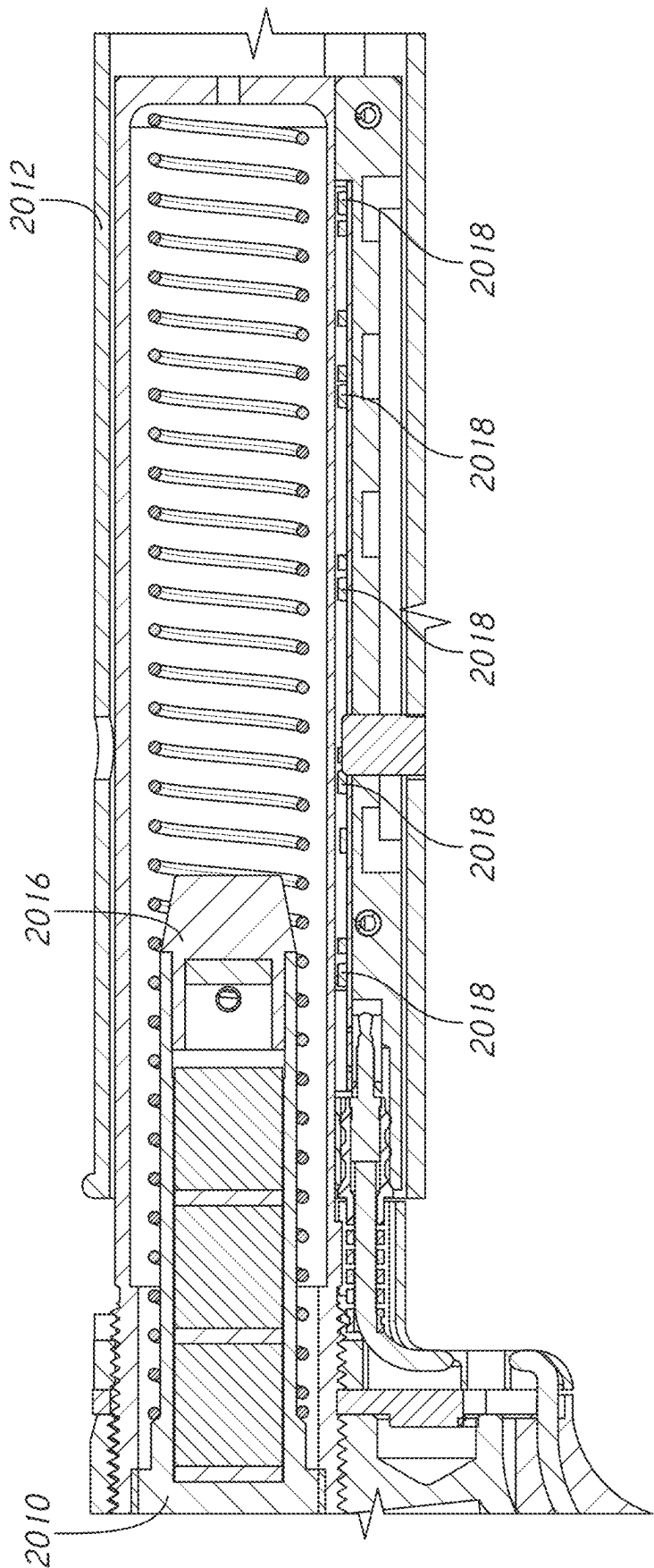


FIG. 25

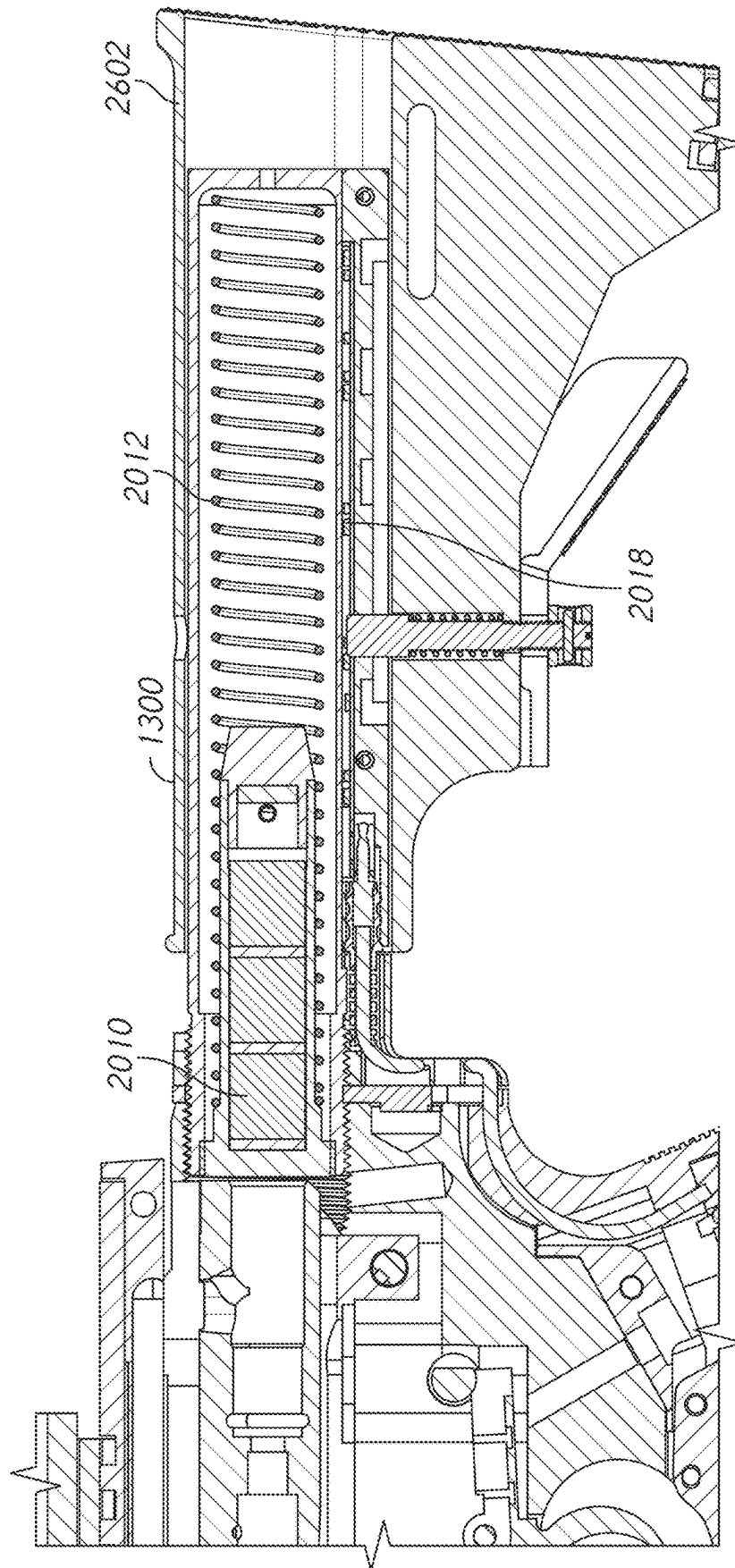


FIG. 26

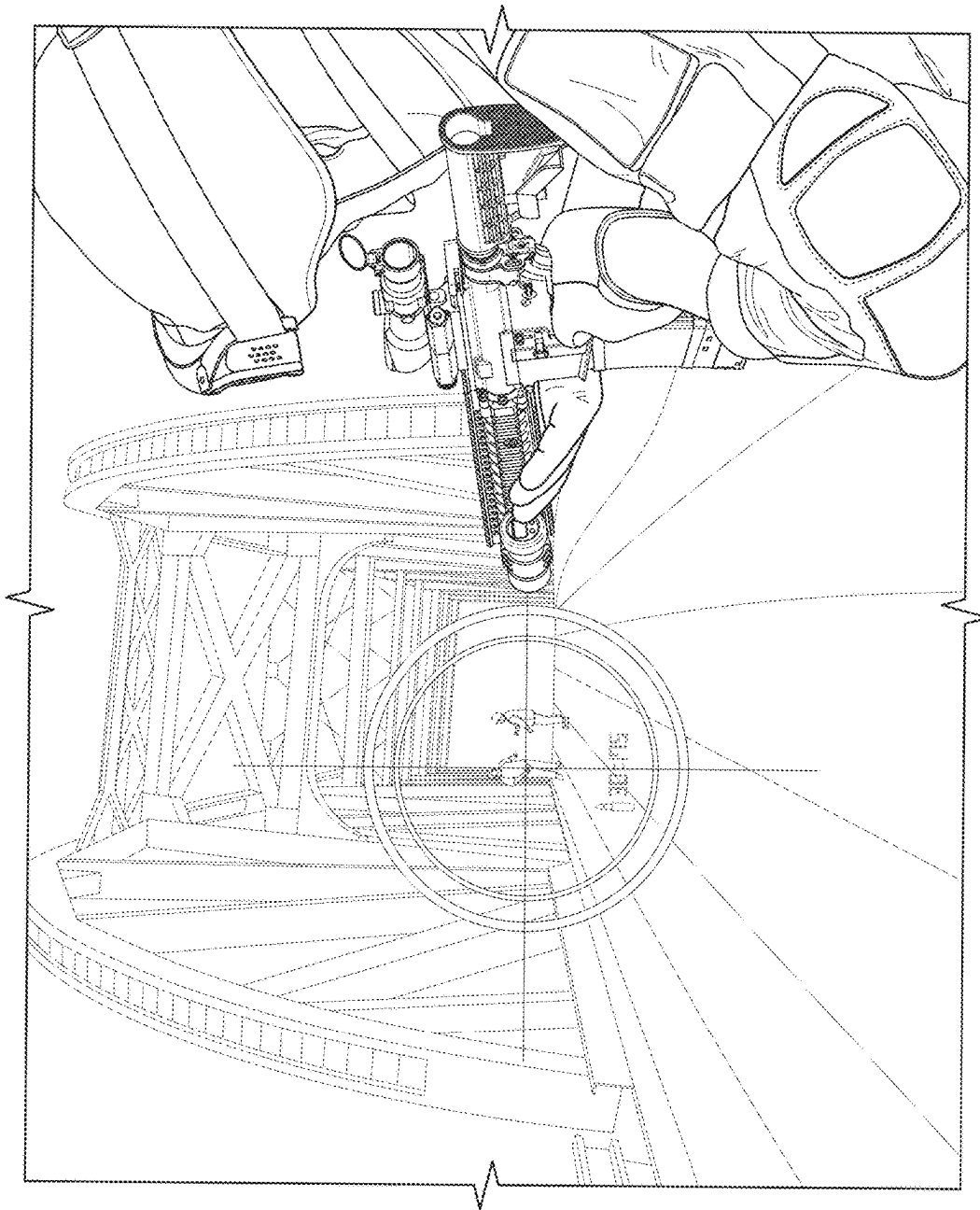


FIG. 27

AMMUNITION COUNT DETERMINATION PROCESS

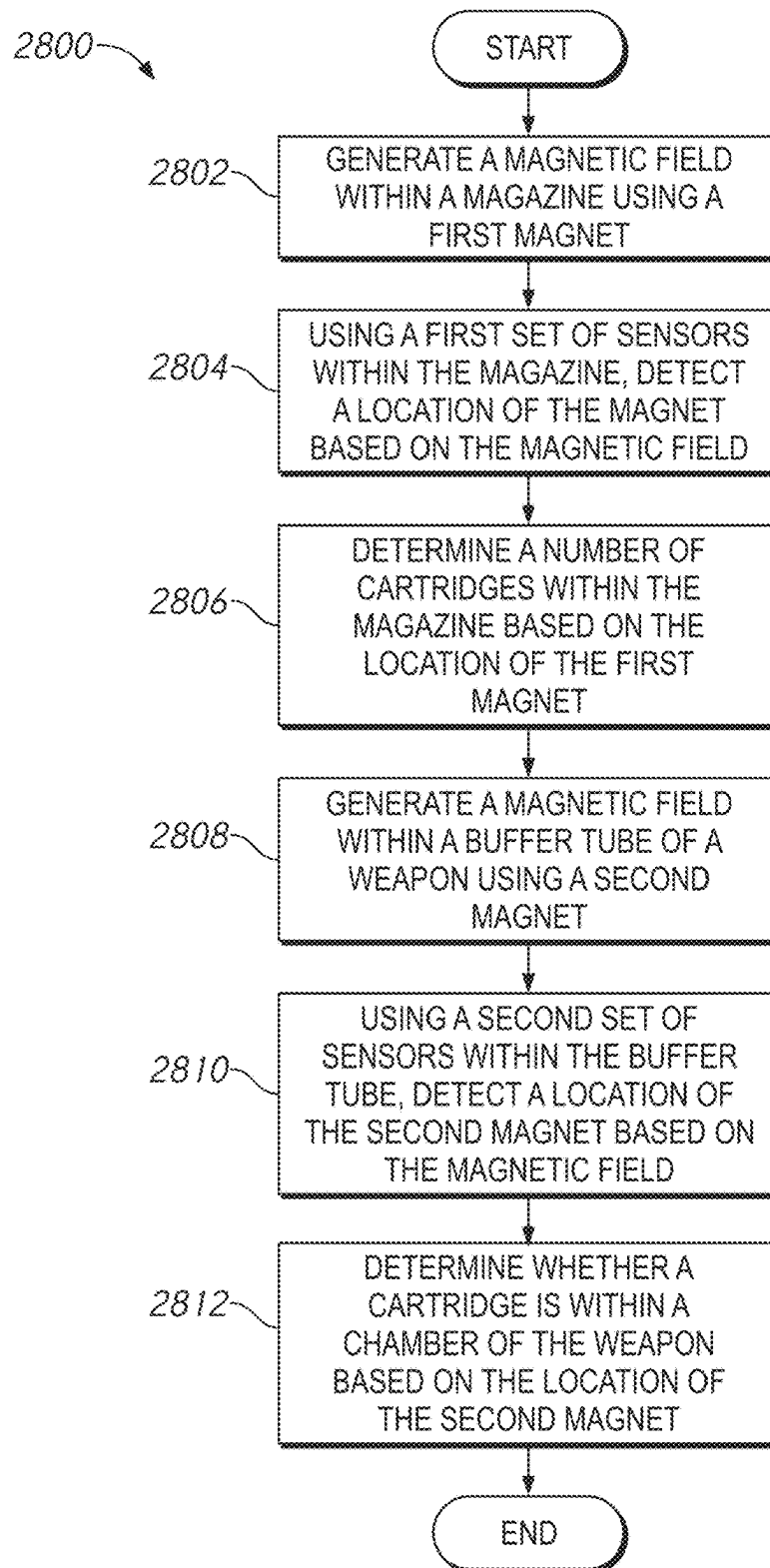


FIG. 28

AMMUNITION COUNT DISPLAY PROCESS

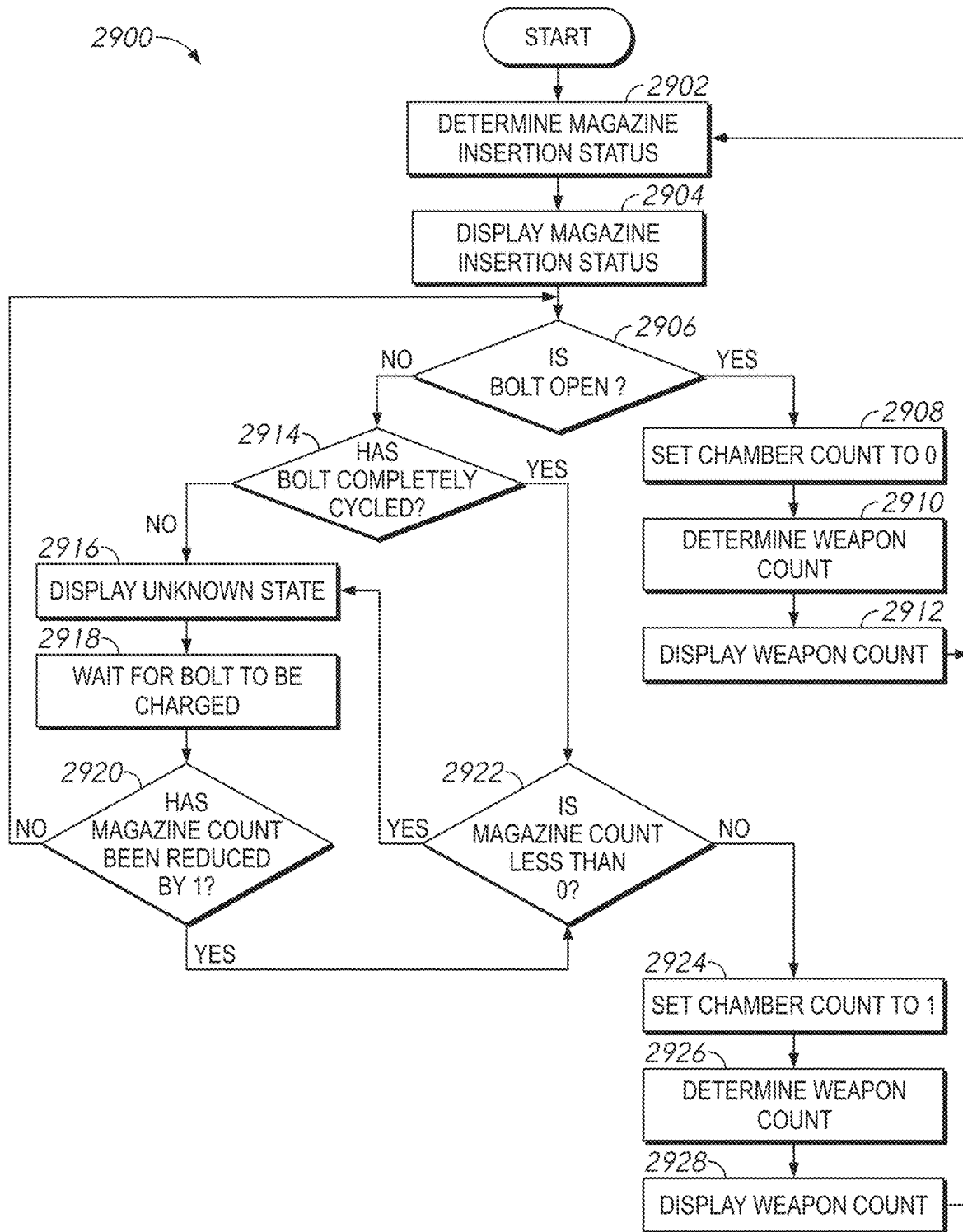
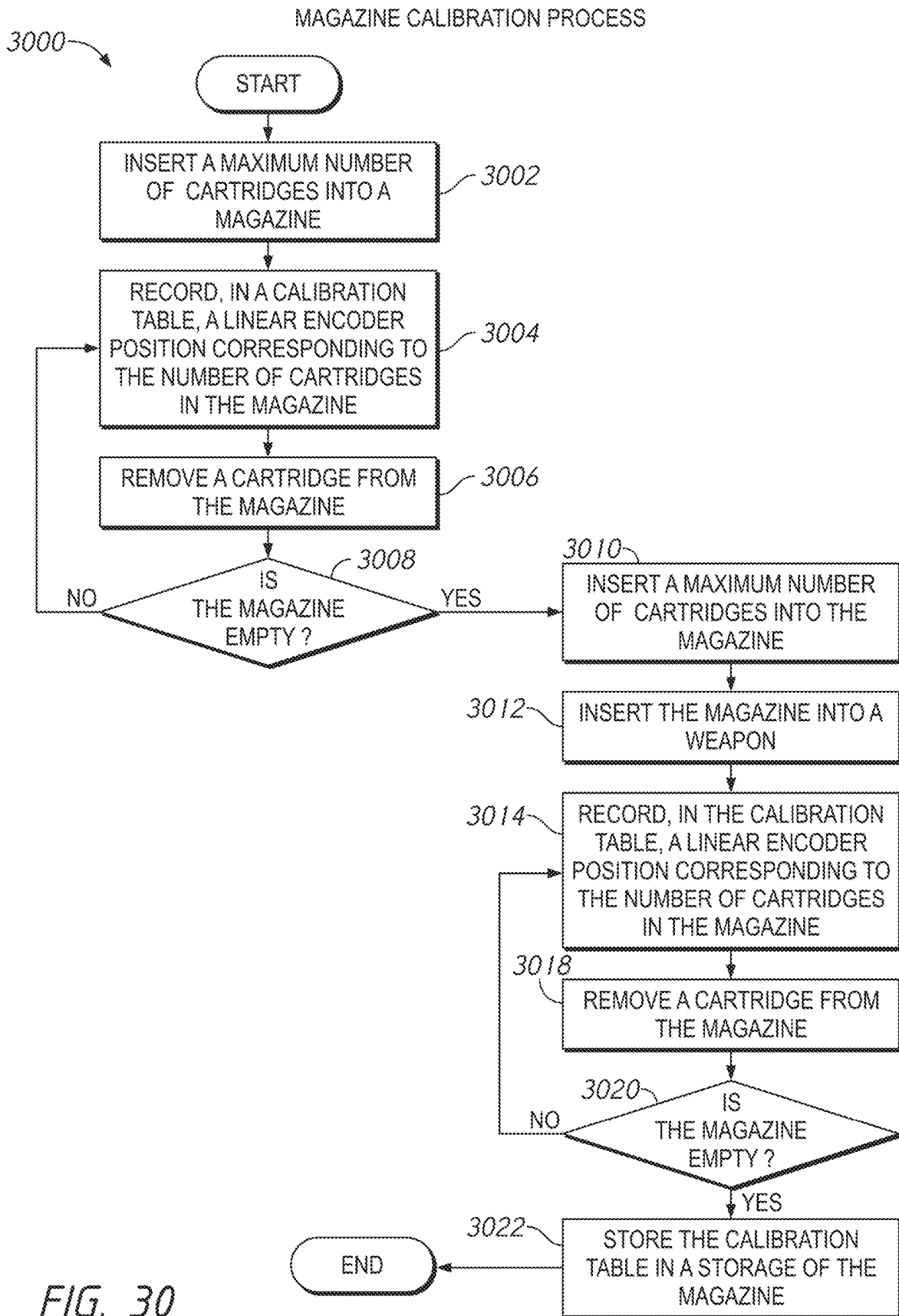
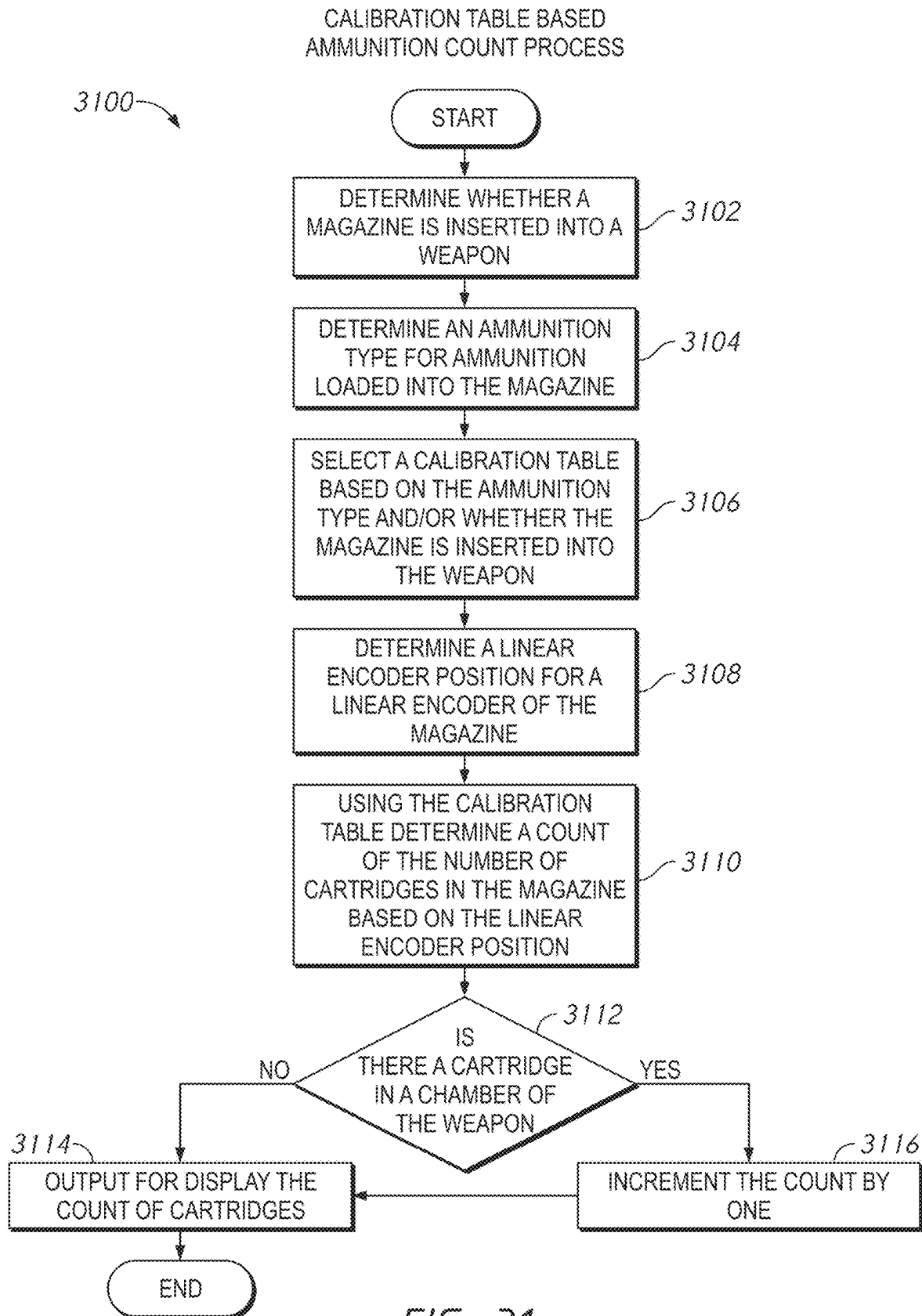


FIG. 29





MAGAZINE RECALIBRATION PROCESS

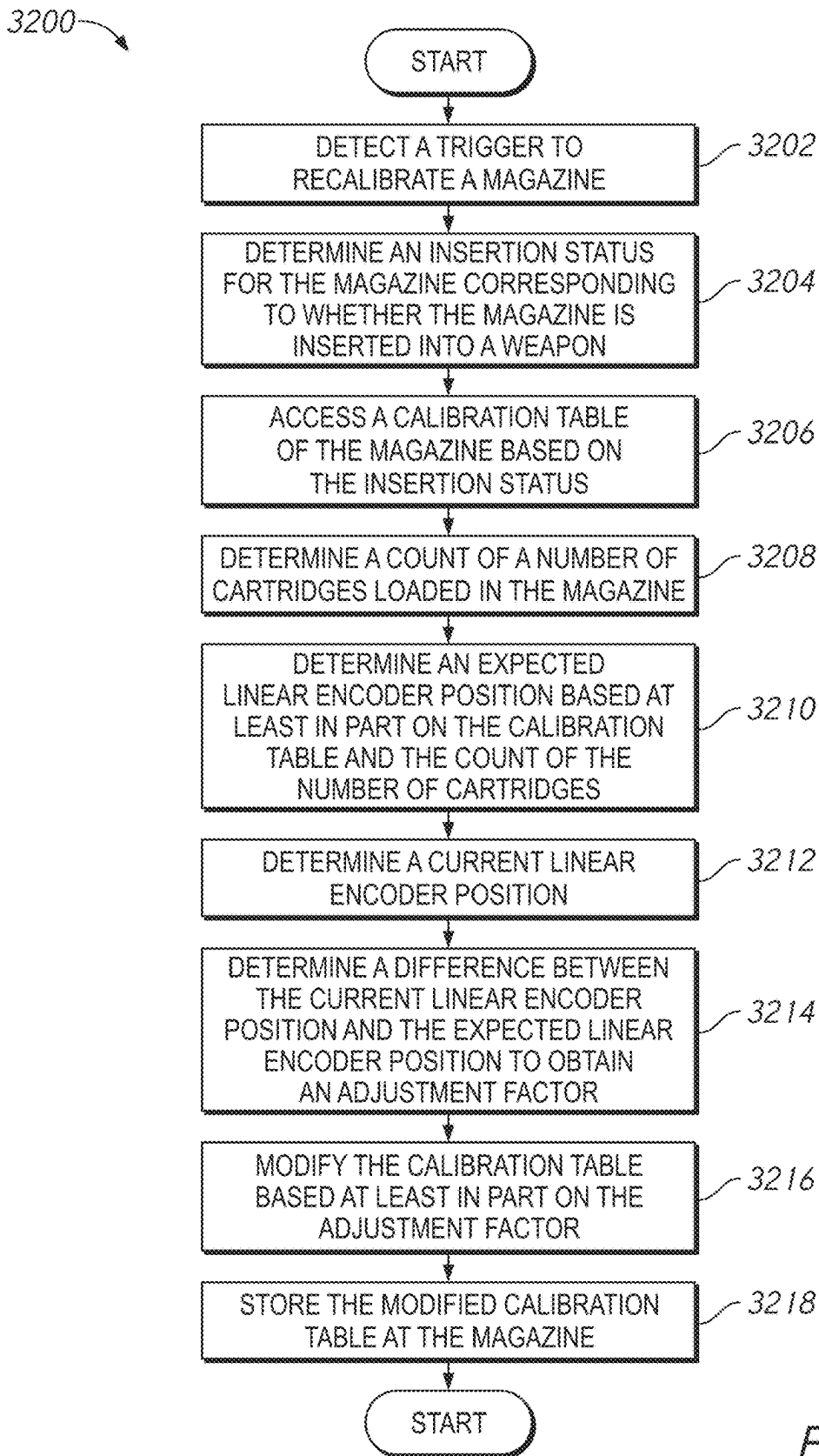


FIG. 32

TOTAL AMMUNITION COUNT PROCESS

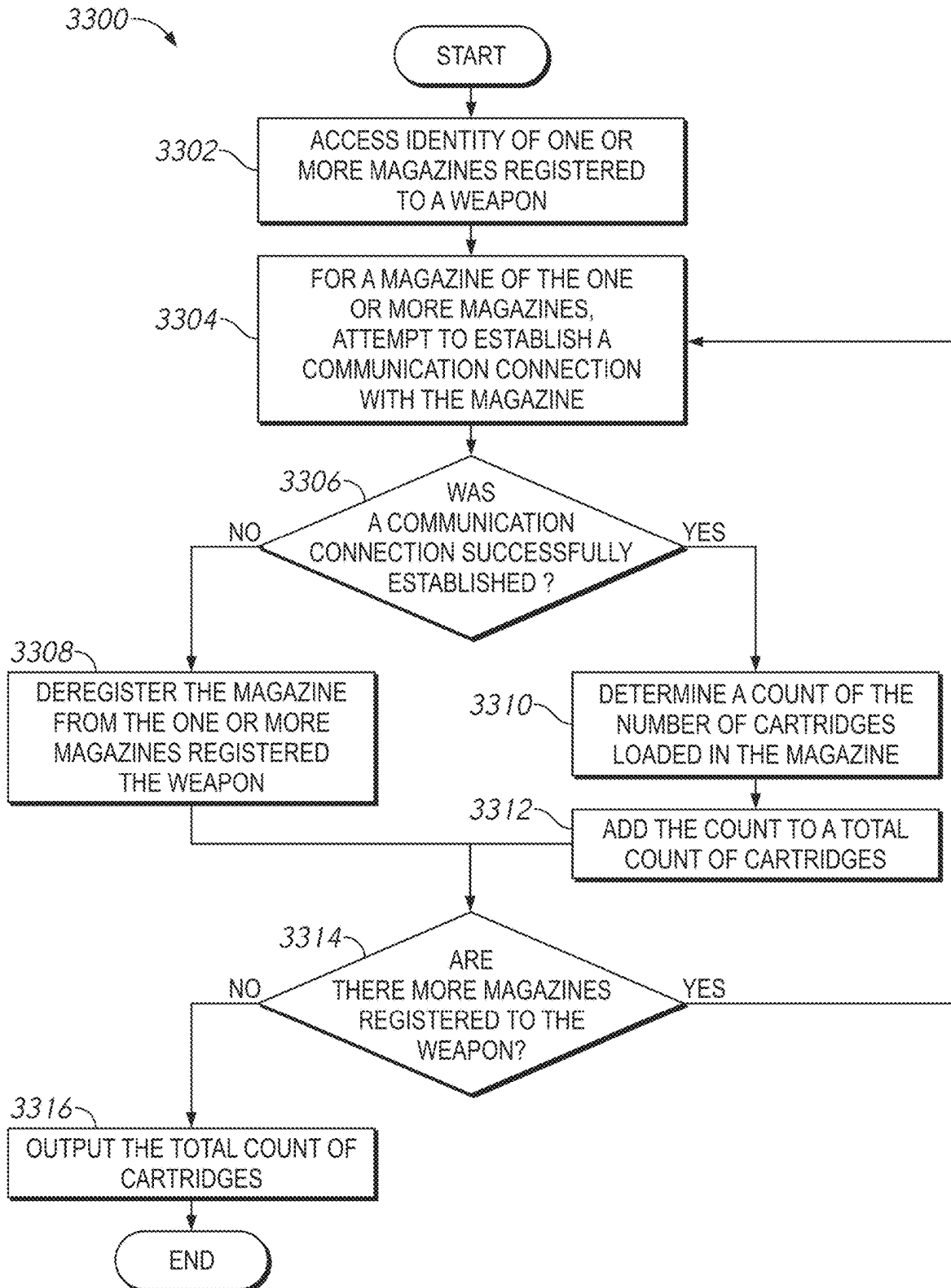


FIG. 33

MAGAZINE REGISTRATION PROCESS

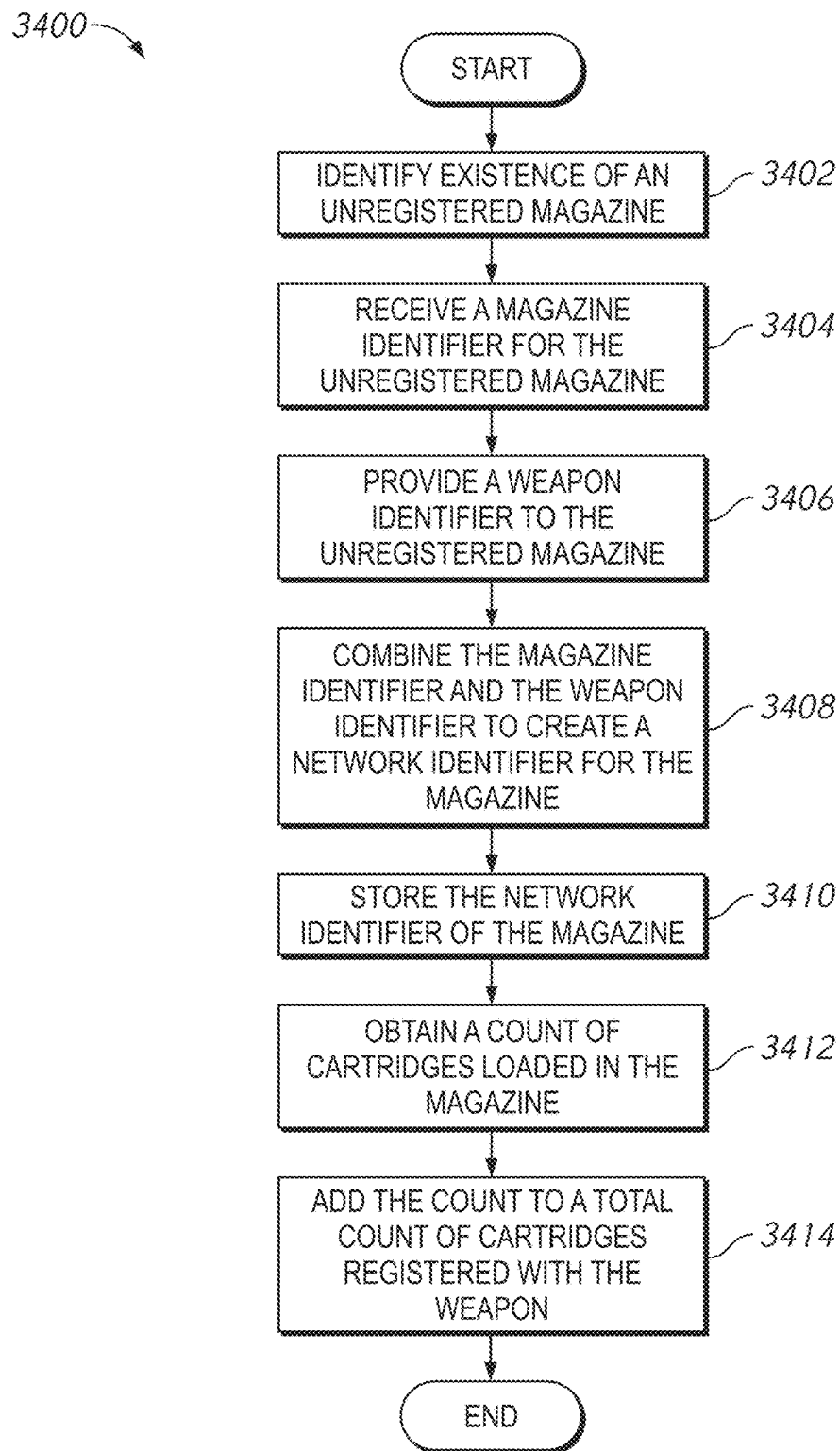


FIG. 34

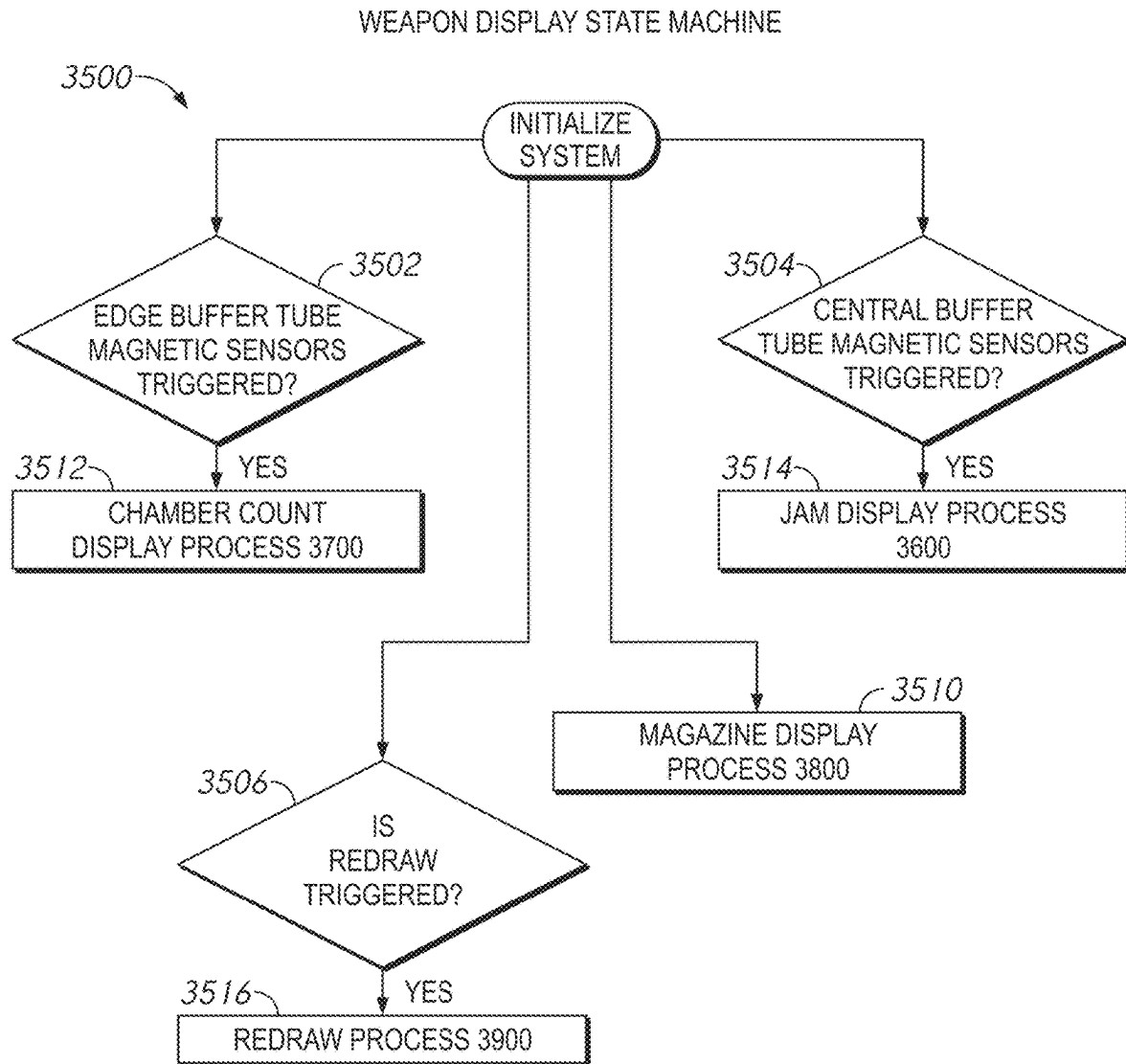


FIG. 35

JAM DISPLAY PROCESS

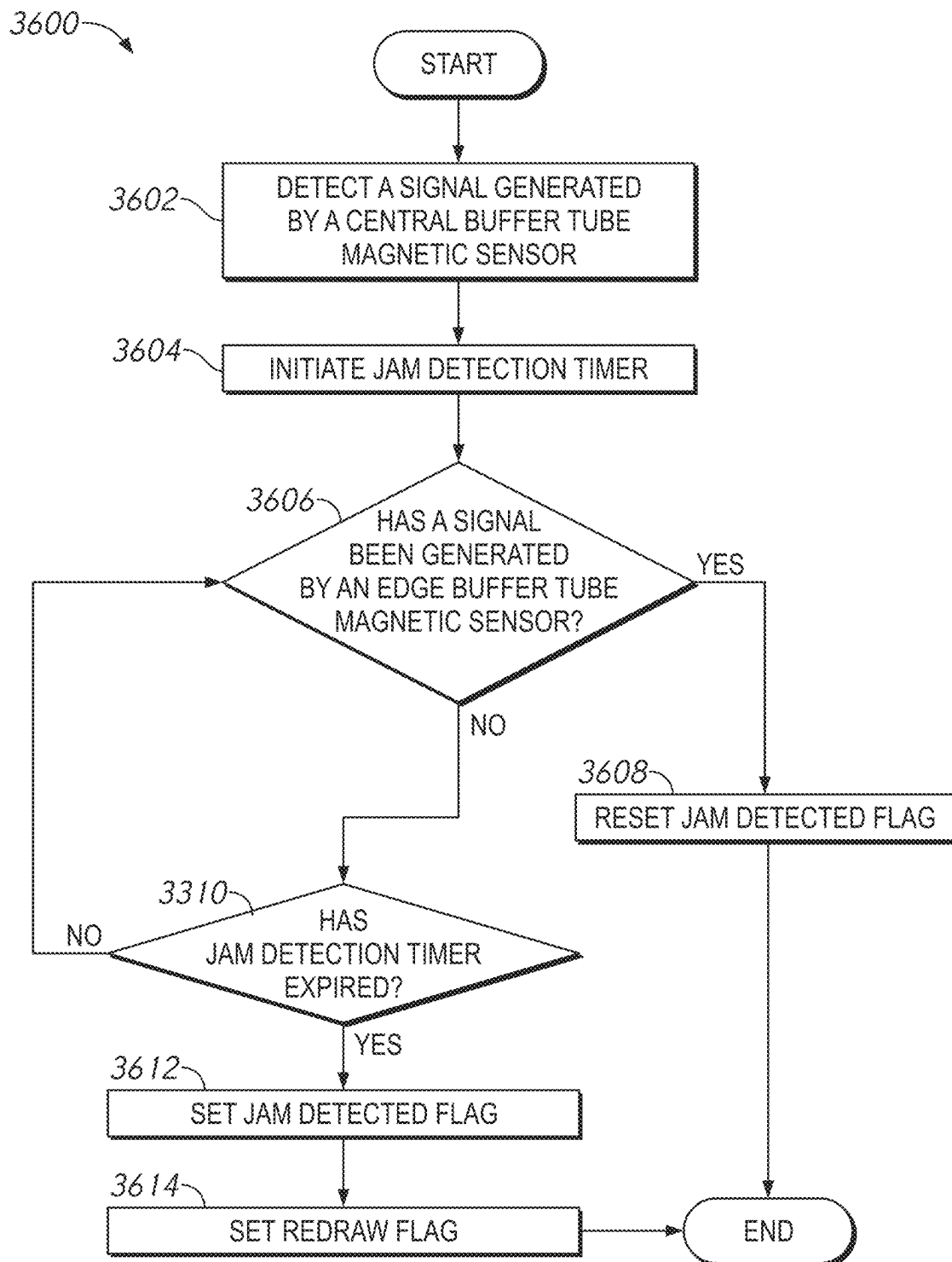


FIG. 36

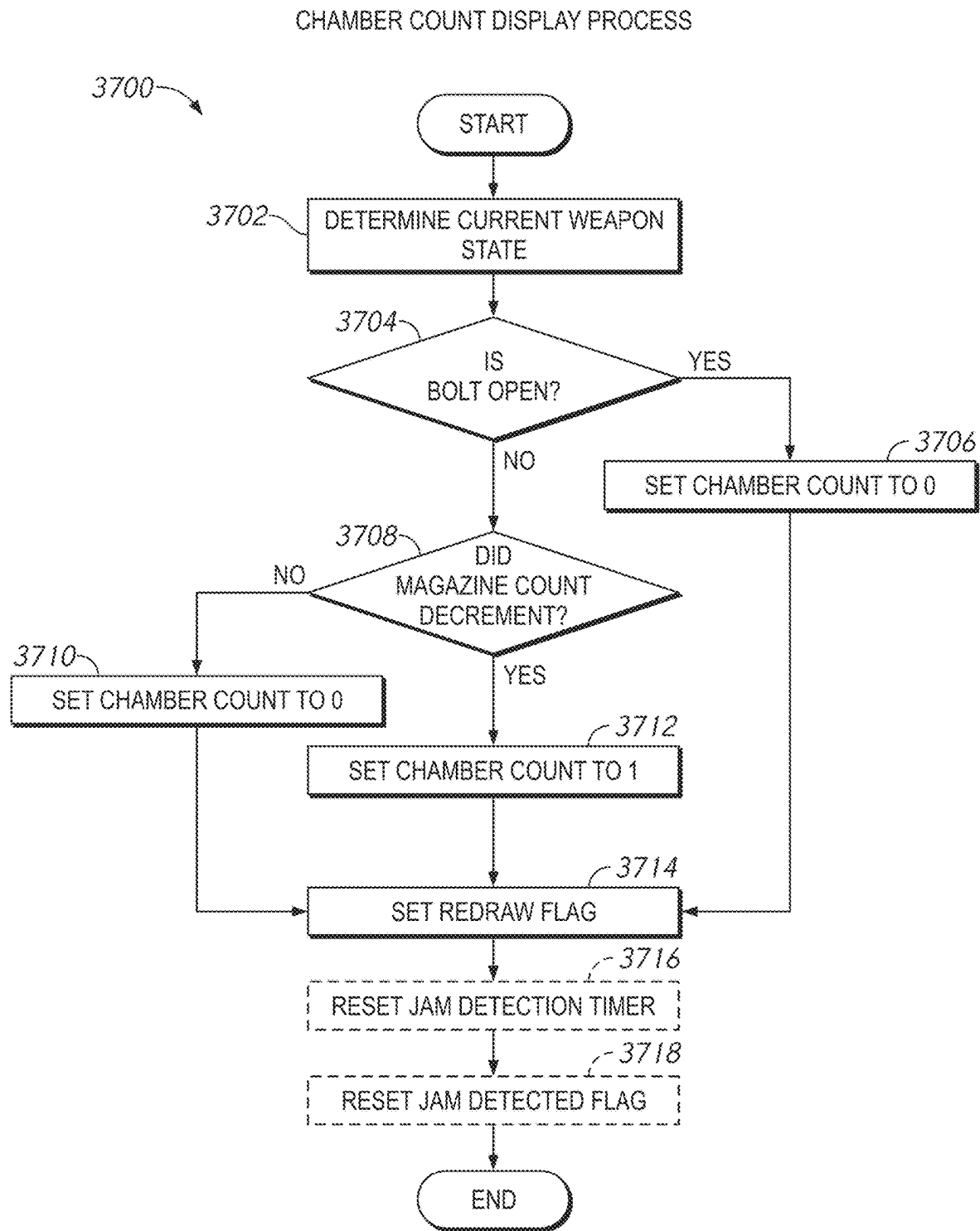


FIG. 37

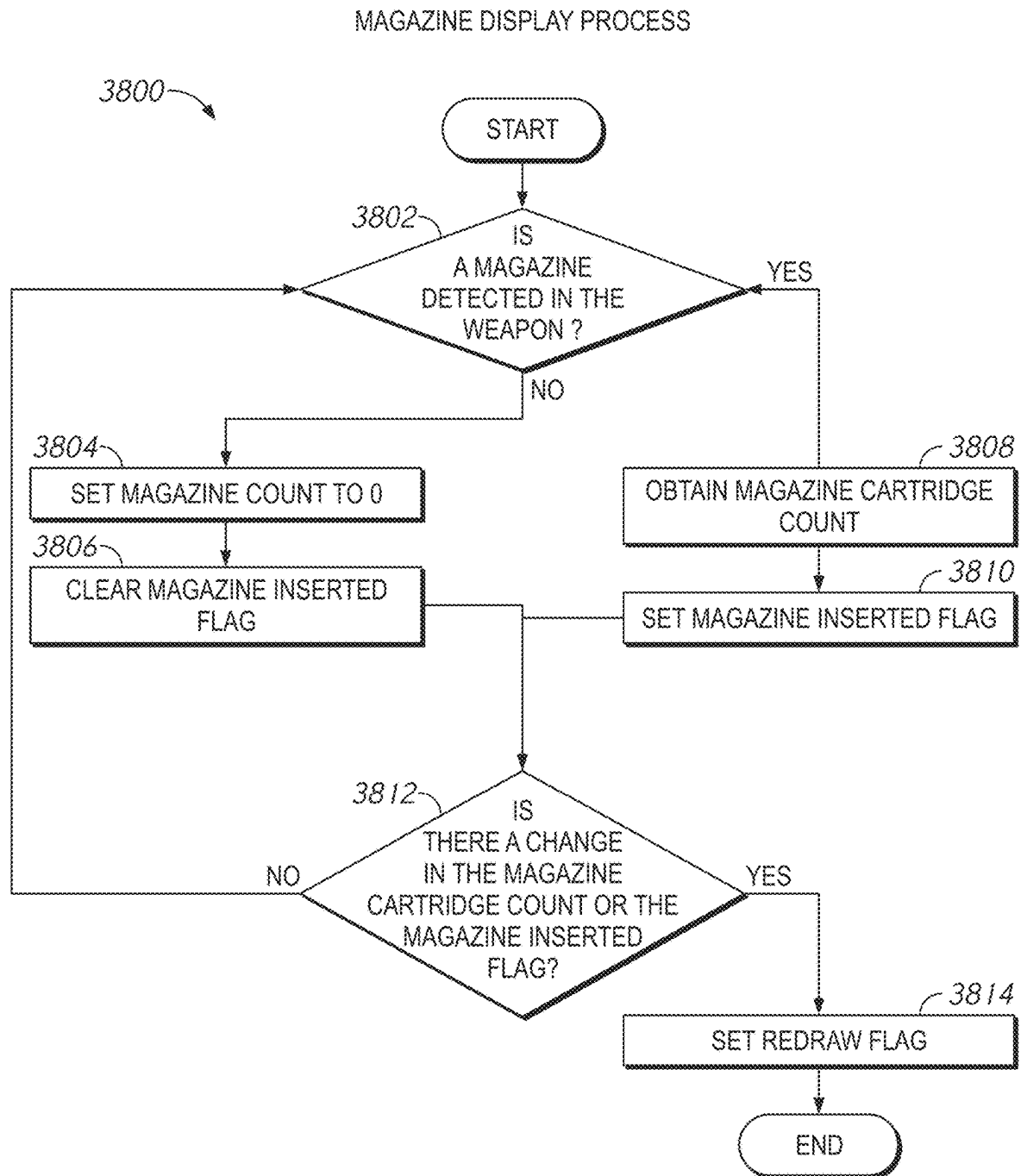


FIG. 38

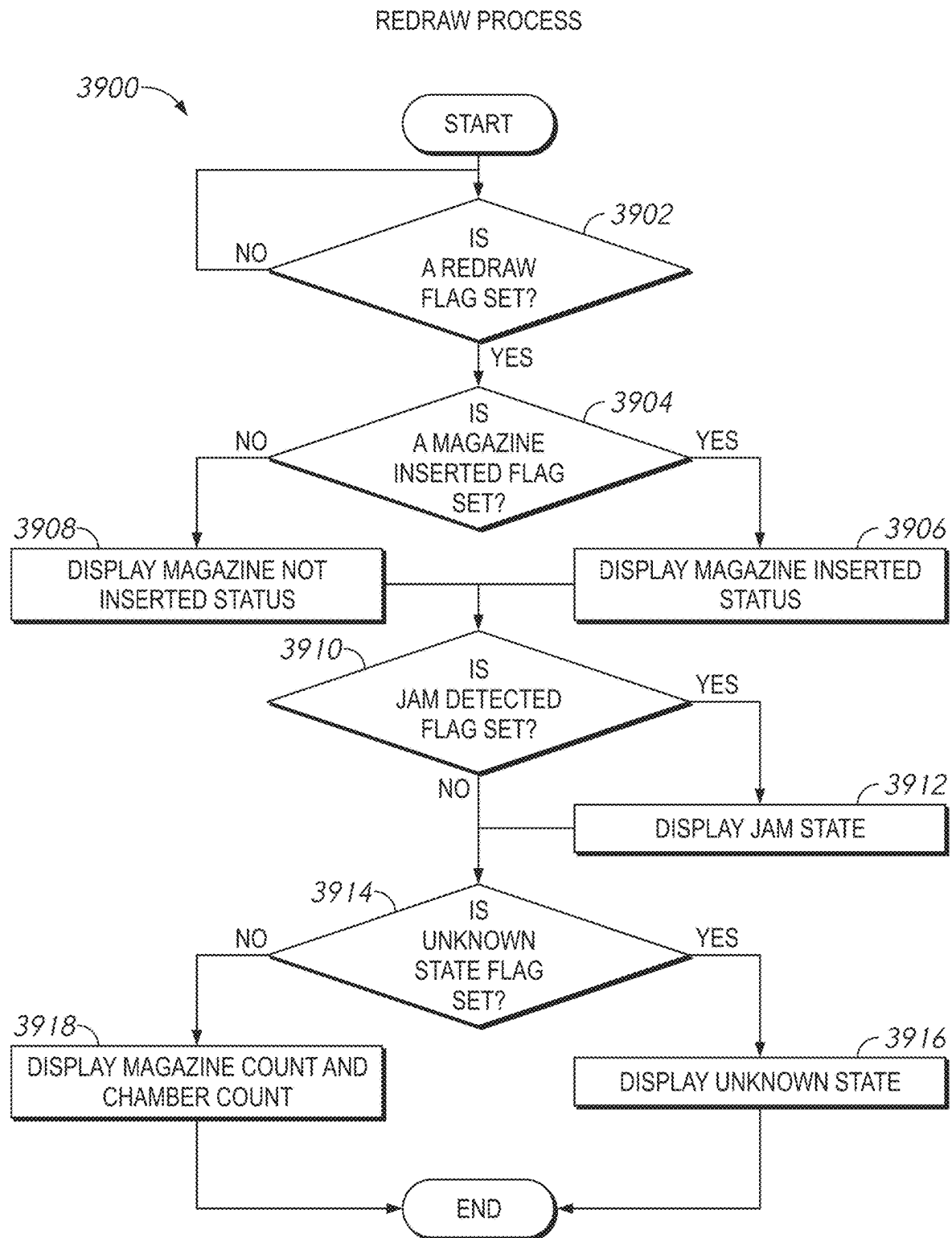


FIG. 39

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FIREARM AMMUNITION AVAILABILITY DETECTION SYSTEM

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/929,648, filed Sep. 2, 2022 and titled "FIREARM AMMUNITION AVAILABILITY DETECTION SYSTEM," the disclosure of which is hereby incorporated by reference in its entirety for all purposes herein, and which is a continuation of U.S. application Ser. No. 17/129,066, filed Dec. 21, 2020 and titled "FIREARM AMMUNITION AVAILABILITY DETECTION SYSTEM," the disclosure of which is hereby incorporated by reference in its entirety for all purposes herein, and which is a continuation of U.S. application Ser. No. 16/746,770, filed Jan. 17, 2020 and titled "FIREARM AMMUNITION AVAILABILITY DETECTION SYSTEM," the disclosure of which is hereby incorporated by reference in its entirety for all purposes herein, and which is a continuation of U.S. application Ser. No. 16/297,288, filed Mar. 8, 2019 and titled "FIREARM AMMUNITION AVAILABILITY DETECTION SYSTEM," the disclosure of which is hereby incorporated by reference in its entirety for all purposes herein, and which claims the benefit of U.S. Provisional Patent Application No. 62/640,451, filed Mar. 8, 2018, and titled "FIREARM AMMUNITION AVAILABILITY DETECTION SYSTEM," the disclosure of which is hereby incorporated by reference in its entirety for all purposes herein. Any and all applications, if any, for which a foreign or domestic priority claim is identified in the Application Data Sheet of the present application are hereby incorporated by reference in their entireties under 37 CFR 1.57.

TECHNICAL FIELD

This disclosure relates to tracking the number of rounds available for a firearm. More specifically, this disclosure relates to determining the number of bullets, shells, or cartridges within a magazine, weapon, and or other container for storing ammunition for a firearm.

BACKGROUND

Many weapons and firearms use magazines to provide access to multiple cartridges without the need to reload the weapons or firearms. Different magazines can hold different numbers of cartridges. Further, some weapons can support different size magazines.

It is often beneficial for a user to know how much ammunition is available to the user. For example, when hunting or in a firefight, it is important for the user to know whether the user has sufficient ammunition. This can often be determined by the user when loading a magazine as the user can count how many cartridges are placed in the magazine. However, when the user is in a situation where he or she is firing a weapon, the user may easily lose track of the amount of ammunition fired and/or remaining.

SUMMARY

The systems, methods and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for the all of the desirable attributes disclosed herein. Details of one or more implementations of

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the subject matter described in this specification are set forth in the accompanying drawings and the description below.

Certain aspects of the present disclosure relate to a magazine configured to hold ammunition. The magazine can include: a housing comprising an ammunition chamber, wherein the ammunition chamber is configured to store one or more cartridges of a first ammunition type up to a first maximum number of cartridges of the first ammunition type; a linear encoder comprising a magnet and a plurality of magnetic sensors within the housing and positioned along a wall of the housing, wherein the plurality of magnetic sensors exceeds the first maximum number of cartridges in number, and wherein the magnet is configured to generate a magnetic field that is at least partially within the ammunition chamber; and a hardware processor configured to determine a count of an amount of cartridges of the first ammunition type within the magazine based at least in part on a signal generated by the linear encoder.

The magazine of the preceding paragraph can include any combination or sub-combination of the following features: where the signal generated by the linear encoder is generated by a magnetic sensor from the plurality of magnetic sensors in response to the magnetic field; where the signal generated by the linear encoder is generated based at least in part on a combination of a first signal generated by a first magnetic sensor and a second signal generated by a second magnetic sensor from the plurality of magnetic sensors; where the signal generated by the linear encoder is generated based at least in part on an interpolation of a first signal generated by a first magnetic sensor and a second signal generated by a second magnetic sensor from the plurality of magnetic sensors; where the plurality of magnetic sensors are positioned in a substantially vertical axis between a feed port of the magazine and a bottom cap of the magazine; where the magnet is disposed on a follower of the magazine such that the magnet moves towards a feed port of the magazine as cartridges are expelled from the magazine and moves towards a bottom cap of the magazine as cartridges are inserted into the magazine; where the plurality of magnetic sensors comprise Hall effect sensors or anisotropic magnetoresistive sensors; where the hardware processor is further configured to determine the count by at least: receiving the signal generated by the linear encoder; accessing an ammunition count table; and determining the count from the ammunition count table based at least in part on the signal; where the ammunition count table is selected from a plurality of ammunition count tables based at least in part on a determination of whether the magazine is inserted into a firearm; where the ammunition count table is selected from a plurality of ammunition count tables based at least in part on one or more of a type of cartridge loaded into the magazine, or a manufacturer of a cartridge loaded into the magazine; where the ammunition count table comprises a lookup table, and wherein the count is determined from the lookup table using a value generated from the signal as an index to the lookup table; and where the ammunition count table is a magazine-specific table that is calibrated for the magazine.

Additional aspects of the present disclosure relate to a magazine configured to hold ammunition. The magazine can include: a housing comprising an ammunition chamber, wherein the ammunition chamber is capable of holding between zero and 'N' cartridges of a first ammunition type, wherein 'N' corresponds to a maximum capacity of the magazine for holding cartridges of the first ammunition type; a hardware processor; and a linear encoder positioned within the housing and configured to provide an electrical

signal to the hardware processor, the electrical signal corresponding to a location of a follower within the ammunition chamber; wherein the hardware processor is configured to: determine a location of the follower within the ammunition chamber based at least in part on the electrical signal; access a calibration table of the magazine; and determine a count of cartridges of the first ammunition type within the magazine based at least in part on the location of the follower and the calibration table.

The magazine of the preceding paragraph can include any combination or sub-combination of the following features: where the ammunition chamber is capable of holding cartridges of a second ammunition type; where a maximum capacity of the magazine for holding cartridges of the second ammunition type differs from the maximum capacity of the magazine for holding cartridges of the first ammunition type; where a size of a cartridge of the first ammunition type differs from a size of a cartridge of the second ammunition type; where the hardware processor is further configured to determine the count of cartridges by accessing an entry in the calibration table corresponding to the location of the follower; where the calibration table is selected from a plurality of calibration tables based at least in part on a determination that the cartridges within the ammunition table are of the first ammunition type; where the magazine further comprises a non-volatile memory configured to store the calibration table; and where the linear encoder is one of: an optical encoder, a magnetic encoder, a capacitive encoder, an inductive encoder, a magneto resistive encoder, or a resistive encoder.

Certain aspects of the present disclosure relate to a firearm or weapon capable of determining a quantity of ammunition registered to the firearm or weapon. The firearm can include a hardware processor configured to: determine an identity of one or more magazines registered to the firearm; communicate, using a transceiver, with each of the identified one or more magazines to obtain an ammunition count for each magazine of the one or more magazines; and aggregate the ammunition count for each magazine of the one or more magazines to obtain a total ammunition count registered to the firearm.

The firearm of the preceding paragraph can include any combination or sub-combination of the following features: where the hardware processor determines the identity of the one or more magazines by accessing a storage device of the firearm to obtain the identity of the one or more magazines; where the identity of each magazine from the one or more magazines comprises a combination of a firearm identifier of the firearm and a magazine identifier of the magazine, and wherein each magazine from the one or more magazines is associated with a different magazine identifier; where the transceiver is configured to use ultra-wide band communication to communicate with each of the identified one or more magazines; where the hardware processor is further configured to communicate with an inserted magazine that is inserted into the firearm to obtain an inserted magazine ammunition count, wherein aggregating the ammunition count for each magazine of the one or more magazines includes adding the inserted magazine ammunition count to the total ammunition count; where the hardware processor communicates with the inserted magazine using an optical transceiver; where the hardware processor communicates with the inserted magazine using inductive, electromagnetic, or electrical communication; where the hardware processor is further configured to: register an unregistered magazine with the firearm; communicate with the unregistered magazine to obtain an ammunition count of the unregistered

magazine; and modify the total ammunition count to include the ammunition count of the unregistered magazine; where the hardware processor is further configured to register the unregistered magazine by: obtaining a magazine identifier for the unregistered magazine from the unregistered magazine; providing a firearm identifier to the unregistered magazine; combining at least the magazine identifier and the firearm identifier to obtain a network identifier for the unregistered magazine; and storing the network identifier for the unregistered magazine in a storage device of the firearm; where the hardware processor is further configured to provide the network identifier to the unregistered magazine, and wherein the firearm identifier is provided to the unregistered magazine as part of the network identifier; where the hardware processor is further configured to deregister a magazine of the one or more magazines registered to the firearm when an attempt to communicate with the magazine is unsuccessful for a threshold period of time; where the hardware processor is further configured to update the total ammunition count registered to the firearm when the magazine is deregistered; and where the transceiver is configured to use radio frequency communication to communicate with each of the identified one or more magazines.

Additional aspects of the present disclosure relate to a method of determining a quantity of ammunition registered to a firearm. The method may include obtaining, by a hardware processor, an identity of one or more magazines registered to the firearm, wherein the hardware processor accesses a storage device of the firearm to obtain the identity of the one or more magazines; communicating, using a transceiver, with each of the identified one or more magazines to obtain an ammunition count for each magazine of the one or more magazines; and aggregating, by the hardware processor, the ammunition count for each magazine of the one or more magazines to obtain a total ammunition count registered to the firearm.

The method of the preceding paragraph can include any combination or sub-combination of the following features: where the identity of each magazine from the one or more magazines comprises a combination of a firearm identifier associated with the firearm and a magazine identifier associated with the magazine, and wherein each magazine from the one or more magazines is associated with a different magazine identifier; where communicating with each of the identified one or more magazines comprises communicating using ultra-wide band communication; where the method further includes obtaining a loaded magazine ammunition count comprising a count of cartridges included in a magazine inserted into a loading port of the firearm, wherein aggregating the ammunition count for each magazine of the one or more magazines includes adding the loaded magazine ammunition count to the total ammunition count; where the method further includes determining whether a cartridge is within a chamber of the firearm, wherein aggregating the ammunition count for each magazine of the one or more magazines includes adjusting the total ammunition count based on the determination of whether the cartridge is within the chamber of the firearm; where the method further includes: registering an unregistered magazine with the firearm; communicating with the unregistered magazine to obtain an ammunition count of the unregistered magazine; and modifying the total ammunition count to include the ammunition count of the unregistered magazine; where registering the unregistered magazine comprises: receiving a magazine identifier for the unregistered magazine; providing a firearm identifier to the unregistered magazine; combining at least the magazine identifier and the firearm identifier to

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obtain a network identifier for the unregistered magazine; and storing the network identifier for the unregistered magazine in the storage device of the firearm; where the network identifier is used to communicate with the unregistered magazine after the unregistered magazine is registered with the firearm; where registering the unregistered magazine occurs in response to insertion of the unregistered magazine into an insertion port of the firearm; where the method further includes: attempting to communicate with a magazine from the one or more magazines registered to the firearm; failing to receive a response from the magazine within a time period subsequent to attempting to communicate with the magazine; and responsive to failing to receive the response from the magazine, removing the identity of the magazine from the storage device; where the method further includes modifying the total ammunition count to remove the ammunition count for the magazine from the total ammunition count; where the method further includes using encrypted communication when communicating with each of the identified one or more magazines; and where the method further includes outputting the total ammunition count for display to a user on a scope of the firearm.

Certain aspects of the present disclosure relate to a magazine capable of updating a calibration table used to determine a quantity of cartridges loaded in the magazine. The magazine can include: a storage device configured to store one or more calibration tables, wherein a calibration table of the one or more calibration tables maps linear encoder position values to cartridge counts corresponding to quantities of cartridges loaded in the magazine; a linear encoder configured to generate one or more signals corresponding to the one or more linear encoder position values; and a hardware processor configured to: detect a trigger to recalibrate the magazine; access the calibration table of the magazine from the storage device; determine a quantity of cartridges loaded in the magazine; receive a signal from the linear encoder corresponding to a linear encoder position value; update the calibration table based at least in part on the signal from the linear encoder; and store the updated calibration table at the storage device, wherein the updated calibration table is useable to automatically determine the quantity of cartridges loaded in the magazine without user action.

The magazine of the preceding paragraph can include any combination or sub-combination of the following features: where the hardware processor is further configured to determine the quantity of cartridges loaded in the magazine based on a user interaction with a user interface; where the hardware processor is further configured to: determine a type of cartridge loaded in the magazine; and select the calibration table from a plurality of calibration tables stored at the storage device based at least in part on the type of cartridge; where the hardware processor is further configured to: determine that the magazine is loaded into a firearm; and select the calibration table from a plurality of calibration tables stored at the storage device based at least in part on the determination that the magazine is loaded into the firearm and/or based at least in part on a determination that a bolt of the firearm is open or closed, wherein values in the calibration table differ from a second calibration table of the plurality of calibration tables, and wherein the second calibration table is selected when it is determined that the magazine is not loaded into the firearm; where the hardware processor is further configured to: determine, for each position value in a set of linear encoder position values in the calibration table, an adjustment factor based at least in part

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on the signal, the set of linear encoder position values comprising one or more position values; and adjust each linear encoder position value of the set of linear encoder position values based on the adjustment factor for the position value; where the adjustment factor varies for at least some of the linear encoder position values of the set of linear encoder position values; where the one or more linear encoder position values correspond to positions of a follower of the magazine with respect to a set of sensors included in the magazine; where the hardware processor is further configured to update the calibration table by: generating a new calibration table based at least in part on the signal value from the linear encoder; and replacing the calibration table with the new calibration table; where the trigger comprises one or more of a passage of time, an operation of a magazine control interface (such as, for example, activation of a button, receipt of a command via a wired or wireless user interface, detection of a control gesture or movement of the magazine, or detection of other user control actions), an occurrence of an error when determining the quantity of cartridges loaded in the magazine; a difference between a value corresponding to the signal and a position value in the calibration table exceeding a threshold, or interaction by a user with a user interface of the magazine; and where the hardware processor automatically recalibrates the magazine after a time period has elapsed, and wherein the trigger comprises the elapse of the time period.

Additional aspects of the present disclosure relate to a method of recalibrating a magazine configured to count a number of cartridges within the magazine. The method may include: detecting, by a hardware processor included in a magazine, a trigger to recalibrate the magazine; accessing, by the hardware processor, a calibration table of the magazine from a storage device of the magazine; determining, by the hardware processor, a count of a number of cartridges loaded in the magazine; receiving, by the hardware processor, a signal from a linear encoder within the magazine; modifying, by the hardware processor, the calibration table based at least in part on the signal from the linear encoder; and storing, by the hardware processor, the modified calibration table at the storage device, wherein the modified calibration table is useable to automatically determine a quantity of cartridges loaded in the magazine without user action.

The method of the preceding paragraph can include any combination or sub-combination of the following features: where the count of the number of cartridges is determined based on a user input; where the count of the number of cartridges is obtained from the storage device of the magazine, wherein the storage device stores a pre-specified count to be used during recalibration of the magazine; where the method further includes determining a type of cartridge loaded in the magazine; and selecting the calibration table based at least in part on the type of cartridge; where the method further includes determining a manufacturer of a cartridge loaded in the magazine; and selecting the calibration table based at least in part on the manufacturer of the cartridge; where the method further includes determining whether the magazine is loaded into a firearm; and selecting the calibration table based at least in part on the determination of whether the magazine is loaded into the firearm; where the method further includes determining a second count of a second number of cartridges loaded in the magazine; and receiving a second signal from the linear encoder within the magazine, wherein the calibration table is modified based at least in part on the signal and the second

signal from the linear encoder; where modifying the calibration table based at least in part on the signal from the linear encoder comprises: determining an adjustment factor for a set of position values in the calibration table based at least in part on the signal, the set of position values comprising one or more position values; and adjusting each position value of the set of position values based on the adjustment factor for the set of position values; where the adjustment factor differs for at least some of the position values of the set of position values; where the set of position values correspond to positions of a magnet of the linear encoder within the magazine corresponding to different quantities of cartridges within the magazine; where the set of position values is a subset of position values in the calibration table; where determining the adjustment factor for the set of position values comprises determining a difference between a value corresponding to the signal and at least one position value from the set of position values in the calibration table; and where the trigger comprises one or more of a passage of time, a command received via a magazine control interface, an occurrence of an error when determining the quantity of cartridges loaded in the magazine; a difference between a value corresponding to the signal and a position value in the calibration table exceeding a threshold, or interaction by a user with a user interface of the magazine.

Certain aspects of the present disclosure relate to an ammunition magazine. The ammunition magazine may include: a hardware processor comprising an optical transceiver interface and an ammunition sensor interface, wherein the ammunition sensor interface is configured to receive sensor outputs used to determine a number of cartridges in the ammunition magazine, and wherein the optical transceiver interface is configured to transmit a digital data signal indicating a count of the number of cartridges in the ammunition magazine; an ammunition sensor positioned and configured to sense the presence of cartridges loaded in the ammunition magazine; and an optical transceiver configured to receive the digital data signal from the hardware processor and generate an optical digital data signal configured to be transmitted through a transmission window of the ammunition magazine.

The ammunition magazine of the preceding paragraph can include any combination or sub-combination of the following features: where the ammunition sensor is part of a linear encoder included in the ammunition magazine; where the optical transceiver comprises: an optical transmitter that converts the digital data signal to the optical digital data signal and transmits the optical digital data signal through the transmission window; and an optical receiver that receives a second optical digital data signal and converts the second optical digital data signal to a second digital data signal that is provided to the hardware processor; where the second digital data signal comprises a command to obtain the count of the number of cartridges in the ammunition magazine; where the second optical digital data signal is received from a second transmission window of the ammunition magazine; where the ammunition magazine further includes a collimating lens configured to collimate the optical digital data signal to be transmitted through the transmission window; and where the ammunition sensor comprises a linear encoder.

Additional aspects of the present disclosure relate to a firearm system. The firearm system may include a magazine configured to hold one or more cartridges; and a firearm comprising: an insertion port configured to accept the magazine, wherein the magazine is configured to be inserted into

the insertion port and to provide the one or more cartridges to the firearm; and a firearm optical port within a housing of the insertion port, the firearm optical port configured to align with a magazine optical port of the magazine when the magazine is inserted into the insertion port and to receive a signal from the magazine optical port.

The firearm system of the preceding paragraph can include any combination or sub-combination of the following features: where the firearm further comprises a hardware processor configured to determine a count of cartridges available in the firearm based at least in part on the signal; where the signal is an optical signal and wherein the firearm further comprises a convertor that converts the optical signal to an electrical signal and provides the electrical signal to the hardware processor; where the convertor comprises a transceiver configured to receive the signal from the magazine optical port and to transmit a second signal to the magazine via the firearm optical port; where the hardware processor is located within a handle of the firearm; where the hardware processor is further configured to determine the count of the cartridges available in the firearm based at least in part on a determination of whether a cartridge exists in a chamber of the firearm; where the firearm further comprises a plurality of magnetic sensors within a buffer tube of the firearm and a magnet attached to a bolt of the firearm; where a hardware processor of the firearm determines whether a cartridge is within a chamber based at least in part on a motion and a position of the bolt as detected by the plurality of magnetic sensors; where the firearm further comprises a battery located within a handle of the firearm; where the firearm further comprises a wireless transceiver configured to communicate with one or more magazines using ultra-wideband communications; where the firearm further comprises a wireless transceiver configured to communicate over a network; where the firearm optical port comprises a transmission window; where the magazine comprises an optical transceiver that converts a digital data signal to an optical digital data signal that is transmitted to the firearm as the signal through the magazine optical port; and where a gap exists between the firearm optical port and the magazine optical port when the magazine is inserted into the firearm.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects and advantages of the embodiments provided herein are described with reference to the following detailed description in conjunction with the accompanying drawings. Throughout the drawings, reference numbers may be reused to indicate correspondence between referenced elements. The drawings are provided to illustrate example embodiments described herein and are not intended to limit the scope of the disclosure. In addition, various features of different disclosed embodiments can be combined to form additional embodiments, which are part of this disclosure. Any feature or structure can be removed or omitted.

FIG. 1 illustrates a cross-section of an embodiment of the inside of a magazine.

FIG. 2 illustrates a line drawing of the embodiment of the cross-section of the magazine illustrated in FIG. 1.

FIG. 3 illustrates an embodiment of the Hall effect sensors within an embodiment of the magazine.

FIG. 4A illustrates an embodiment of the alignment pin within an embodiment of the magazine.

FIG. 4B illustrates another embodiment of the Hall effect sensors within an embodiment of the magazine without an alignment pin.

FIG. 5 illustrates an embodiment of the follower within an embodiment of the magazine.

FIG. 6 illustrates a line drawing of the embodiment of the follower illustrated in FIG. 5.

FIG. 7 illustrates an embodiment of the outside of a magazine.

FIG. 8A illustrates a line drawing of another perspective of the outside of the magazine illustrated in FIG. 7.

FIG. 8B illustrates an embodiment of a magazine with a control interface positioned near the egress of the magazine in accordance with certain embodiments.

FIG. 8C illustrates the magazine of FIG. 8B inserted into a weapon.

FIG. 9A illustrates an embodiment of a non-contact optical connector included as part of the magazine.

FIG. 9B illustrates an embodiment of a signal converter within a magazine and a signal converter within a weapon that enables optical communication between the magazine and the weapon.

FIG. 10 illustrates an embodiment of a display included as part of the magazine.

FIG. 11A illustrates an embodiment of the electronic circuit and display included as part of the magazine.

FIG. 11B illustrates an alternative view of the embodiment of the electronic circuit and display of FIG. 11A.

FIG. 12 illustrates an embodiment of a display circuit for a display.

FIG. 13 illustrates an embodiment of a weapon with the magazine inserted.

FIG. 14 illustrates a modified view of the embodiment of the weapon of FIG. 13.

FIG. 15 illustrates a line drawing of the embodiment of the weapon of FIG. 13 without the inserted magazine.

FIG. 16 illustrates a line drawing of the embodiment of the weapon of FIG. 13 and an un-inserted magazine.

FIG. 17 illustrates an embodiment of the weapon handle and magazine.

FIG. 18 illustrates an embodiment of the ammunition status display attached to a rail of the weapon.

FIG. 19 illustrates an embodiment of the ammunition status display separate from the weapon.

FIG. 20 illustrates a cross-section of an embodiment of the weapon with the magazine installed.

FIG. 21 illustrates a line drawing of the cross-section of the embodiment of the weapon of FIG. 20 with the magazine installed.

FIG. 22 illustrates a line drawing of a cross-section of an embodiment of the handle and magazine inserted into the weapon.

FIG. 23 illustrates a line drawing of another perspective of the cross-section of the embodiment of the handle and magazine inserted into the weapon of FIG. 22.

FIG. 24A illustrates an embodiment of a non-contact optical connector included in the lip of the insertion point for inserting a magazine into the weapon.

FIG. 24B illustrates a portion of the magazine with a transceiver and a magnetic sensor in accordance with certain embodiments.

FIG. 24C illustrates a portion of the weapon with a transceiver 2302 and a magnet in accordance with certain embodiments.

FIG. 25 illustrates an embodiment of a buffer tube within a weapon.

FIG. 26 illustrates a line drawing of an embodiment of a buffer tube and stock of a weapon.

FIG. 27 illustrates an example use case of certain embodiments described herein.

FIG. 28 presents a flowchart of an embodiment of an ammunition count determination process.

FIG. 29 presents a flowchart of an embodiment of an ammunition count display process.

FIG. 30 presents a flowchart of an embodiment of a magazine calibration process.

FIG. 31 presents a flowchart of an embodiment of a calibration table based ammunition count process.

FIG. 32 presents a flowchart of an embodiment of a magazine recalibration process.

FIG. 33 presents a flowchart of an embodiment of a total ammunition count process.

FIG. 34 presents a flowchart of an embodiment of a magazine registration process.

FIG. 35 presents a flowchart of an embodiment of a weapon display state machine.

FIG. 36 presents a flowchart of an embodiment of a jam display process.

FIG. 37 presents a flowchart of an embodiment of a chamber count display process.

FIG. 38 presents a flowchart of an embodiment of a magazine display process.

FIG. 39 presents a flowchart of an embodiment of a redraw process.

DETAILED DESCRIPTION

The headings provided herein, if any, are for convenience only and do not necessarily affect the scope or meaning of the claimed invention. For the purpose of the present disclosure, the terms mobile devices and wireless devices are used interchangeably.

INTRODUCTION

In a number of situations, it can be beneficial to maintain an accurate count of available ammunition within a magazine of a weapon. For example, a soldier will often want to know how much ammunition is available to him or her. Although the soldier may be aware of the amount of ordinance assigned to the soldier, during a firefight, the soldier may lose track of how many shots have been taken and how much ammunition remains.

Certain embodiments described herein provide an improved ammunition tracking system that determines the amount of ammunition within a magazine and outputs the information for display to a user. The system includes a magnet and a set of Hall effect sensors that determine a location of the magnet within the magazine. Based at least in part on the determination of the location of the magnet, the system determines the amount of ammunition within the magazine.

Further, in certain embodiments disclosed herein, the system can track the total amount of ammunition available to a user (e.g., a soldier, policeman, or hunter). The system can register one or more magazines to a user or a particular weapon. The system can then maintain a count of ammunition available to the user or the weapon based on the ammunition within the one or more magazines. Advantageously, in certain embodiments, by monitoring ammunition within each magazine as well as the number of magazines available, a user can be presented with an accurate count of the total ammunition available to the user without the user needing to track the amount of ammunition fired or used.

In addition, in certain embodiments, a magazine installed or inserted within the weapon can communicate a count of the ammunition within the magazine to the weapon and/or to

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one or more accessories installed on the weapon or within proximity of the weapon. This communication can be accomplished using an optical transceiver, and/or radio included in the magazine. Advantageously, in certain embodiments, using the optical transceiver eliminates cables that can be susceptible to damage during manufacture and/or use.

Additional features and embodiments of the magazine and weapon system are described below with respect to the figures.

Example Magazine

FIG. 1 illustrates a cross-section of an embodiment of the inside of a magazine 100 that may be used with certain embodiments disclosed herein. Further, FIG. 2 illustrates a line drawing of the embodiment of the cross-section of the magazine 100 illustrated in FIG. 1.

With respect to FIG. 1, the magazine 100 can include a housing 102 that includes a chamber 104 for holding one or more bullets or cartridges 106 of ammunition. It should be understood that the features disclosed herein can be used with any type of magazine capable of holding any type of ammunition.

As illustrated, the magazine includes a follower 108 that is configured to push the cartridges 106 towards an egress point of the magazine 100. The follower 108 includes a magnet 110. As cartridges 106 are removed from the magazine (e.g., when a cartridge 106 is fired), the follower 108 moves towards the egress point of the magazine 100 shrinking the size of the chamber 104 that holds cartridges 106 within the magazine 100. Consequently, as the follower 108 moves, so does the magnet 110, thereby altering the location of the magnet 110 within the magazine 100.

The location of the magnet 110 can be used to determine the number of cartridges 106 within the housing 102 of the magazine 100. To determine the magnet's 110 location, the magazine 100 includes a number of Hall effect sensors 112. The Hall effect sensors 112 can detect the location of the magnet 110 using the Hall effect. The Hall effect relates to the production of a voltage difference across an electrical conductor when a magnetic field is perpendicular to a current in the conductor. Thus, as the magnet 110 approaches a Hall effect sensor 112 within the magazine 100, a voltage will be produced that can be measured by electronic circuitry within, and/or external to, the magazine 100. Based on the detected voltage, the location of the magnet 110 can be determined and consequently, the location of the follower 108. This information can be used to determine the quantity of ammunition within the chamber 104. In certain embodiments, the Hall effect sensors 112 may be replaced by any type of transducer, electrical, magnetic, or electromagnetic sensor that produces or modifies an electrical signal or measurable electrical property (e.g., a voltage, a current, or a resistance) based at least in part on a detected magnetic field from a magnet. For example, in some embodiments, an anisotropic magnetoresistive or anisotropic magnetoresistance (AMR) sensor may replace or complement the Hall effect sensors 112. In some embodiments, the microelectromechanical systems (MEMS) magnetic field sensor may be used instead of, or in addition to, the Hall effect sensors 112. Other examples of sensors that may be used in place of or in addition to the Hall effect sensors 112 may include sensors that measure or detect negative magnetoresistance, giant magnetoresistance, tunnel magnetoresistance, or extraordinary magnetoresistance. In some embodiments, the change in the type of sensor 112 may be paired with a change of the type of magnet 110.

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The combination of the Hall effect sensors 112 and the magnet 110 may create a linear encoder. The set of sensors 112 may be used to encode or identify particular locations within the magazine 100. Based on the location of the magnet 110 with respect to the set of sensors 112, the number of cartridges loaded within the magazine 100 can be determined. The position or location of the magnet 110 may change as the number of cartridges within the magazine 100 changes. Based on the change in location of the magnet 110, as determined by the one or more Hall effect sensors 112 that generate electrical signals, the number of cartridges within the magazine 100 can be determined. In some embodiments, each, or all, of the sensors 112 may generate an electrical signal based on the sensors relative location to the magnet 110. In some such cases, the location of the magnet 110 may be determined based on the respective signals generated by the set of sensors 112. In some embodiments, the one or more electrical signals generated by the one or more sensors 112 may be compared to a table that is stored in the circuitry or memory of the magazine 100. Based on the comparison between the generated electrical signals and the data stored in the table, the location of the magnet 110 may be determined. In some embodiments, the signals generated by the sensors 112 are compared to data stored within the table. In other embodiments, values that are generated or determined based on the signals generated by the sensors 112 are compared to data stored within the table.

Although the linear encoder is primarily described herein as being formed from a magnet and a set of magnetic sensors, the linear encoder is not limited as such. For example, the linear encoder may use an optical, capacitive, inductive, or resistive system. For instance, with an optical system, the magnet may be replaced with a light source, such as an LED. The magnetic sensors may be replaced with light sensors that are capable of determining the linear encoder position based on the one or more light sensors that detect light from the light source. A capacitive system may similarly be implemented to detect a capacitance between an element attached to the follower and an element aligned with the cartridges in the magazine. Further, the linear encoder may be an absolute encoder or an incremental encoder.

In some embodiments, the number of sensors 112 included within the magazine corresponds to the maximum number of cartridges that can be loaded into the magazine 100. Thus, if the bottommost sensor that is closest to the butt of the magazine 100 generates an electrical signal, it can be determined that the magazine includes the maximum number of cartridges. In another example, if the sensor 112 closest to the feed point or ingress/egress point of the magazine 100 generates an electrical signal, it can be determined that the magazine 100 is empty. It should be understood that multiple sensors 112 may generate a signal. The number of cartridges within the magazine 100 may be determined by interpolating or averaging the signals, or by selecting the sensor 112 that generates the strongest electrical signal.

However, with some magazines, a one-to-one correspondence between the number of sensors 112 and the maximum number of cartridges that the magazine 100 can hold is insufficient to accurately determine the number of cartridges within the magazine 100. There are several reasons why having a one-to-one correspondence between the number of sensors 112 included in the magazine 100 and the number of cartridges the magazine 100 can hold when fully loaded is insufficient. For example, one reason why having a one-to-one correspondence between sensors and cartridges is that, even for a particular size cartridge, there may be variation in

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cartridge size between manufacturers such that the alignment between the cartridges within the magazine 100 and the sensors 112 may vary for different manufacturers of the same ammunition type. As another example, some magazines are capable of holding different types of cartridges, which may vary in size and may result in a different position of the magnet 110 for the same number of cartridges. For instance, the magazine 100 may be capable of holding both .458 SOCOM rounds and .223 rounds, which differ in size, thereby causing a location of the magnet 110 with respect to the sensors 112 to differ for the same number of cartridges.

In some cases, even if the same type of ammunition from the same manufacturer is consistently used with the magazine 100, wear and tear of the magazine 100 over time may cause the alignment of the magnet 110 with respect to the sensors 112 to change for the same number of cartridges. For example, the rigidity of the plastic or the strength of the spring in the follower may change over time. In some cases environmental conditions, such as temperature, may cause a change in the magazine 100 that affects the alignment of the magnet 110 with the sensors 112 for a particular number of cartridges inserted into the magazine 100. Further, with many weapons, inserting the magazine 100 into the weapon may cause pressure to be applied to the cartridges within the magazine 100. This pressure may shift the cartridges compared to when the magazine 100 is not inserted into the weapon, thereby modifying the alignment of the magnet 110 with the sensors 112. As such, in certain embodiments, the determination of the number of cartridges within the magazine 100 may change based on whether the magazine 100 is inserted into the weapon resulting in an inaccurate count of the cartridges either when the magazine 100 is inserted into the weapon or when the magazine 100 is not inserted into the weapon.

For at least the above reasons, in some cases having a one-to-one correspondence between the number of sensors included in the magazine 100 and the maximum number of cartridges that the magazine 100 can support may be insufficient. Thus, in certain embodiments, the magazine 100 may include more sensors 112 than the total number of cartridges that the magazine 100 can hold or is designed to hold. Advantageously, in certain embodiments, by including a greater number of sensors in the magazine 100 than the number of cartridges the magazine 100 can hold when at capacity, the accuracy of the measurements made by the linear encoder of the magazine 100 may be increased, thereby reducing or eliminating the problems associated with having a one-to-one correspondence between the sensors and the number of cartridges that the magazine 100 can support. To determine the number of cartridges within the magazine 100, the linear encoder may further include a hardware processor that can compare the signals generated by the sensors 112 with the calibration table stored in a memory to determine the number of cartridges within the magazine 100. The hardware processor may generate a value, such as a coordinate value indicating a relative location of the magnet 110 with respect to a coordinate system associated with the sensors 112. The hardware processor may compare the value to values within the calibration table to identify a number of cartridges corresponding to the value. In certain embodiments, the hardware processor may interpolate signal values from the sensors 112 to more accurately identify the location of the magnet 110, and consequently the number of cartridges loaded within the magazine 100.

In certain embodiments, the number of sensors within the magazine may match the number of cartridges the magazine

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is capable of holding. In some such embodiments, a linear encoder may be used to determine the number of cartridges in the magazine. In certain embodiments, the linear encoder may eliminate the problems described above when the granularity of sensors matches the maximum ammunition capacity of a magazine. The location of the magnet in the magazine with respect to the sensors in the magazine may be determined based on the electrical signals received from two or more sensors. This location may be mapped to a calibration table to determine the number of cartridges within the magazine. By mapping the location to the calibration table, instead of using the exact position of the magnet with respect to the sensors to determine an ammunition count, a range of positions may be used to determine the ammunition count enabling the linear encoder to account for wear and tear of the magazine and differences in cartridge sizes. Thus, instead of determining that a magazine includes X cartridges because sensor X detects the magnet, it can be determined that because the magnet is between sensors X and Y, the magazine includes X cartridges. With magazines that require a one-to-one correspondence between the sensor and the ammunition count, a change in the position of the magnet due to, for example, wear and tear may cause an error in determining the ammunition count in the magazine. However, with a magazine that includes a linear encoder, the amount of ammunition in the magazine can be determined even when the magnet does not align with a particular sensor.

As described above, the number of cartridges 106 in the magazine 100 can be determined based on the location of the magnet 110 with respect to the one or more Hall effect sensors 112. Thus, to determine the number of cartridges 106, it is desirable that the Hall effect sensors 112 maintain particular positions within the magazine 100. However, during manufacture and/or over time, the position of the Hall effect sensors 112 may shift. For example, vibrations caused by the firing of the weapon within which the magazine 100 is loaded may result in movement of the Hall effect sensors 112. To maintain the position of the Hall effect sensors 112, one or more alignment pins 114 can be used to maintain the position of the circuit board that includes the Hall effect sensors 112.

In some embodiments, the use of an alignment pin is unnecessary, and the alignment pin may be omitted. For example, the Hall effect sensors may be secured or built into the housing of the magazine 100 such that the sensors 112 do not shift over time. Alternatively, or in addition, the calibration table may be updated or modified over time using a calibration or recalibration process. The calibration process may be used to determine the location of the magnet 110 with respect to the sensors 112. By calibrating or recalibrating the magazine 100, the table that identifies the number of cartridges loaded within the magazine 100 based on a determination of the location of the magnet 110 with respect to the sensors 112 may be updated. Thus, changes in location of the magnet 110 with respect to the sensors 112 for a particular number of cartridges loaded within the magazine 100 may be captured by the calibration process, thereby reducing or eliminating inaccuracies in the measurement of the number of cartridges loaded into the magazine that may be introduced by wear and tear of the magazine or variation in ammunition loaded into the magazine. This calibration processes described in more detail below with respect to FIG. 32.

As is described further herein, the amount of cartridges within the magazine can be presented to a user via a display included in the magazine 100. However, in some cases, it

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may be desirable to transmit information associated with the magazine, such as the number of cartridges within the magazine or the state of a battery used to power the circuitry included in the magazine, to the weapon or another system. In some embodiments, the magazine **100** can include an optical transceiver **116** or radio to transmit the information to the weapon or another device. Further, the magazine **100** may include a digital to optical signal adapter enabling the conversion of a digital signal created by the electronic circuitry **122** to an optical signal for transmission by the optical receiver **116**. Moreover, the magazine **100** may include an optical to digital signal adapter enabling the conversion of an optical signal received at the optical transceiver **116** capable of being processed by electronic circuitry **122**.

The enlarged portion **120** of the bottom of the magazine **100** illustrates additional details of the magazine **100**. As illustrated within the enlarged portion **120**, the magazine includes electronic circuitry **122** that receives one or more electrical signals from the one or more Hall effect sensors **112**. Based on the one or more electrical signals, the electronic circuitry **122** determines the position of the magnet **110** and/or follower **108** within the magazine **100** and consequently, the number of cartridges **106** within the magazine **100**. The electronic circuitry **122** may include a hardware processor or programmable hardware, such as a field programmable gate array (FPGA), which may be configured to interpret the signals received from the sensors **112** and determine the location of the magnet **110** and consequently the number of cartridges within the magazine. Further, the hardware processor or programmable hardware of the electronic circuitry **122** may be configured to perform one or more of the processes described herein.

Further, the magazine **100** includes a display **124**, which as illustrated may be a 7 segment light emitting diode (LED) display. However, it should be understood that the display **124** is not limited to a 7 segment LED display and can include any type of display that can output, among other things, the number of cartridges within the magazine **100** to a user.

The electronic circuitry **122** and/or the Hall effect sensors **112** can be powered by a battery **126** that can be included within the magazine **100**. In some embodiments, the battery **126** is rechargeable using, for example, wireless charging. Alternatively, and/or in addition, the battery may be replaced by opening a cap **128**. The cap **128** can be a sealing cap that is configured to prevent any moisture or detritus from entering the magazine **100**.

Further, the magazine **100** may include a control interface **130** (such as, for example, a control button) that can be used to activate or deactivate the system for measuring the cartridges in the magazine **100**. Moreover, the control interface **130** may be used to modify the brightness of the display **124**. In some embodiments, the control interface **130** can be used to show or hide the display of the number of cartridges within the magazine **100**. In some cases, a user may determine that light from the display **124** is undesirable. For example, in some combat situations, it may be desirable to reduce as much light as possible emanating from the soldier. Advantageously, in certain embodiments, by being able to control the display of the ammunition within the magazine **100**, or other information, a user can determine whether or not the display **124** is active. Moreover, in some cases, the control interface **130** can be used to control whether ammunition information is displayed on an alternative display, such as via a scope attached to the weapon or a helmet worn by the user. In some embodiments, the control interface **130**

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is disposed on a portion of the magazine **100** that is inserted into the firearm. In such embodiments, the control interface **130** can only be accessed by the user when the magazine **100** is not in the firearm. This configuration can help reduce or eliminate false readings that may occur if the control interface **130** is operated while the magazine is inserted in the firearm. In certain embodiments, the control interface **130** comprises one or more of a control panel, a control button, a wired control interface, a wireless control interface, or a combination of interfaces. For example, the control interface **130** may include a cell phone, laptop, or other wireless device. A user may interact with the control interface **130** to request a count of ammunition within one or more magazines. Further, a user may interact with the control interface **130** to trigger calibration or recalibration of one or more magazines to update a calibration or linear encoder table used to determine a number of cartridges within a magazine.

FIG. 3 illustrates an embodiment of the Hall effect sensors **112** within an embodiment of the magazine **100**. As illustrated, the Hall effect sensors **112** may include a plurality of sensors **112** that can each provide a signal to the electronic circuitry **122** (see FIG. 1) of the magazine **100**. In some cases, each Hall effect sensor **112** provides a signal to the electronic circuitry **122**. In other cases, a subset of the Hall effect sensors **112**, such as those within a threshold distance of the magnet **110**, provide a signal to the electronic circuitry **122**.

The Hall effect sensors **112** may be included as part of a circuit board **302**. The circuit board **302** may be designed to precisely place, within a threshold degree of tolerance, the Hall effect sensors **112** with respect to the location of the magnet **110** when the magazine **100** is loaded with a particular amount and type (e.g., size) of ammunition. Alternatively, or in addition, the electronic circuitry **122** may be calibrated based on the position of the Hall effect sensors **112** and the magnet **110**. In some embodiments, the circuit board **302** may be a flexible circuit board. Advantageously, in certain embodiments, by using a flexible circuit board, the hall circuit board **302** can be shaped to match the curvature of the magazine **100** and to maintain the Hall effect sensors **112** at a particular distance from the magnet **110** when the magnet is positioned perpendicular to a particular Hall effect sensor **112**.

As previously mentioned, one or more alignment pins **114** may be used to position the circuit board **302** within the magazine **100**. FIG. 4A illustrates an embodiment of the alignment pin **114** within an embodiment of the magazine **100**. Although the alignment pin **114** is illustrated as a rod, it should be understood that other shapes may be used. For example, the alignment pin **114** may include fingers or other protrusions from the pin **114** that can be used to grip or further hold the circuit board **302** in place. Further, in the embodiments of FIG. 4A, the Hall effect sensors **112** are positioned at the rear of the magazine **100** and the magnet **110** is positioned at the rear.

In some embodiments, the alignment pin **114** is omitted. FIG. 4B illustrates another embodiment of the Hall effect sensors **112** within an embodiment of the magazine without an alignment pin. As illustrated in FIG. 4B, in some implementations, the circuit board **302**, and associated Hall effect sensors **112**, may be positioned on a side of the housing of the magazine as opposed to the back of the magazine behind the cartridges. In some embodiments, the Hall effect sensors **112** may be positioned near the front **402** of the magazine, where the front is defined as the direction in which the cartridges are ejected when fired from a firearm. As described in more detail below, by positioning the Hall effect

sensors **112** near the front of the magazine, another Hall effect sensor may be positioned at the rear of the magazine for determining when the magazine is inserted into a weapon or firearm without the additional Hall effect sensor detecting the magnet **110** positioned near or adjacent to the front edge **402** (e.g., the edge in which the cartridges are pointing when loaded within the magazine **100**) of the magazine **100**. Further, the Hall effect sensors **112** may be positioned along the vertical or y-axis of the magazine to form a vertical column of sensors. Many magazines are not completely rectangular in shape, but are instead curved to some degree. Accordingly, in certain embodiments, the Hall effect sensors may be positioned substantially vertically along an axis that matches the curvature of the magazine.

In certain embodiments, a magnet attached to a follower **108** may align in the horizontal or x-axis of the magazine with one or more of the Hall effect sensors **112** when one or more cartridges are loaded in the magazine. In certain implementations, one or more of the Hall effect sensors **112** may align with the magnet attached to the follower **108** when no cartridges are loaded in the magazine.

FIG. **5** illustrates an embodiment of the follower **108** within an embodiment of the magazine **100**. Often, followers include a solid protrusion from the follower body that pushes cartridges within the magazine towards the egress of the magazine. In certain embodiments of the follower **108**, the solid protrusion is replaced by a spring-loaded plunger **502**. Advantageously, in certain embodiments, replacing the solid protrusion with the spring-loaded plunger **502** creates a space for inclusion of the electronic circuitry **122**. Thus, in certain embodiments, the magazine **100** can include the electronic circuitry **122** for automatically and electronically monitoring the number of cartridges within the magazine with minimal or no increase in the size of the magazine **100** compared to magazines that are not capable of electronically monitoring the number of cartridges within the magazine. FIG. **6** illustrates a line drawing of the embodiment of the follower illustrated in FIG. **5**.

Example Magazine Housing

FIG. **7** illustrates an embodiment of the outside of a magazine **100**. FIG. **8A** illustrates a line drawing of another perspective of the outside of the magazine illustrated in FIG. **7**. As previously described, the magazine **100** may include a display **124** and an optical transceiver **116**. One or more of the display **124** and the optical transceiver **116** may be part of the housing **102**. Alternatively, the display **124** and/or the optical transceiver **116** may be inserted into gaps or spaces within the housing **102**.

In some embodiments, the magazine **100** may include a machine-readable code that includes a unique identifier for the magazine. This machine-readable code may be a bar code, a matrix code, a quick response (QR) code, or any other type of machine-readable code. Alternatively, or in addition, the magazine **100** may include a radio frequency identification (RFID). Advantageously, in certain embodiments, by including a machine-readable code and/or tag on the magazine **100**, the magazine **100** can be registered with a weapon enabling a user, such as a soldier, to track the total ammunition available to the user. In some embodiments, the magazine **100** may be registered with the weapon upon insertion of the magazine **100** within the weapon. When the magazine is loaded into the weapon, circuitry within the weapon may communicate with circuitry within the magazine to register the magazine with the weapon.

Tracking the total ammunition enables a user to monitor ammunition while in a particular environment. For example, a soldier in the field can monitor ammunition without

needing to remember how many magazines are in the soldier's gear or how many magazines or cartridges the soldier has used. In some embodiments, the machine-readable code and/or RFID tag can be modified to reflect the amount of ammunition remaining in the magazine. Alternatively, or in addition, an indicator may provide a status of the magazine. For example, when the magazine is empty or has less than a threshold number of cartridges, the indicator may turn red informing the user that the cartridge is empty or has below a threshold number of cartridges. Thus, for example, when a soldier is in the field, he or she can easily determine whether a magazine is a loaded magazine or a spent magazine that is now empty or is close to empty.

In some use cases, a user may register a magazine that is currently registered with one weapon with another weapon. For example, a first soldier may take a magazine registered to the soldier's weapon and give it to a second soldier, who may then register the magazine with his or her weapon. In certain embodiments, when the second soldier registers the magazine with his or her weapon, the magazine may be deregistered from the first soldier's weapon. In some embodiments, the magazine may send a signal to the first soldier's weapon to indicate that the magazine should be deregistered from the first soldier's weapon. In other embodiments, the weapon of the first soldier may deregister the magazine from the weapon and the weapon is no longer able to communicate with the magazine. In some cases, the weapon may no longer be able to communicate with the magazine because the magazine moves outside of communication range of the weapon. Alternatively, or in addition, the weapon may no longer be able to communicate with the magazine because when the magazine is registered with the second weapon, the second weapon changes an identifier of the magazine. For example, each weapon may provide an identifier to the magazine that is based at least in part on the weapon's identifier. Thus, when a magazine is registered to a weapon, the magazine's identifier may change and the magazine may no longer respond to attempts to communicate with the magazine from another weapon with which the magazine was previously registered.

In some embodiments, the magazine **100** may include a carbon fiber casing. The use of a carbon fiber casing can reduce the weight of the magazine **100** offsetting any added weight from the additional sensors and electronics. Further, in some embodiments, the magazine **100** may include a rubber sleeve. The rubber sleeve can improve the strength and durability of the magazine **100** as well as provide additional protection for the added electronics to help prevent damage during impacts (e.g., if the magazine is dropped).

As illustrated in FIG. **8A**, the control interface **130** may be a button or other interface element positioned near the base of the magazine. A user may interact with the control interface **130** to cause a cartridge count to be displayed on a display of the magazine **100**, on a display of a weapon, or any other display that can present the cartridge count.

In some cases, the magazine may be inserted into a traditional weapon, or a weapon that does not support one or more of the embodiments described herein. In some such cases, the magazine **100** may present an inaccurate count of the cartridges within the magazine **100** when the user interacts with the control interface **130** because, for example, the pressure applied to the cartridges in the magazine by the weapon or because of the inability to determine when a cartridge is within a chamber of the weapon. In some cases, the weapon may support embodiments disclosed herein, but the battery may be drained or the circuitry of the

weapon may be off. In such cases, an inaccurate cartridge count may also be displayed because the status of the bolt and/or chamber may be unknown.

FIG. 8B illustrates an embodiment of a magazine with a control interface **130** positioned near the egress of the magazine in accordance with certain embodiments. In particular the controller interface **130** is positioned at a location on the magazine **100** that is inaccessible when the magazine **100** is inserted into a weapon. Advantageously, by moving the control interface **130** to a location the magazine that is inaccessible when the magazine **100** is inserted into the weapon, a user cannot interact with the control interface **130** and therefore, cannot obtain a cartridge count without removing the magazine **100** from the weapon. Thus, a magazine is prevented from displaying an inaccurate cartridge count in cases where the magazine is inserted into a weapon that does not support embodiments disclosed herein, or in cases where the weapon does support the embodiments disclosed herein, but in which the battery of the weapon is drained or dead, or in which the circuitry of the weapon is inactive or turned off. Further, the control interface **130** may be positioned within a recess **820** preventing the weapon from interacting with the control interface **130** when the magazine is inserted into the weapon.

FIG. 8C illustrates the magazine of FIG. 8B inserted into a weapon. As illustrated, the control interface **130** is not accessible when the magazine **100** is inserted into the weapon. Accordingly, a user cannot interact with the control interface **130** to obtain a cartridge count within the magazine. It should be understood that the cartridge or ammunition count may be obtained by either removing the magazine, or interacting with a weapon that includes the embodiments disclosed herein for obtaining the ammunition count from the magazine and for determining the status of the chamber and/or bolt of the weapon.

Example Non-Contact Connector

In some embodiments, the magazine **100** can communicate with a weapon using a non-contact connector, such as an optical connector. In some embodiments, wireless radio frequency communication or electrical communication may be used between the magazine and the weapon. However, in some embodiments, the weapon may be fired at a faster rate than the speed of the RF communication leading to an inaccurate ammunition count being displayed to the user. Further, electrical communication connections may become damaged by the environment. Advantageously, in certain embodiments, using an optical connector may enable faster communication compared to radio communication. Further, optical connectors can be cleaned using water and are less susceptible to damage by the environment or rough handling compared to electrical connectors.

FIG. 9A illustrates an embodiment of a non-contact optical connector **116** included as part of the magazine. The optical transceiver **116** may be positioned at a location that enables the optical transceiver **116** to line up or mate with an optical transceiver included in a weapon that is designed to be used with the magazine **100**, or for which the magazine **100** is designed. The optical transceiver **116** is configured to communicate using light and thus, can mate with another optical transceiver without making physical contact. However, in some cases, the optical transceiver **116** may be in physical contact with another optical transceiver to minimize or eliminate potential interference from ambient light.

Further, while there are advantages to using an optical transceiver **116**, such as the elimination of breaks in a wire-based communication system, in some embodiments the optical transceiver **116** may be supplemented by or

replaced by other types of transceivers, such as a wireless transceiver, or a wire-line transceiver configured to communicate with the weapon when metal contacts included in the transceiver of the magazine contact corresponding metal contacts in a transceiver of the weapon. In some embodiments, the magazine **100** may communicate with a weapon using inductive, electromagnetic, or electrical communication. For example, an alignment of inductive elements may enable a data and/or power transfer between the weapon and the magazine **100**. As another example, when the magazine is inserted into the weapon, pins or other contact points of a transceiver on the magazine may align with corresponding pins or contact points on the weapon enabling electrical communication.

In some cases, the transceiver **116** may use near field communications (NFC) and/or radio frequency communication. Alternatively, or in addition, the transceiver **116** may use ultra-wide band communication to reduce the impact of interference on the communications and/or to reduce the chance of eavesdropping. The ultra-wide band communication may communicate over a larger bandwidth than conventional narrowband communication or carrier wave transmission. In some embodiments, the ultra-wide band communication may occur over a bandwidth exceeding 500 MHZ. In some embodiments, the transceiver **116** may use spread spectrum communication. Further, in some embodiments, the transceiver **116** may use encrypted communications or a secure channel during communication. In some embodiments, the transceiver **116** may use Bluetooth®, Wi-Fi®, the RADIUS protocol, or any other type of narrow band or wide band communication protocols.

FIG. 9B illustrates an embodiment of a signal converter **900** within a magazine and a signal converter **902** within a weapon that enables communication between the optical transceiver **116** of the magazine and an optical transceiver of a weapon, such as the optical transceiver **1602** described with respect to FIG. 16 below. In certain embodiments, the signal converter **900** may be or may be included as part of the optical transceiver **116**.

The signal converter **900** can be configured to convert between electrical digital signals received from a magazine processor **906** and optical digital signals, which can be transmitted optically to a signal converter **902** of a weapon. It should be understood that the reverse process is possible as well. In other words, the signal converter **902** of the weapon may communicate optically with the signal converter **900** of the magazine.

An electrical signal can be passed from a magazine processor **906** to an optical transmitter **910**. This electrical signal may be a digital signal used to provide information about the magazine **100** to a weapon. For example, the magazine may communicate a cartridge count to the weapon. The optical transmitter **910** can be configured to convert an electrical digital signal into a corresponding optical digital signal. The optical transmitter **910** can convert electrical signals to optical signals using appropriate techniques, such as, for example, by outputting an optical signal proportional to the input electrical current. The optical transmitter **910** can be any suitable component for converting electrical digital signals to optical digital signals, such as, for example, HXT4101A-DNT manufactured by GigOptix, Inc. of San Jose, CA. The output of the optical transmitter **910** is an optical digital signal that can be coupled to a collimating lens **920**.

In some embodiments, an input optical digital signal passes through a focusing lens **921** configured to substantially focus a collimated optical signal onto an optical

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receiver **912**. The focused optical digital signal can be substantially directed and focused onto the optical receiver **912** configured to convert an optical digital signal into a corresponding electrical digital signal. The corresponding electrical digital signal can be provided to the magazine processor **906**. This electrical digital signal may be an acknowledgement of the ammunition count, a command to obtain a magazine count, an updated calibration table for a linear encoder of the magazine, or any other command or data that the weapon may supply to a magazine. The optical receiver **912** can convert optical signals to electrical signals using any appropriate technique such as, for example, outputting an electrical current that is proportional to the input power of the optical signal. The optical receiver can be any suitable component for converting optical digital signals to electrical digital signals, such as, for example HXR-4101A-DNT-T manufactured by GigOptix, Inc. of San Jose, CA.

The output optical digital signal output by the optical transmitter **910** can be collimated by collimating lens **920**. The collimated optical signal may pass through an output gap **922** before passing through a focusing lens. The collimated optical signal passes through an input gap **923** before being focused by focusing lens **921**. In some embodiments, the output and input gaps **922** and **923** can be about 1 mm between lens elements. In some embodiments, the gaps **922** and **923** may be greater than or equal to about 2 mm, less than or equal to about 1 mm, about 0.5 mm, or about 1.5 mm. In some embodiments, the gaps **922** and **923** can have differing distances between lens elements, such as, for example, there can be about 1.5 mm between lens elements **920** and **926** in output gap **922** and there can be about 0.8 mm between lens elements **921** and **927** in input gap **923**. In some embodiments, the non-contact optical connection can include transparent windows **928** and **929** of the signal converter **900** that are configured to have an exterior surface that is substantially aligned with an exterior surface of the corresponding transparent windows **928** and **929** of the signal converter **902**. The transparent windows **928** and **929** can be configured to be substantially transmissive for wavelengths that correspond to wavelengths of light used in the optical-transceiver **918** of the magazine and the weapon. The transparent windows **928** and **929** can be treated with coatings to make them more durable, scratch resistant, hydrophobic, polarized, filtered, and the like. The transparent windows **928** and **929** can provide a protective surface for the lens elements **920**, **921**, **926** and **927**. The transparent windows **928** and **929** can provide a surface that is cleaned with relative ease to maintain optical coupling between components of the magazine and the weapon.

In some embodiments, the signal converters **900** and **902** may include power connectors and power transmission lines that can optionally be used to supply power to the magazine and/or to the weapon. In some such embodiments, the power may be used to power a processor in the magazine or the weapon. Alternatively, or in addition, the power may be used to charge a battery in the magazine or the weapon.

In certain embodiments, the magazine processor **906** of the magazine may be the electronic circuitry **122** or may be included in the electronic circuitry **122** of the magazine **100**. The magazine processor **906** may be an FPGA, a microprocessor, or a custom processor configured to at least determine the ammunition count for cartridges inserted into the magazine **100**. Further, the magazine processor **906** may control a display of the magazine **100**.

The firearm processor **908** of the weapon may be or may be included in the electronic circuitry **2002** described in more detail in FIG. **20**. The firearm processor **908** may be an

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FPGA, a microprocessor, or a custom processor configured to perform one or more of the embodiments described herein. Further, the firearm processor **908** may determine a number of cartridges registered to the firearm or weapon via the inclusion of the cartridges in one or more magazines registered with the firearm or weapon. In some embodiments, the firearm processor **908** may determine whether a cartridge is loaded within a chamber of the firearm. In addition, the firearm processor **908** may determine whether the firearm is jammed.

Example Magazine Display

FIG. **10** illustrates an embodiment of a display **124** included as part of the magazine. The display **124** is positioned at the bottom of the side of the magazine **100** that faces the user so that the user can view the display **124** while creating minimal interference with the chamber that includes the cartridges and to prevent blockage from the weapon when the magazine **100** is inserted into the weapon. However, the location of the display **124** is not limited as such and may be positioned elsewhere on the magazine **100**. For example, the display **124** may be positioned closer to the optical transceiver **116** (see FIG. **7**) so that the display **124** is closer to eye level with a user when the user is using the weapon. In some cases, the display **124** may extend outwards from the housing **102** providing room for installation of the display **124** without interfering with the placement of cartridges within the magazine **100**.

As illustrated in FIG. **10**, the display **124** may be inset. Advantageously, in certain embodiments, the inset display **124** can improve visibility to the user while reducing visibility of the display **124** by other observers, such as enemy combatants. Further, the display **124** may comprise an LED display, an LCD display, an OLED display, a touchscreen display, or any other type of display. Moreover, the display **124** may display one or more data items to a user. For example, the display **124** may display a number of cartridges within the magazine, a number of cartridges fired from the magazine, a number of shots fired by the weapon, a number of magazines available to the user, a jam state of the weapon, whether a cartridge is within a chamber of the weapon, and any other information relating to the status of the magazine loaded in the weapon, magazines available to the user, and/or the weapon itself. In some embodiments, the display **124** may be optional and/or may supplement an additional display associated with the weapon and/or other gear (e.g., a helmet or goggles) of the user.

In certain embodiments, the output of the number of cartridges within a magazine, or a number of cartridges available to a user across magazines, maybe an auditory output that is received via a user's radio or headset. In some cases, the user may receive both a visual and audio indicator of the number of cartridges available.

Example Magazine Circuitry

FIG. **11A** illustrates an embodiment of the electronic circuitry **122** and display **124** included as part of the magazine. As illustrated, the electronic circuit **122** may be fit into the base or cap **128** of the magazine. FIG. **11B** illustrates an alternative view of the embodiment of the electronic circuitry **122** and display of FIG. **11A**. As previously illustrated with respect to FIG. **1**, the battery **126** may be or may be shaped as an AA or AAA battery. Alternatively, the battery may be a disc or circular shaped battery, similar to a watch battery in shape. The battery may be a lithium-ion battery, an alkaline battery, a nickel cadmium battery, a nickel metal hydride battery, or any other type of battery. In some cases, the battery may be a rechargeable battery. In embodiments where the battery is a rechargeable battery, the battery may

be recharged by placing the magazine on a recharger or a charging pad. In some embodiments, the battery may be removed from the magazine for replacement or charging purposes. In other embodiments, the battery may not be removed from the magazine. For example, in some cases, the electronics of the magazine may be housed in a water-proof housing, and the battery may be sealed within the housing.

FIG. 12 illustrates an embodiment of a display circuit 1200 for a display. The display may be part of a magazine, a weapon, a scope, or other component of a weapon or weapon system. The display circuit 1200 may be included as part of the display or may be in communication with the display. In some embodiments, the display circuit 1200 may be included as part of the electronic circuitry of the weapon, such as the electronic circuitry 2002 described below with respect to FIG. 20.

The display circuit 1200 may include a plurality of light emitting diodes (LEDs) 1202. Alternatively, or in addition, these light emitting diodes 1202 may be included as part of the display and may be in electrical communication with the display circuit 1200. The LEDs 1202 may include ultra-bright LEDs that are visible when a user is in full daylight. The ultra-bright LEDs 1202 may have a very bright output enabling them to be viewable during the daytime. For example, ultra-bright LEDs 1202 may have a luminosity of 100 millicandelas (mcd), 200 mcd, 300 mcd, 500 mcd, or more, or any range between the foregoing. Further, the ultra-bright LEDs 1202 may be dimmed to a very low level enabling them to be viewed through night vision goggles without interfering with the user's ability to use the night vision goggles. For example, the ultra-bright LEDs 1202 may be configured to consume milliwatts of power during the daytime and may be configured to consume microwatts of power when a user is wearing night vision goggles. Further, the ultra-bright LEDs 1202 may be configured to output light and varying degrees of magnitude. Thus, there may be one brightness level during a sunny day, another brightness level during a cloudy day, another brightness level at night, and another brightness level when a user is wearing night vision goggles. In some embodiments, the LEDs 1202 may be dual-mode or multi-mode LEDs that are capable of functioning at different levels of brightness based at least in part on an amount of power received and/or on a control signal. In other words, the LEDs 1202 may provide different brightness outputs based at least in part on the amount of power received.

In certain embodiments, the LEDs 1202 are configured to provide a seven segment display. Further, the LEDs 1202 may display multiple numbers. In some embodiments, the LEDs 1202 may be configured to display characters and/or symbols instead of or in addition to the numbers.

The display circuit 1200 may control the LEDs 1202 to make them viewable during the daytime, at nighttime, or as previously described when a user is wearing a vision goggles. To enable the different viewable modes, the display circuit 1200 includes multiple resistor paths or other current driving device that are in communication with a battery 1210. For example, the display circuit 1200 may include two resistor paths: a first path that includes the resistor 1204 and a second path that includes resistor 1206. The resistor 1204 may be relatively small resistor (e.g., a 10, 20, 50, or 100 ohm (Ω) resistor, or any value between the foregoing resistances) compared to the resistor 1206 which may be a relatively large resistor (e.g., a 1, 2, 10, or 50 mega-ohm ($M\Omega$) resistor, or any value between the foregoing resistances). Further, the display circuit 1200 may include a

switch 1208 that may connect the battery 1210 to the light emitting diodes 1202 via the path that includes the resistor 1206 or the path includes the resistor 1204.

In certain embodiments, when a user is not wearing night vision goggles, the LEDs 1202 may be connected to the battery 1210 via the path that includes the smaller resistor 1204 resulting in a brighter output compared to when the LEDs 1202 are connected to the battery 1210 via the path that includes the larger resistor 1206. In contrast, when it is determined that the user is wearing night vision goggles, LEDs 1202 may be connected to the battery 1210 via the path includes the larger resistor 1206 resulting in a dimmer output compared to the when the LEDs 1202 are connected to the battery 1210 via the path includes the smaller resistor 1204. In some embodiments, the display circuit 1200 may include additional resistor paths or other driver circuitry that connect the battery 1210 to the light emitting diodes 1202 depending on the brightness of the ambient light and/or whether the user is wearing night vision goggles.

The switch 1208 may select the resistor paths to connect the battery 1210 to the light emitting diodes 1202 based on a control signal received from a controller 1212. The controller 1212 may configure the switch 1208 based on input from a user interface switch 1214. The user interface switch 1214 may enable a user to turn off the light emitting diodes 1202, identify that the user is wearing night vision goggles, or select a brightness auto adjustment mode that automatically selects a resistor path so as to adjust the brightness of the LEDs 1202 based on an ambient light detected by the light sensor 1216.

In some embodiments, the display circuit 1200 may support pulse width modulation (PWM). In some such embodiments, the controller 1212 may include a pulse width modulation (PWM) controller 1218. PWM may be used to set or adjust a duty cycle for the power supplied to the LEDs 1202. In other words, instead of supplying power from the battery 1210 to the LEDs 1202 at a constant level, the PWM controller 1218 can alternate between supplying power and not supplying power to the LEDs 1202. For example, when the display controlled by the display circuit 1202 is activated or turned on, the PWM controller 1218 can alternate equally between supplying and not supplying power to the LEDs 1202 (e.g., 50% duty cycle) for a repeating set of time periods (e.g., a set of clock cycles). As another example, the PWM controller 1218 may supply power 75% of the time (e.g., 75% duty cycle) or 25% of the time (e.g., 25% duty cycle) for a repeating set of time periods (e.g., a set of clock cycles). Advantageously, in certain embodiments, by using PWM, greater control over the brightness of the LEDs 1202 can be achieved. Further, a greater variety of brightness levels can be achieved. Thus, the LEDs 1202 can be made bright or dim based on the amount of sunlight. Further, the LEDs 1202 can be dimmed to very low levels when the user is wearing night vision goggles. In some embodiments, the LEDs 1202 may be dimmed to a level not visible to the human eye, but that is visible to a user wearing night vision goggles. In certain embodiments, the PWM controller 1218 may implement PWM by adjusting the switch 1208 between an open position (as illustrated in FIG. 12) and a closed position with the path that includes the larger resistor 1206 (when in night vision mode) or the smaller resistor 1204 (when not in night vision mode).

In some embodiments, the display circuit 1200 may include a controller for controlling an augmented reality display. In some embodiments, the output of the display may be projected onto a scope generating an augmented reality display interface. In other embodiments, the augmented

reality display may be output via another display of the user, such as via the user's goggles or night vision helmet.

In certain embodiments, the magazine may include an accelerometer, a gyroscope, and/or other types of motion sensors that can detect when a user has picked up the magazine or when the magazine is in motion. When it is determined that the magazine is in motion, the magazine circuitry can be transitioned from an off-state to an on-state or from a sleep-mode to an on-state. Transitioning to an on-state may include turning on a processor of the magazine to enable a determination of cartridge count within the magazine. However, in certain embodiments, the display 124 of LEDs 1202 remain unpowered to prevent the emission of light at an undesired time.

Example Weapon System

FIG. 13 illustrates an embodiment of a weapon 1300 with the magazine 100 inserted. FIG. 14 illustrates a modified view of the embodiment of the weapon of FIG. 13. Referring to FIG. 13, the weapon may optionally include a scope 1302 and/or a rail mounted display 1304. In certain embodiments, the scope 1302 and/or display 1304 may present a user with some or all of the information that can be displayed on the display 124. Further, in some cases, the scope 1302 and/or display 1304 may display additional and/or complementary information from what is displayed on the display 124 and/or another display. For example, the display 124 may display the amount of cartridges within the magazine 100, the scope 1302 may display the amount of cartridges within the loaded magazine and the number of magazines available to the user, and the display 1304 may display the number of shots fired.

Electronics for controlling the additional displays, such as the scope 1302 and the display 1304 may be located within the handle of the weapon 1300. The handle may include a control 1306 for activating or deactivating electronics included in the weapon 1300. Further, the control 1306 may be used to cycle between different display options including the data to display and the display sources (e.g., the scope 1302 or the display 1304). In some cases, the control 1306 may also control the display 124 of the magazine 100.

To display data and to control the various display sources, the weapon 1300 may include one or more optical transceivers, similar to the optical transceiver 116 of the magazine 100. Cables or wires may be used to communicate between the electronics within the handle of the weapon 1300 and various connection points in the weapon 1300 that enables accessories connected to the weapon 1300 to serve as display devices. For example, a cable may connect the electronics to an optical transceiver that communicates with a transceiver on the scope 1302.

FIG. 15 illustrates a line drawing of the embodiment of the weapon of FIG. 13 without the inserted magazine. FIG. 16 illustrates a line drawing of the embodiment of the weapon of FIG. 13 and an un-inserted magazine. As can be seen with respect to FIG. 16, the weapon includes a lip 1602 that extends from the insertion port 1604 where the magazine 100 is inserted into the weapon 1300. The lip 1602 and the magazine 100 may be designed to align the optical transceiver 116 of the magazine 100 and an optical transceiver that may be included in the lip 1602 of the weapon 1300. The mating of the optical transceiver 116 and the transceiver in the lip 1602 of the weapon 1300 can be seen in FIG. 17, which illustrates an embodiment of the weapon handle of the weapon 1300 and the magazine 100. In some embodiments, the optical transceiver 116 may be configured to communicate using infrared communications. Altern-

tively, or in addition, the optical transceiver 160 may be configured to communicate using other optical or frequency communication bands.

In certain embodiments, as previously described with respect to the optical transceiver 116, the transceiver included in the lip 1602 may be a contactless transceiver. Further, in some implementations of the weapon 1300, the transceiver included in the lip 1602 may be a non-optical transceiver, such as a near field communications (NFC) reader.

In some embodiments, the weapon 1300 may further include an optical scanner that can scan or otherwise access a machine-readable code included on the magazine 100. The weapon 1300 can register one or more magazines by scanning the machine-readable code included on each magazine. Alternatively, the weapon 1300 can access an RFID tag on the magazine to register the magazine. Once the magazine has been registered with the weapon 1300, the weapon 1300 can monitor the status of the magazine, such as whether the magazine has been inserted into the weapon, whether the magazine is empty, or the number of cartridges included in the magazine. In some cases, the weapon 1300 may use near-field communication, Bluetooth™, Wi-Fi™, or any other type of communication to communicate with one or more magazines within a particular distance of the weapon 1300.

In some embodiments, the weapon 1300 can receive one or more status signals from one or more magazines registered with the weapon. Advantageously, in certain embodiments, the weapon 1300 can aggregate the information received in the status signals to determine a status of a user's total ammunition. For example, the weapon 1300 can determine and present to the user a total quantity of ammunition available to the user, the total number of magazines available to the user, the number of loaded and/or empty magazines in the user's gear, etc.

FIG. 18 illustrates an embodiment of the ammunition status display 1304 attached to a rail 1802 of the weapon 1300. FIG. 19 illustrates an embodiment of the ammunition status display 1304 separate from the weapon 1300. It should be understood that the display 1304 may be attached to other locations of the weapon 1300. The display 1304 may include a number of controls 1804 and 1806 for configuring the data that is presented on the display 1304. For example, the display 1304 may present a count of shots fired, a count of ammunition available, and/or a count of ammunition within a loaded magazine. Further, the display 1304 may include circuitry for detecting when shots have been fired. For example, the display 1304 may include circuitry for detecting vibrations within the barrel 1808 of the weapon 1300 and/or audio from a bullet being fired. Using the detected vibrations and/or audio, the display 1304 can count shots fired.

Example Weapon Internals

FIG. 20 illustrates a cross-section of an embodiment of the weapon 1300 with the magazine 100 installed. FIG. 21 illustrates a line drawing of the cross-section of the embodiment of the weapon of FIG. 20 with the magazine installed. Referring to FIG. 20, as illustrated, the handle 2000 may include electronic circuitry 2002 that can process data received from the magazine 100. This electronic circuitry 2002 may be powered by a battery 2004 that can be installed in the handle. In certain embodiments, the battery 2004 may also power the electronic circuitry included in the magazine 100. In some such embodiments, the battery 126 of the magazine 100 may be optional or omitted. In other embodiments, the battery 2004 may be used to charge the battery

126 and/or vice versa. The battery within the weapon used to power the electronic circuitry **2002** may be of the same type as the battery in the magazine **100**. In other cases, the battery used for the weapon may differ from the battery used for the magazine **100**.

In addition to processing data relating to the amount of cartridges within the magazine **100**, the electronic circuitry **2002** can track a number of magazines available to a user; the status of the magazines, such as cartridges per magazine, empty or full status, etc.; the number of rounds fired since a particular time, such as per day, during a particular mission, since manufactured, etc.; and the position of a bolt **2010** within a buffer tube **2012**. In some embodiments, the electronic circuitry **2002** may include a transmitter for transmitting some or all of the data to a display that is separate from the weapon **1300**. For example, the data may be transmitted to a heads up display (HUD) within a helmet or goggles of a user. In some embodiments, data may be transmitted to a command center enabling a commanding officer to monitor ammunition count of soldiers, or other users, in the field or during a mission.

In certain embodiments, electronic circuitry **2002** of the weapon may be configured to determine a number of shots fired by the weapon. The electronic circuitry **2002** may determine the number of shots fired by tracking a change in the cartridges available to the user. For example, if it is determined that a magazine has 12 cartridges at the first point in time and at a second point in time it is determined that the magazine has 5 cartridges, the electronic circuitry **2002** may determine that seven cartridges have been fired by the weapon. However, if electronic circuitry **2002** determines a change in the cartridges available in a magazine that is not inserted into the weapon, the electronic circuitry **2002** may determine that a user has added or removed cartridges without firing the cartridges from the weapon. Similarly, if electronic circuitry **2002** determines that the magazine is no longer registered to the weapon, the total count of cartridges available to the user may be reduced without increasing a shot count for the weapon. In some embodiments, the electronic circuitry **2002** may determine the number of shots fired based at least in part on the number of measured movements of a bolt in the weapon. The bolt may include a cylinder, a rod, or other movable portion of the weapon located within a buffer tube. In certain embodiments, the movement of the bolt can be used to open or close the chamber of the weapon. Further, movement of the bolt may facilitate loading a cartridge into a chamber and/or expelling a spent cartridge from the chamber or the weapon. If the bolt is moved, but a trigger is not pulled, electronic circuitry **2002** may determine that a shot was not fired despite movement of the bolt. Similarly, if the bolt is moved, but a magazine has not been inserted into the weapon or a magazine inserted into the weapon has no cartridges, the electronic circuitry **2002** may determine that a shot was not fired despite movement of the bolt.

In some embodiments, the electronic circuitry **2002** may be configured to communicate with different magazines to obtain a total count of magazines registered with the weapon. The electronic circuitry **2002** may include a wireless transceiver to communicate with the magazines registered with the weapon. This wireless transceiver may use ultra-wideband communications to reduce the possibility of interference by other systems or users and/or to reduce the possibility that the communications are captured by other systems or users. In some embodiments, the communication may be encrypted or use a secure channel. Further, communication between the weapon and systems of another user or

a command center may be encrypted. Thus, in some instances, electronic circuitry **2002** of the weapon of one user may use encrypted ultra-wide band communication to communicate with a system of another user or a computing system at a command center.

In some implementations of the magazine **100**, data may be transmitted from the magazine **100** to a display that is separate from the magazine **100** and the weapon **1300**. For example, as described above with respect to the handle electronics **2002**, data from the magazine **100** may be transmitted to a HUD in a helmet or eyewear.

As previously described, the magazine **100** and/or weapon **1300** can track ammunition within the magazine **100**. Further, the electronic circuitry **2002** can determine whether a cartridge or bullet is loaded into a chamber **2014** of the weapon **1300**. The weapon **1300** may include a magnet **2016** positioned within the buffer tube **2012**. Further, the weapon **1300** may include one or more Hall effect sensors **2018** within the buffer tube **2012**. The Hall effect sensors **2018** can be used to determine a location of the magnet **2016** within the buffer tube **2012**. Using the location of the magnet **2016**, the electronic circuitry **2002** can determine whether a cartridge has been loaded into the chamber **2014**. Similar to the electronic circuitry **122**, the electronic circuitry **2002** may include a hardware processor, or programmable hardware, and may be configured to perform one or more of the processes described herein.

In some cases, a user may insert a magazine **100** that includes the features disclosed herein for determining a cartridge count into a traditional weapon that does not include the features disclosed herein. Although the magazine **100** can still determine the cartridge count in the magazine, the weapon may be unable to communicate whether a cartridge is loaded into the buffer tube. To prevent a user from obtaining a false reading of the cartridge count within the weapon (e.g., the cartridges in the magazine combined with the cartridge, if any, in the buffer tube), a button or other deactivation trigger may be positioned on the magazine **100**. When the magazine **100** is inserted into a weapon that does not include the capability of detecting a cartridge in the buffer tube, the button or other deactivation trigger may be activated causing the electronics of the magazine **100** to be deactivated. This button may be positioned such that when the magazine **100** is inserted into the weapon, a portion of the weapon presses against the button. For example, the button may be positioned near the top of the magazine such that the button is inserted within the weapon along with the magazine and is pressed against an inner surface of the insertion port of the weapon.

Accordingly, a user will not obtain a cartridge count of cartridges within the weapon. To determine the number of cartridges within the magazine, a user can eject the magazine **100** and reactivate the circuitry to determine the cartridge count within the magazine **100**. The circuitry may automatically be reactivated when the magazine is ejected as the button, or other deactivation trigger, will no longer be in contact with the weapon.

A weapon configured to detect a cartridge in a chamber may be configured to not press against the button of the magazine thereby preventing the cartridge count circuitry from being deactivate. For example, the insertion port may include a gap or notch that prevents the button on the magazine from being depressed. As an alternative, the circuitry of the weapon may communicate its functionality to the circuitry of the magazine. Upon receipt of a message from the weapon that it is capable of detecting a cartridge in the chamber, the magazine may activate its cartridge count-

ing capabilities. In some embodiments, the control interface of the magazine can be positioned on a portion of the magazine that is inaccessible to the user when the magazine is inserted into the firearm.

In some embodiments, the electronic circuitry **2002** may track a number of times a weapon has been fired. Accordingly, the electronic circuitry **2002** may provide maintenance information to a user or maintenance warnings alerting the user when one or more portions of the weapon should be serviced. In some cases, the electronic circuitry **2002** may track a number of times a weapon is by an fired since the weapon, or a portion thereof, was last serviced.

The electronic circuitry **2002** may be capable of processing auxiliary data received from auxiliary sources and/or output to auxiliary systems. For example, the weapon **1300** may include a receiver or transceiver that can receive information from a range finder or a global positioning system (e.g., GPS). This information may be provided on a display, such as one integrated with a scope of the weapon. Thus, a user can view the cartridges available to the user in one or more magazines, a position of the user within a geographic area, and a distance to a target. Advantageously, by displaying the auxiliary information on a scope of the weapon, a user can obtain or view the information without removing his or her focus from a potential target. In another example, the weapon **1300** may communicate information obtained by the weapon **1300** or determined by the electronic circuitry **2002** to a radio of the user for transmission to another user, such as a unit leader or commander. For instance, an ammunition count may be communicated over the radio to a user.

FIG. **22** illustrates a line drawing of a cross-section of an embodiment of the handle **2000** and magazine **100** inserted into the weapon **1300**. FIG. **23** illustrates a line drawing of another perspective of the cross-section of the embodiment of the handle and magazine inserted into the weapon of FIG. **22**. With reference to FIG. **22**, the handle **2000** can further include a cap **2202** that can be opened to replace the batteries **2004** that power the electronic circuit **2002**. Further, the handle **2000** may include optical connections **2204** that communicate data to one or more optical transceivers that provide the data to one or more displays (e.g., the display **1304** or scope **1302**) for display to a user.

FIG. **23** illustrates a line drawing of another perspective of the cross-section of the embodiment of the handle and magazine inserted into the weapon of FIG. **22**. FIG. **23** further illustrates the alignment of the optical transceiver **116** of the magazine **100** and the optical transceiver **2302** located in the lip **1602** of the weapon **1300**. As illustrated by the line drawing, when inserted into the weapon **1300**, the magazine **100** is aligned such that the optical transceiver **116** aligns with the optical transceiver **2302**. Further, the spacing between the magazine **100** and the weapon **1300** may be sufficiently small to prevent ambient light from interfering with the optical connection.

FIG. **24A** illustrates an embodiment of a non-contact optical connector **2302** included in the lip **1602** of the insertion point for inserting a magazine **100** into the weapon **1300**. This non-contact optical connector **2302** can function as a transceiver for receiving data from a magazine **100** when the magazine **100** is inserted into the weapon **1300**. Although typically the optical transceiver **116** transmits data and the optical transceiver **2302** receives data, in some embodiments, the optical transceiver **116** may also receive data and/or the optical transceiver **2302** may also transmit data. Advantageously, in certain embodiments, by enabling the weapon **1300** to transmit data to the magazine **100**, the

display **124** of the magazine **100** can display data gathered by the weapon **1300**, such as shot count or total magazines available.

In certain embodiments, it is desirable to confirm whether the magazine **100** is inserted into the weapon **1300**. For example, a different calibration table for a linear encoder may be used when the magazine is inserted into the weapon **1300**. In some cases, the magazine **100** can determine it is positioned within the weapon **1300** by communicating via the optical transceiver. But in some cases, the magazine **100** may not be able to communicate using the optical transceiver because, for example, the weapon is deactivated or the battery within the weapon **1300** is uncharged or dead. One method of determining whether the magazine is inserted into the weapon **1300** is to use a magnetic sensor, such as a Hall effect sensor that can detect when it is within a threshold distance of a magnet positioned within the insertion port of the weapon **1300** or at a location that is within a threshold distance of the magazine **100** when the magazine **100** is inserted into the insertion port of the weapon **1300**.

FIG. **24B** illustrates a portion of the magazine **100** with a transceiver **116** and a magnetic sensor **2410**. The magnetic sensor **2410** may be a Hall effect sensor. Because the magnetic sensor **2410** is collocated with the transceiver **116**, the magnetic sensor **2410** may be aligned with a corresponding transceiver **2302** when the magazine **100** is inserted into a weapon **1300**.

FIG. **24C** illustrates a portion of the weapon **1300** with a transceiver **2302** and a magnet **2420**. Because the magnet **2420** is collocated with the transceiver **2302**, the magnet **2420** may be aligned with the magnetic sensor **2410** and the transceiver **116** when the magazine **100** is inserted into the weapon **1300**. Thus, the magazine **100** can determine from a signal generated by the magnetic sensor **2420** when the magazine is inserted into the weapon **1300**. The magazine **100** can then determine whether to load a calibration table associated with an inserted magazine or a calibration table associated with an uninserted magazine when determining an ammunition count.

Further, as the magnetic sensor **2410** is located at the rear **2425** of the magazine **100**, while the magnetic sensors **112** of the linear encoder and magnet **110** are located near the front edge **402** of the magazine, the magnetic field generated by the magnet **110** will not be detected, or will not generate a strong enough signal in the magnetic sensor **2410** to cause an incorrect determination of insertion status. Similarly, as the magnet **2420** in the weapon **1300** is located adjacent to the rear of the magazine when the magazine is inserted into the weapon, the magnetic field generated by the magnet **2420** will not be detected by the sensors **112** or will not generate a strong enough signal in the magnetic sensors **112** to cause an incorrect determination of ammunition count.

FIG. **25** illustrates an embodiment of a buffer tube **2012** within a weapon **1300**. As previously described, the buffer tube may have a set or series of Hall effect sensors **2018**. In certain embodiments, the Hall effect sensors **2018** may be supplemented and/or replaced by other types of sensors, such as optical sensors. Using the Hall effect, the sensors **2018** can detect the motion or action of a bolt **2010** based on the movement of a magnet **2016** attached to the bolt **2010**. Based on the position of the bolt **2010**, a hardware processor included, for example, in the handle of the weapon, can determine whether a cartridge has been loaded into a chamber of the weapon. Thus, advantageously, the weapon can determine the total ammunition count within the weapon based on a summation of the ammunition in a magazine

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loaded into the weapon and the amount of cartridges within the chamber of the weapon. Although the number of cartridges within a chamber of the weapon is typically 0 or 1, embodiments disclosed herein can be adapted for use with weapons that may load multiple cartridges into a chamber or multiple chambers of the weapon, such as with a combination gun that has chambers configured to load different types of ammunition (e.g., a combination shotgun and rifle). In some embodiments, the movement of the bolt **2010** can be detected based on principles other than the Hall effect. For example, if the sensors **2018** are light sensors, the movement of the bolt **2010** can be detected based on the change of light around the sensors **2018**. This light may filter in via the barrel of the weapon and/or the chamber. Alternatively, or in addition, the light may be based on a light source (e.g., an LED) attached to the bolt **2010**.

FIG. **26** illustrates a line drawing of an embodiment of a buffer tube **2012** and stock **2602** of a weapon **1300**. As illustrated, the stock **2602** may be at least partially hollow allowing for the buffer tube **2012** to extend into the stock **2602** and providing room for the bolt **2010** to move into the stock **2602**. In some embodiments, the bolt **2010** may be extended compared to other bolts to enable the bolt to travel through the extended buffer tube. Advantageously, by extending the buffer tube **2012** into the stock **2602**, additional space may exist in the weapon **1300** compared to weapons that use a solid stock. This additional room may be used to insert the sensors **2018**. Further, using a hollow stock enables the weapon **1300** to maintain its weight despite the addition of the magnet, sensors, electronic circuitry, and/or batteries. It should be understood that in certain embodiments, the weapon **1300** can implement features of the present disclosure without modification to the stock of the weapon. For example, the sensors may be positioned between the chamber and the stock.

Example Use Case

FIG. **27** illustrates an example use case of certain embodiments described herein. In this non-limiting example use case, a soldier is approaching enemy combatants on a bridge. The soldier may consider engaging the enemy combatants, but does not wish to engage without a minimum quantity of ammunition loaded in the soldier's weapon. To ensure a minimum ammunition quantity in the weapon, the soldier could replace the magazine with a new magazine. However, this could result the soldier carrying a number of partially filled magazines without knowing how much ammunition is available in total to the soldier.

Alternatively, using embodiments disclosed herein, the user can look at a display, such as via the scope and see the total ammunition count within the weapon including the magazine and chamber of the weapon (e.g., 30 cartridges in the illustrated example). Further, the soldier can be presented with a total ammunition count available to the soldier via additional magazines the soldier is carrying (e.g., 175 cartridges in the illustrated example). Alternatively, the numbers illustrated via the display in the scope may represent the cartridges available in the weapon out of the total capacity of the magazine.

As illustrated in FIG. **27**, the information presented to the soldier is presented in as part of a HUD and/or augmented reality display observable through a scope on the weapon. Thus, when the soldier looks through the scope, the soldier can see the targets in front of the weapon, and also see information provided by the weapon to the scope regarding the cartridges available to the user in, for example, the loaded magazine.

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The use case presented in FIG. **27** is one non-limiting example use case. Other uses cases are possible. For example, a police force can use embodiments disclosed herein to monitor the ammunition available to its officers. Further, a hunter could use embodiments disclosed herein to monitor the ammunition available during a hunt.

Moreover, in certain embodiments, the weapon can include a transmitter for transmitting ammunition information to a command post or other location monitored by users (e.g., commanders) associated with the carrier of the weapon. Alternatively, or in addition, the weapon may communicate with another system of the user, such as the user's helmet, which may communicate the ammunition information to the command post. Advantageously, the ability to monitor ammunition at a command post or other location enables a user, such as a commander or police captain, to monitor the ammunition of a user (e.g., a soldier or policeman) to determine whether reinforcements are needed, additional ammunition is needed, or if an unexpected firefight is occurring. For instance, a police captain can determine if a traffic officer has fired his weapon by receiving an alert that an amount of ammunition carried by an officer has changed. As another example, a field commander can determine that soldiers have been ambushed while guarding supplies based on an unexpected change in ammunition.

Example Ammunition Count Determination Process

FIG. **28** presents a flowchart of an embodiment of an ammunition count determination process **2800**. The process **2800** can be implemented by any system that can determine a count of the number of cartridges within a magazine. For example, the process **2800**, in whole or in part, can be implemented by electronic circuitry included in the magazine, such as the electronic circuitry **122** included in the magazine **100**. This electronic circuitry **122** can include hardware, such as a hardware processor, that can perform the process **2800**. In some embodiments, the electronic circuitry **122** includes application-specific hardware configured to perform the process **2800**. In other embodiments, the hardware may include a computer processor programmed with special instructions configured to perform the process **2800**. Further, the electronic circuitry **122** may include control circuitry for controlling one or more of the magnet **110** and/or the sensors **112**. In some embodiments, some or all of the process **2800** may be performed by electronic circuitry in the weapon, such as the electronic circuitry **2002**. The electronic circuitry **2002** may include some or all of the embodiments of the electronic circuitry **122**. To simplify discussion and not to limit the disclosure, portions of the process **2800** will be described with respect to particular systems, such as the electronic circuitry **122** or **2002**. However, it should be understood that operations of the process **2800** may be performed by other systems. For example, operations described as being performed by the electronic circuitry **122** may alternatively be performed by the electronic circuitry **2002**.

The process **2800** begins at block **2802** wherein, for example, the electronic circuitry **122** generates a magnetic field within a magazine **100** using a first magnet **110**. In certain embodiments generating the magnetic field may include producing an electric current in a wire that surrounds the magnet, such as with an electromagnet. In other embodiments, the magnetic field is generated by the magnet **110**. In some such cases, the electronic circuitry **122** may not be involved in generating the magnetic field. However, electronic circuitry **122** may be involved in determining the strength of the magnetic field.

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Using a first set of sensors **112** within the magazine **100**, the electronic circuitry **122** at block **2804** detects a location of the magnet **110** based at least in part on the magnetic field measured by the set of sensors **112**. In certain embodiments, the location of the magnet **110** is determined based on the particular sensor from the set of sensors that detects the magnetic field of the magnet **110**. In other embodiments, the location of the magnet **110** is determined based on the strength of the magnetic field as detected by a plurality of the sensors **112**.

At block **2806**, the electronic circuitry **122** determines a number of cartridges within the magazine **100** based at least in part on the location of the first magnet as determined at the block **2804**. For example, if it is determined that the location of the first magnet **110** is at a particular location, the electronic circuitry **122** can determine that there are a particular number of cartridges within the magazine **100** based on a correspondence between the locations of the magnet **110** within the magazine **100** and the quantity of cartridges remaining within the magazine **100**.

In some embodiments, the electronic circuitry **122** may access a number of tables stored in a memory of the electronic circuitry **122** that identify a corresponding between one or more signals generated by sensors included in the magazine **100** and the number of cartridges loaded in the magazine **100**. In some embodiments, the electronic circuitry **122** may perform an interpolation process to determine a position of a magnet within the magazine **100** based on a plurality of signals received from sensors of the magazine **100**.

At block **2808**, the electronic circuitry **2002** generates a magnetic field within a buffer **2012** of a weapon **1300** using a second magnet **2016**. In certain embodiments, the block **2808** may include one or more of the embodiments described with respect to the block **2802**.

Using a second set of sensors **2018** within the buffer **2012**, the electronic circuitry **2002** at block **2810** detects a location of the second magnet **2016** based at least in part on the magnetic field generated at block **2808**. In certain embodiments, the block **2810** may include one or more of the embodiments described with respect to the block **2804**.

At block **2812**, electronic circuitry **2002** determines whether a cartridge is within a chamber **2014** of the weapon **1300** based at least in part on the location of the second magnet **2016**. Determining whether a cartridge is within the chamber **2014** may include determining whether a bolt **2010** has fully cycled or is within a particular position as determined based at least in part on the position of the magnet **2016** within the buffer tube **2012** and/or the stock **2602**. In certain embodiments, the block **2812** may include one or more of the embodiments described with respect to the block **2806**. In certain embodiments, the number of cartridges within the magazine and/or whether a cartridge exists within a chamber of the weapon may be output on a display on the magazine, on the weapon, on another device of a user of the weapon, or on a device in another location, such as a command station. In some embodiments, the count of ammunition or cartridges in the weapon may be displayed using the example process **2900** described below with respect to FIG. **29**.

In certain embodiments, one or more of the blocks **2808** through **2812** may be optional or omitted. For example, in some embodiments, the number of cartridges within the magazine may be determined without the magazine being inserted into a weapon.

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Example Ammunition Count Display Process

FIG. **29** presents a flowchart of an embodiment of an ammunition count display process **2900**. The process **2900** can be implemented by any system that can display a count of the number of cartridges within a weapon including a chamber of the weapon and a magazine inserted into the weapon. For example, the process **2900**, in whole or in part, can be implemented by electronic circuitry included in the magazine **100**, such as the electronic circuitry **122**, and/or electronic circuitry in the weapon **1300**, such as the electronic circuitry **2002**. To simplify discussion and not to limit the disclosure, portions of the process **2800** will be described with respect to particular systems, such as the electronic circuitry **122** or **2002**. However, it should be understood that operations of the process **2800** may be performed by other systems. For example, operations described as being performed by the electronic circuitry **122** may alternatively be performed by the electronic circuitry **2002** and vice versa.

The process **2900** begins at block **2902** wherein, for example, the electronic circuitry **2002** determines a magazine insertion status. Determining the magazine insertion status may include detecting whether a signal is received at the optical receiver **2302** of the weapon **1300**. At block **2904**, a display of the weapon **1300**, such as the display **1304**, displays the magazine insertion status.

At decision block **2906**, electronic circuitry **2002** determines whether the bolt **2010** is open. Determining whether the bolt **2010** is open may include determining the position of the magnet **2016** as described with respect to the process **2800**. If it is determined at decision block **2906** that the bolt **2010** is open, electronic circuitry **2002** determines that the chamber **2014** is empty and may set a chamber counter corresponding to whether a cartridge exists within the chamber **2014** to **0**. At block **2910**, electronic circuitry **2002** determines a weapon count, or a count of the cartridges within the weapon **1300**. In some cases, determining a count of the cartridges within the weapon **1300** includes determining a number of cartridges within a magazine inserted into the weapon **1300**. Thus, the count of cartridges within the weapon, or the weapon count, may include the number of cartridges within the chamber **2014** and the number of cartridges within a magazine **100**, when inserted into the weapon **1300**. In cases where no magazine is inserted into the weapon, the weapon count will be equal to the number of cartridges, if any, within the chamber **2014**.

At block **2912**, a display, such as a HUD display of a scope **1302**, displays the total count of cartridges within the weapon **1300**. The count of cartridges within the weapon **1300** may be displayed for a particular period of time or until there is a change in a count of cartridges, such as by a removal or insertion of a magazine or the firing of the weapon. Alternatively, or in addition, the count of cartridges may be displayed in response to an interaction by a user with a user interface, such as a user pressing a button on the weapon or on the display device. In some cases, the count of cartridges may be displayed on multiple displays, such as on a display attached to the rail of the weapon, a display integrated into the magazine, and/or a display generated in the scope **1302**. Upon completion of, or in parallel with, operations associated with the block **2912**, the process **2900** may return to the block **2902**. In some embodiments, the process **2900** advances from the block **2912** to the block **2902** in response to detecting a change in the status of the weapon, such as the removal or addition of a magazine, a firing of a cartridge, or a detection of a change in the position of the bolt **2010**.

If it is determined at the decision block **2906** that the bolt **2010** is not open, electronic circuitry **2002** determines at the

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decision block **2914** whether the bolt **2010** has completely cycled. In some cases, when the bolt is completely cycled, a cartridge is inserted into the chamber **2014**. In certain embodiments, the decision block **2914** may also include determining whether a magazine is inserted into the weapon. Determining whether the bolt has completely cycled includes determining a position of the magnet **2016** within a buffer tube **2012** as previously described with respect to the process **2800**.

If it is determined at the decision block **2914** that the bolt is not completely cycled, a display of the weapon **1300**, such as the scope **1302**, displays a message or indicator corresponding to an unknown state at block **2916**. At block **2918**, electronic circuitry **2002** waits until it detects that the bolt is charged, or completely cycled indicating that cartridge may have been loaded into the chamber **2014**. In some embodiments, if the bolt does not completely cycle for a threshold period of time, it may be determined that the weapon is jammed. If it is determined that the weapon is jammed, a display of the weapon, or other display in communication with the weapon, such as an augmented reality display of a user's helmet, may output a weapon jammed indicator.

At decision block **2920**, electronic circuitry **2002** determines whether the magazine count has been reduced by 1. The magazine count corresponds to the number of cartridges within the magazine **100**. In some embodiments, the decision block **2920** may include determining whether the magazine count has been reduced by some value other than one. If it is determined that the magazine count has not been reduced by one, the process **2900** returns to the decision block **2906**.

If it is determined at the decision block **2920** that the magazine count has been reduced by one or if it is determined at the decision block **2914** that the bolt has completely cycled, the process **2900** proceeds to the decision block **2922**. At decision block **2922**, electronic circuitry **2002** determines whether the magazine count is less than zero. If it is determined that the magazine count is less than zero, indicating negative cartridges within the magazine, then the process **2900** proceeds to the block **2916** where a message or indicator corresponding to an unknown state is displayed.

In some embodiments, an unknown state indicator may be triggered in other circumstances. For example, when an error occurs in the electronic circuitry **2002** or if a conventional magazine is inserted into the weapon that does not include the features disclosed herein, an unknown state flag may be set and displayed to the user. In some embodiments, if a conventional magazine is inserted into the weapon, a display of the weapon may indicate that the magazine does not support the cartridge counting features, or other features, disclosed herein.

If it is determined that the decision block **2922** that the magazine count is not less than zero, and the process **2900** proceeds to the block **2924** where the chamber count is set to one. At block **2926**, the electronic circuitry **2002** determines the weapon count indicating the number of cartridges within the weapon **1300**. In certain embodiments, the block **2926** can include one or more of the embodiments of the block **2910**. At block **2928** a display of the weapon **1300** displays the weapon count or the number of cartridges within the weapon **1300** inclusive of both the chamber and a magazine, if any, installed within the weapon **1300**. In certain embodiments, the block **2928** can include one or more embodiments of the block **2912**.

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Example Magazine Calibration Process

FIG. **30** presents a flowchart of an embodiment of a magazine calibration process **3000**. The process **3000** can be implemented by any system that can calibrate an ammunition count system of a magazine. For example, the process **3000**, in whole or in part, can be implemented by electronic circuitry included in the magazine **100**, such as the electronic circuitry **122**, and/or electronic circuitry in the weapon **1300**, such as the electronic circuitry **2002**. To simplify discussion and not to limit the disclosure, portions of the process **3000** will be described with respect to particular systems, such as the electronic circuitry **122** or **2002**. However, it should be understood that operations of the process **3000** may be performed by other systems. For example, one or more operations of the process **3000** may be performed by a computing device that is configured to communicate with the magazine **100**.

In some embodiments, the process **3000** is performed by a manufacturer of the magazine **100**. The process may be an automated process performed as part of the process of manufacturing the magazine **100**. Alternatively, the process **3000** may be performed in whole or in part by a user calibrating the magazine **100** prior to sale or distribution, or by a user that has purchased or otherwise obtained the magazine **100**. For example, a user can instruct the magazine **100** to enter a calibration mode before performing a calibration process **3000**. In some embodiments, the calibration process **3000** can be repeated for a magazine. Repeating the process **3000** can improve the calibration if the magazine **100** has become worn and/or if a different type of ammunition is loaded into the magazine. Thus, in some embodiments, some or all of the operations of the calibration process **3000** can be used as a recalibration process and/or be used in place of some or all of the operations of the process **3200** described with reference to FIG. **32**.

The process **3000** begins at block **3002** where a maximum number of cartridges is inserted into a magazine. The maximum number of cartridges may be the maximum number of cartridges that the magazine is designed or configured to hold. In some embodiments the maximum number of cartridges may differ based on the manufacturer of the cartridges or the type of cartridges. The insertion of the cartridges into the magazine may be performed by an automatic magazine loader or other automated machine. Further, the loading of the cartridges into the magazine may be performed as part of the manufacturing process for the magazine. In some embodiments, the cartridges may be manually loaded into the magazine.

At block **3004**, a linear encoder position corresponding to the number of cartridges in the magazine is recorded in a calibration table. The linear encoder position may be determined from one or more signals received from one or more magnetic sensors, such as Hall effect sensors **112**, by a hardware processor. This hardware processor may be included within the electronic circuitry of the magazine, such as the circuitry **122**. Determining the linear encoder position may include determining a position of a magnet, such as a magnet attached to a follower of the magazine. The detected position of the magnet, as determined by the signals received from the one or more Hall effect sensors **112**, is associated with the number of cartridges inserted into the magazine. The relationship between the linear encoder position, or the detected position of the magnet, and the number of cartridges in the magazine is stored in the calibration table. Thus, after loading the magazine with a maximum number of cartridges, the linear encoder position recorded at the block **3004** is associated with the maximum number of cartridges can be loaded into the magazine. In some embodi-

ments, the operations of the block **3004** may be performed one more time after determining that the magazine is empty in order to store a linear encoder position for when the magazine is empty.

At block **3006**, a cartridge is removed from the magazine. The cartridge may be removed from the magazine by an automated machine. For example, a machine used during the manufacturing process of the magazine may be used to add and/or remove cartridges from the magazine during the magazine calibration process **3000**. Alternatively, in certain embodiments, the cartridge may manually be removed from the magazine.

At decision block **3008**, it is determined whether the magazine is empty. In certain embodiments, a machine used during manufacturing of the magazine determines whether the magazine is empty. Alternatively, or in addition, electronic circuitry of the magazine determines whether the magazine is empty based at least in part on a number of cartridges that have been ejected from the magazine or a location of the linear encoder. In some embodiments, a user indicates that the magazine is empty.

If the magazine is not empty, the process returns to the block **3004** where an updated linear encoder position corresponding to the updated number of cartridges in the magazine is recorded in the calibration table. The operations associated with the blocks **3004**, **3006**, and **3008** may be repeated until it is determined that the magazine is empty.

If it is determined at the decision block **3008** that the magazine is empty, a maximum number of cartridges is inserted into the magazine at the block **3010**. The block **3010** may include one or more of the embodiments described with respect to the block **3002**.

At block **3012**, the magazine is inserted into a weapon. The magazine may be inserted into a weapon as part of an automated manufacturing process for the magazine. Alternatively, the magazine may be inserted into the weapon by a user.

At block **3014**, a linear encoder position corresponding to the number of cartridges in the magazine is recorded in the calibration table. The block **3014** may include one or more of the embodiments described with respect to the block **3004**. In some embodiments, the block **3014** may include recording the linear encoder position in a different calibration table in the calibration table of the block **3004**. Alternatively, the linear encoder positions recorded at the block **3014** may be recorded in a different portion of the calibration table. In some embodiments, the linear encoder positions recorded at the blocks **3004** and **3014** may be marked or otherwise tagged to identify that the linear encoder positions are associated, respectively, for the magazine not being inserted into the weapon and with the magazine being inserted into the weapon.

At block **3018**, a cartridge is removed from the magazine. Removing a cartridge from the magazine may include cycling a bolt in the weapon to load a cartridge from the magazine into a chamber of the weapon and/or to expel a cartridge from a chamber of the weapon. In some embodiments, the magazine may be removed from the weapon, a cartridge may then be removed from the magazine, and the magazine may be re-inserted into the weapon. In some such embodiments, the block **3018** may include one or more of the embodiments previously described with respect to the block **3006**.

At decision block **3020**, it is determined whether the magazine is empty. The decision block **3020** may include one or more of the embodiments previously described with respect to the decision block **3008**.

If the magazine is not empty, the process returns to the block **3014** where a new linear encoder position is recorded in the calibration table for the current number of cartridges within the magazine. The operations associated with the blocks **3014**, **3018**, and **3020** may be repeated until it is determined that the magazine is empty.

If it is determined at the decision block **3020** that the magazine is empty, the calibration table is stored in a storage of the magazine at block **3022**. In some embodiments, the operations of the block **3014** may be performed one more time after determining that the magazine is empty in order to store a linear encoder position for when the magazine is empty. The storage of the magazine may be a non-volatile storage. The calibration table may be identified as a default or manufacturer default calibration table in the non-volatile storage. In some embodiments, the calibration table may be stored in a storage that is external or independent of the magazine. For example, copies of the calibration table may be stored in a manufacturer database.

Advantageously, in certain embodiments, by recording linear encoder positions for each load state of the magazine corresponding to the quantity of cartridges in the magazine, it is possible during use of the magazine to determine the number of cartridges within the magazine. Thus, a user can determine the number of cartridges within the magazine without manually counting the cartridges or keeping track of the number of shots fired. Further, by generating calibration table for when the magazine is inserted into a weapon and when the magazine is not inserted into a weapon, it is possible to more accurately track a number of cartridges loaded within the magazine. Typically, different pressures are applied to cartridges within the magazine when the magazine is loaded into the weapon compared to the when the magazine is not loaded into the weapon. These different pressures may cause the cartridges within the magazine to move. Thus, using the same calibration table or linear encoder positions stored within the calibration table for when the magazine is both inserted and not inserted may lead to an inaccurate count of cartridges loaded within the magazine. By separately recording the linear encoder positions for when the magazine is loaded into the weapon and when the magazine is not loaded into the weapon, a more accurate ammunition count can be determined for the magazine.

In some embodiments, the pressure on the cartridges in the magazine varies based on a position of the bolt of the weapon. For example, when the bolt is drawn back into the buffer tube of the stock and the chamber is open, the pressure applied to the cartridges in the magazine inserted into the weapon may differ from the pressure applied to the cartridges when the bolt is pushed forward towards the barrel of the weapon and slides over the top cartridge, or cartridge closest to the feed point, of the magazine. In some embodiments, the pressure applied to the cartridges in the magazine when the magazine is not inserted into the weapon may be the same as the pressure applied to the cartridges when the magazine is inserted into the weapon, but the bolt is slid back, or the chamber is open. The movement of the cartridges because of the additional pressure that may be applied to the cartridges by the bolt when it is slid forward over the cartridges in the magazine may be 1, 2, or 3 millimeters, or any range of values between the foregoing distances. In some embodiments, the movement of the cartridges may be more or less and may depend, for example, on how large the bolt is relative to the buffer tube and/or the strength of the spring within the magazine. In certain embodiments, the process **3000** may include creating

a first calibration table for when the magazine is inserted into the weapon and the bolt is closed and a second calibration table for when the magazine is inserted into the weapon and the bolt is open. The number of cartridges within the magazine may then be determined with reference to the first calibration table when the bolt is closed and with reference to the second calibration table when the bolt is open. In some cases, the second calibration table may also be used when the magazine is not inserted into the weapon. Alternatively, a third calibration table may be used for determining a number of cartridges in the magazine for when the magazine is not inserted into the weapon.

In certain embodiments, the block **3012** may include identifying the weapon that the magazine is loaded into. In some cases, different weapons may have different impacts on the magazine and the relative position of the linear encoder for a particular number of cartridges loaded within the magazine. Thus, in certain embodiments, different calibration tables may be generated for the magazine for different weapons. In such cases, the process **3000**, or parts thereof, may be repeated for different weapons and each calibration table generated for each weapon may be stored separately along with an identifier of the type of weapon in the storage or non-volatile memory of the magazine.

Further, in certain embodiments, a particular magazine may be capable of supporting different types of ammunition or the same type of ammunition manufactured by different manufacturers. In some such cases, the different types of ammunition, or ammunition manufactured by different manufacturers, may differ in the impact on the position of the linear encoder when a particular number of cartridges are loaded in the magazine. In some such cases the process **3000** may be repeated for each type of ammunition that is capable of being loaded into the magazine and/or for each manufacturer's version of a particular type of ammunition. For example, a particular magazine may be capable of holding .458 SOCOM, .223, 5.56, .50 Beowulf, or 6.5 Grendel ammunition. Because the size of the different types of ammunition may vary, the position of the linear encoder for a particular number of cartridges in the magazine may vary. Thus, in certain embodiments, the process **3000** may be repeated for each type of ammunition capable of being loaded into a particular magazine to create a calibration table, or other data structure, for each type of supported ammunition type. The identity of the ammunition or manufacturer may be stored with the calibration table corresponding to the ammunition type or manufacturer.

Example Calibration Table Based Ammunition Count Process

FIG. **31** presents a flowchart of an embodiment of a calibration table based ammunition count process. The process **3100** can be implemented by any system that can determine an amount of ammunition in a magazine or in a weapon with the magazine inserted using a calibration table. For example, the process **3100**, in whole or in part, can be implemented by electronic circuitry included in the magazine **100**, such as the electronic circuitry **122**, and/or electronic circuitry in the weapon **1300**, such as the electronic circuitry **2002**. To simplify discussion and not to limit the disclosure, portions of the process **3100** will be described with respect to particular systems, such as the electronic circuitry **122** or **2002**. However, it should be understood that operations of the process **3100** may be performed by other systems. For example, one or more operations of the process **3100** may be performed by a computing device that is configured to communicate with the magazine **100**.

The process **3100** begins at block **3102** where, for example, a hardware processor included in the electronic circuitry **122** determines whether a magazine **100** is inserted into a weapon **1300**. This hardware processor may be a field programmable gate array (FPGA) processor, a general-purpose processor, an application specific integrated circuit (ASIC), a microcontroller, a single board computer, or any other type of processor or computing device that may be used to determine the amount of ammunition in the magazine and/or in the weapon using a calibration table. The hardware processor may determine that the magazine is inserted into the weapon by communicating with electronic circuitry **2002** within the weapon. Alternatively, or in addition, the hardware processor of the electronic circuitry **122** may determine that the magazine is inserted into the weapon **1300** based at least in part on pressure detected on a pressure sensor of the magazine **100**. In some embodiments, the hardware processor of the electronic circuitry **122** may determine that the magazine is inserted into the weapon **1300** based at least in part on one or more signals received from one or more magnetic sensors within the magazine **100**. In some implementations the magazine determines whether it is inserted into the weapon **1300** based on input from a user.

At block **3104**, the hardware processor determines an ammunition type for cartridges or ammunition loaded into the magazine. Alternatively, or in addition, a user may provide input that identifies an ammunition type for cartridges or ammunition loaded into the magazine. Determining the ammunition type may include determining a type of the ammunition loaded into the magazine and/or the manufacturer of the ammunition loaded into the magazine. In some cases, determining the ammunition type for cartridges loaded in the magazine may include determining the ammunition type for cartridges that are capable of being loaded into the magazine, but which may not currently be loaded into the magazine because, for example, the magazine has yet to be loaded with ammunition or all of the ammunition that was loaded into the magazine has been fired. In some cases, ammunition type is determined automatically by scanning a code, such as a quick response (QR) code or other type of machine-readable code on a cartridge or on a box that included the cartridge. In some embodiments, the block **3104** is optional or omitted. For example, in some embodiments, the magazine **100** is capable of being loaded with only one type of ammunition or variation between manufacturers of a particular type of ammunition is sufficiently small enough to not impact the ability of the hardware processor to count the number of cartridges within the magazine.

At block **3106**, the hardware processor selects a calibration table based at least in part on the ammunition type and/or whether the magazine is inserted into the weapon. In some embodiments, the calibration table may be selected based at least in part on whether the bolt of the weapon is open or closed. The calibration table may be loaded from a non-volatile memory within the magazine **100** and/or within the weapon **1300**. The calibration table may be selected from one or more calibration tables generated for the magazine. In some embodiments, the magazine **100** includes a single calibration table because, for example, the magazine **100** supports being loaded by only a single type of ammunition or because calibration tables were only generated for a single type of ammunition. In cases where the magazine **100** can only be loaded by a single type of ammunition, the calibration table may be selected at the block **3106** based on whether the magazine is inserted into the weapon. In some

embodiments, the portion of the calibration table accessed may be based on whether the magazine **100** is inserted into the weapon.

At block **3108**, the hardware processor determines a linear encoder position for a linear encoder of the magazine **100**. Determining the linear encoder position may include determining a position of a magnet attached to a follower of the magazine. In some embodiments, the magnet is not attached to the follower, but moves as the follower moves. The follower may move towards an entry or egress point of the magazine as cartridges are expelled from the magazine. Conversely, the follower may move towards the base of the magazine as cartridges are loaded into the magazine. Thus, as cartridges are loaded or expelled from the magazine, a magnet or linear encoder position may change. In certain embodiments, the hardware processor determines the linear encoder position by mapping one or more signals received from one or more magnetic sensors to data stored in the calibration table. As the signals produced by the one or more magnetic sensors vary based on a location of a magnet with respect to the magnetic sensors, the linear encoder position can be determined based on the signals received from the one or more magnetic sensors.

At block **3110**, hardware processor uses the calibration table to determine a count of the number of cartridges in the magazine based at least in part a linear encoder position. Determining the count of the number of cartridges in the magazine may include comparing the linear encoder position to the calibration table to determine a number of cartridges corresponding to the linear encoder position. In some embodiments, the hardware processor may use the linear encoder position as an index for accessing the calibration table to determine a number of cartridges associated with the linear encoder position. In some embodiments, the hardware processor may determine a range of values stored in the calibration table that includes the linear encoder position. A number assigned to the range of values indicate the number of cartridges associated with the linear encoder position. In some embodiments, the linear encoder position correlates to the number of cartridges loaded in the magazine. For example, if the linear encoder position is identified as five, indicates that there are five cartridges loaded within the magazine. In other embodiments, the linear encoder position does not directly correlate to the number of cartridges within the magazine, but is instead mapped within the calibration table to the corresponding number of cartridges within the magazine. For example, a linear encoder position between 5.1 and 5.3 may be mapped to a count of seven cartridges within the magazine, and a linear encoder position between 5.3 and 5.7 may be mapped to a count of the cartridges within the magazine.

At decision block **3112**, the hardware processor determines whether there is a cartridge in a chamber of the weapon. Determining whether there is a cartridge in the chamber of the weapon may include determining whether an electrical signal is received from one or more magnetic sensors within the buffer tube of the weapon. In certain embodiments, the hardware processor of the magazine receives the electrical signal from the one or more magnetic sensors of the buffer tube to determine a position of the bolt and consequently, whether there is a cartridge within a chamber of the weapon. In other embodiments, a hardware processor of the weapon itself, such as a hardware processor within a handle of the weapon, determines whether there is a cartridge in the chamber of the weapon. In certain embodiments, the decision block **3112** is optional or omitted. For example, if it is determined that the magazine is not inserted

into the weapon, the decision block **3112** may be omitted. In other embodiments, the decision block **3112** may be performed by the weapon regardless of whether a magazine is inserted into the weapon. It can typically be determined whether a cartridge is in the chamber of the weapon based on magnetic sensor from the plurality of magnetic sensors that provides a signal to the hardware processor. Thus, is generally unnecessary to have a calibration table for the weapon and to determine whether a cartridge is in the chamber. However, in certain embodiments, the calibration table may be generated for the weapon to facilitate determining whether a cartridge is in the chamber.

If it is determined at the decision block **3112** that there is not a cartridge in the chamber of the weapon, the hardware processor outputs for display a count of the cartridges at the block **3114** that is included in the magazine. The block **3114** may include providing a count of the cartridges to a display or a display controller for output to a user. For example, the count of the cartridges may be provided to the controller **1212** which may cause the cartridge count to be displayed by the light emitting diodes **1202**.

If it is determined at the decision block **3112** that there is a cartridge in the chamber of the weapon, the hardware processor increments the count by one at the block **3116**. In certain embodiments, the count may be incremented by more than one. For example, if the weapon is a multi-chambered weapon or is a weapon that loads multiple cartridges at a time into a corresponding number of chambers, the count may be incremented by one for each chamber that is determined to be loaded. The incremented count of cartridges may then be output for display at the block **3114**. Further, in some embodiments, the block **3114** may output separately or together a count of the number of cartridges in the magazine and a count of the number of cartridges within a chamber of the weapon. In some embodiments, the number of cartridges within the magazine and/or chamber is stored within a memory at the block **3114** instead of or in addition to being displayed. For example, in some embodiments, the count of cartridges is stored within a memory and is only displayed upon request by a user or in response to an event, such as the firing of a cartridge, a loading of the magazine, or the clearing of a jam in the weapon.

Example Magazine Recalibration Process

FIG. **32** presents a flowchart of an embodiment of a magazine recalibration process. The process **3200** can be implemented by any system that can recalibrate a magazine or an ammunition count system of the magazine. For example, the process **3200**, in whole or in part, can be implemented by electronic circuitry included in the magazine **100**, such as the electronic circuitry **122**, and/or electronic circuitry in the weapon **1300**, such as the electronic circuitry **2002**. To simplify discussion and not to limit the disclosure, portions of the process **3200** will be described with respect to particular systems, such as the electronic circuitry **122** or **2002**. However, it should be understood that operations of the process **3200** may be performed by other systems. For example, one or more operations of the process **3200** may be performed by a computing device that is configured to communicate with the magazine **100**.

The process **3200** begins at block **3202** where, for example, a hardware processor included in the electronic circuitry **122** detects a trigger to recalibrate the magazine **100**. The trigger may include interaction by a user with a user interface, such as a pressing a button or a powering on of circuitry within the magazine. Alternatively, or in addition, the trigger may be related to a passage of time. For example, the magazine may be recalibrated once a day, once

a week, once a month, or some other determined time. In some cases, the trigger may be the loading or unloading of the magazine a particular number of times. The amount of force applied to cartridges within the magazine may differ for each weapon. In some cases, even weapons of the same manufacturer may apply a different amount of force to the cartridges within the magazine because, for example, of manufacturing tolerances or wear and tear of the weapon. Thus, in some embodiments, the process **3200** may be triggered or initiated each time a user uses a different weapon with the magazine. In some cases, when the magazine **100** is first obtained by a user, or at any other time the user selects, the user may initiate the process **3200**. In some embodiments, the trigger may be a command received in response to a user interacting with a user interface on a magazine, on a weapon, or on any type of wireless device that can communicate with the magazine. For example, a user may access a user interface on a smartphone or a laptop to request recalibration of a set of magazines. The wireless device may communicate wirelessly with each magazine registered to the user, or a subset of identified magazines, to trigger the recalibration process for the magazines.

At block **3204**, the hardware processor determines an insertion status for the magazine corresponding to whether the magazine is inserted into a weapon. The insertion status may be determined based at least in part on the ability of the hardware processor in the magazine to communicate with electronic circuitry of the weapon. In some cases, a mechanical switch in the magazine may be triggered based on the insertion of the magazine into the weapon, thereby indicating whether or not the magazine is inserted into the weapon. In other cases, an optical or electrical switch is triggered based on the insertion of the magazine into the weapon, thereby indicating whether or not the magazine is inserted into the weapon. In some cases, a user may indicate the insertion status of the magazine by interacting with a user interface on the magazine or on a system in communication with the magazine.

At block **3206**, hardware processor accesses a calibration table of the magazine based at least in part on the insertion status. The calibration table may be accessed from a memory of the magazine. In some embodiments, the calibration table may be accessed from a memory of the weapon. In some embodiments, the calibration table may be selected from a plurality of calibration tables based at least in part on the insertion status of the magazine, and identity of a type of cartridges loaded into the magazine, or a manufacturer of the cartridges loaded into the magazine.

At block **3208**, the hardware processor determines a count of a number of cartridges loaded in the magazine. The count of the number of cartridges loaded into the magazine may be identified by a user interacting with a user interface of the magazine. Alternatively, in certain embodiments, the number of cartridges loaded into the magazine is assumed to be a particular number of cartridges required for recalibrating the magazine. For example, a user may be instructed to load the magazine with a particular number of cartridges corresponding to the maximum number of cartridges may be loaded into the magazine. In some embodiments, the process **3200**, or certain operations thereof, may be repeated using different numbers of cartridges within the magazine. For example, the process may be repeated using zero cartridges, the maximum number of cartridges, the number of cartridges corresponding to the magazine being half-full, or a number of cartridges corresponding to the magazine being one third full. In some embodiments, a user may be instructed, such as via a user interface of the magazine, to

load a particular number of cartridges into the magazine. The user may be instructed to load different numbers of cartridges into the magazine across multiple performances of portions of the process **3200** to generate the modified calibration table.

At block **3210**, the hardware processor determines an expected linear encoder position based at least in part on the calibration table and the count of the number of cartridges. Determining the expected linear encoder position may include determining in the calibration table where the linear encoder is expected to be within the magazine for the identified number of cartridges loaded into the magazine. In some cases, the hardware processor may use the number of cartridges as an index value to access the calibration table to determine the expected or anticipated linear encoder position for the linear encoder of the magazine.

At block **3212**, the hardware processor determines a linear encoder position. In some embodiments, the block **3212** may include one or more of the embodiments previously described with respect to the block **3108**.

At block **3214**, the hardware processor determines a difference between the current linear encoder position and the expected linear encoder position to obtain an adjustment factor. In certain embodiments, the difference between the current linear encoder position and the expected linear encoder position is zero and consequently the adjustment factor is zero. However, in other embodiments, the difference between the current linear encoder position and expected linear encoder position is nonzero. The difference between the two encoder positions may be nonzero because, for example, the ammunition used to calibrate the magazine initially and the ammunition used during the recalibration process **3200** may differ in type or a manufacturer. In some cases, the difference may be attributable to tolerances for the ammunition manufactured by a manufacturer. In some cases, the magazine is loaded with a default calibration table created using a model magazine for some magazine other than the current magazine being recalibrated. In some such cases variations may occur during the manufacturing process resulting in the linear encoder position for a particular number of cartridges loaded in the magazine differing from that of the default calibration table. Moreover, in some cases, the structure of the magazine may change over time due to wear and tear of the magazine. For example, the stiffness of the magazine housing or the springiness of the spring may change over time as the magazine is used or is exposed to different environmental factors. In some such cases, the position of the linear encoder for a particular number of cartridges loaded in the magazine may change over time as the condition of the magazine evolves over time.

At block **3216**, the hardware processor modifies the calibration table based at least in part on the adjustment factor. In some embodiments, the hardware processor may modify the calibration table only if the adjustment factor exceeds a threshold. In other embodiments, the hardware processor modifies the calibration table based on any non-zero adjustment factor. In some cases, the adjustment factor is the same for the entire calibration table. In other words, the linear encoder position may be modified based on the adjustment factor for each entry associated with each number of cartridges loaded within the magazine. Alternatively, in certain embodiments, the adjustment factor may vary for different amounts of cartridges loaded within the magazine. For example, the adjustment factor for the linear encoder may be smaller when the magazine is empty or has a fewer number of cartridges loaded compared to when the magazine has a greater number of cartridges loaded. For instance,

he adjustment factor for the linear encoder may be zero when the magazine is empty, 1 μm when one cartridge is loaded in the magazine, 2 μm when two cartridges are loaded in the magazine, 4 μm when three cartridges are loaded in the magazine, and 1 mm when four cartridges are loaded in the magazine. In some embodiments, the hardware processor may modify the calibration table associated with the magazine being inserted into the weapon and the calibration table associated with the magazine not being inserted into the weapon based on the adjustment factor. In other embodiments, the process 3200 is repeated once for the magazine being inserted the weapon and once the magazine not being inserted into the weapon.

At block 3218, the hardware processor stores the modified calibration table at the magazine. The modified calibration table may replace the existing calibration table or may be stored as an additional calibration table. The calibration table may be stored in a memory of the magazine.

Example Process of Obtaining a Total Ammunition Count

FIG. 33 presents a flowchart of an embodiment of a total ammunition count process 3300. The process 3300 can be implemented by any system that can obtain a total ammunition count available to a user across one or more magazines registered to a weapon of the user. For example, the process 3300, in whole or in part, can be implemented by electronic circuitry included in the magazine 100, such as the electronic circuitry 122, and/or electronic circuitry in the weapon 1300, such as the electronic circuitry 2002. To simplify discussion and not to limit the disclosure, portions of the process 3300 will be described with respect to particular systems, such as the electronic circuitry 122 or 2002. However, it should be understood that operations of the process 3300 may be performed by other systems. For example, one or more operations of the process 3300 may be performed by a computing device that is configured to communicate with a weapon 1300 or one or more magazines.

Performance of the process 300 may occur in response to a trigger. This trigger may be a request from a user, a passage of time, the addition or removal of a magazine from a set of magazines registered to a weapon or to a user, or the insertion or removal of a magazine from an insertion port of the weapon.

The process 3300 begins at block 3302 where, for example, a hardware processor included in the electronic circuitry 2002 accesses the identity of one or more magazines registered to a weapon. The identity of the one or more magazines may be obtained from a memory, such as a non-volatile memory, located within the weapon. This memory may be located within a handle of the weapon, or a stock of the weapon. In some embodiments, the memory may be located within hardware other than the weapon carried by the user, such as a helmet or personal communication device included in the user's kit. The identity of the one or more magazines may include a network address for accessing the one or more magazines via an ad hoc network. The network address may be formed from a combination of an identifier of the weapon and an identifier of the magazine.

In some such embodiments, the weapon may serve as a router or hub that can communicate with the one or more magazines. This communication may be performed using ultra-wide band (UWB) communication. Further, the range of communication may be limited to a particular range of the user carrying the weapon creating a small personal network for the user. For example, the range of communication may be limited to 100 meters, 50 meters, 25 meters, 10 meters, or 3 meters or less. In some embodiments, although the

transceiver in the weapon may be capable of communicating over a larger distance, the power supplied may be sufficiently small to limit communication to 1 to 2 meters to prevent eavesdropping or interference. Communication range may be limited based on the amount of power supplied to the networking equipment of the weapon. In some embodiments, the router or hub may be included and other equipment carried by the user, such as a helmet of the user. In some embodiments, the weapon may implement a low probability of intercept/low probability of detect (LPI/LPD) protocol to reduce eavesdropping or interference by malicious users.

At block 3304, the hardware processor, using a transceiver, attempts to establish a communication connection with a magazine from the one or more magazines identified at the block 3302. In some embodiments, the operations associated with the block 3304 may be performed in parallel or substantially in parallel for a plurality of magazines of the one or more magazines. In some cases, a magazine inserted into the weapon may be omitted from the operations of the block 3304 because, for example, a count of cartridges loaded in the magazine inserted into the weapon may be obtained via alternative communication channels between the magazine and the weapon and/or alternative processes available for the magazine inserted into the weapon. For example, an optical communication connection may be established between the weapon and the magazine inserted into the weapon enabling the weapon to obtain the cartridge count for the magazine inserted into the weapon by requesting the cartridge count from the hardware processor of the magazine. As another example, the magazine may automatically provide the cartridge count to the weapon when inserted into the weapon via an optical connection to the weapon. In some embodiments, communication between the magazine and the weapon is encrypted or occurs over a secure channel. Further, the attempt to establish the communication connection with the magazine may occur over a particular time period. For example, the weapon may attempt to communicate with a magazine for 5, 10, 15, or 30 seconds or a minute.

At decision block 3306, the hardware processor determines whether a communication connection was successfully established with the magazine. The hardware processor may determine whether the attempt to establish the communication connection was successful based at least in part on whether the transceiver receives an acknowledgment packet or response from the magazine. In some cases, the hardware processor determines whether the attempt to establish a communication connection was successful based on whether the acknowledgment packet or other response from the magazine is received within a particular time period.

If it is determined at the decision block 3306 that a communication connection was not successfully established with the magazine, the hardware processor deregisters the magazine from the one or more magazines registered to the weapon at block 3308. Deregistering the magazine may include removing the identification of the magazine in a memory of the weapon. Alternatively, or in addition, deregistering the magazine may include marking an identifier of the magazine in the memory as no longer available. Further, in certain embodiments, deregistering the magazine may include removing a number of cartridges associated with the deregistered magazine from a total count of cartridges available to the user.

If it is determined that the decision block 3306 that a communication connection was successfully established, the hardware processor determines at the block 3310 a count of

the number of cartridges loaded in the magazine. Determining the count of the number of cartridges loaded in the magazine may include requesting account information from the magazine. Requesting the account information of the number of cartridges loaded in the magazine may cause the magazine to access its memory to determine the count of cartridges and to transmit the count information to the weapon. Alternatively, or in addition, requesting the count information of the number of cartridges loaded in the magazine may cause the magazine to perform a cartridge count process, such as the process 3100.

At block 3312, the hardware processor adds the count obtained at the block 3310 to a total count of cartridges. The total count of cartridges may represent the total number of cartridges available to a user. Alternatively, or in addition, the total count of cartridges may represent the total number of cartridges loaded into all of the magazines registered with the user or registered to the weapon. In other words, if there are five magazines registered to the weapon or to the user, the total count of cartridges may represent the cumulative number of cartridges loaded into the five registers combined.

At the decision block 3314, the hardware processor determines whether there are more magazines registered to the weapon. If there are more magazines registered to the weapon, the process 3300 returns to the block 3304 where the hardware processor repeats the operations of the block 3304 for a different magazine of the one or more magazines.

If there are no more magazines registered to the weapon, the hardware processor outputs the total count of cartridges at the block 3316. The block 3316 may include one or more of the embodiments previously described with respect to the block 3114. In some embodiments, the total count of cartridges may be displayed on a display of the magazine inserted into the weapon, on a scope attached to the weapon, on a heads up display of the helmet, or any other display accessible to the user. In some embodiments, the total count of cartridges may be transmitted wirelessly to another user other than the user carrying the weapon or otherwise associated with the weapon. For example, the total cartridge count may be transmitted to a display or system of a squad commander or other supervising user of the user carrying the weapon. In some embodiments, the total cartridge count may be transmitted to another location, such as a command center or other monitoring location. In some cases, the total count of cartridges is not output for display or is only output for display upon a request by a user. In some such cases, the total count of cartridges may be stored in a nonvolatile memory of the weapon or in other nonvolatile memory available to the user, such as non-volatile memory may be included in a helmet (e.g., a helmet with a heads up display) of the user.

In some embodiments, the total cartridge count may also include the cartridges loaded in a magazine that is inserted into the weapon. In some such cases, a count of the cartridges loaded in the magazine is inserted into the weapon may be determined using the process 3100. To determine the count of cartridges loaded in the magazine that is inserted into the weapon may be added to the total count of cartridges available to a user across all magazines registered to the weapon. Further, in some embodiments, the total count of cartridges may include a cartridge in a chamber of the weapon.

In some embodiments, the process 3300 or a modified version of the process 3300 may be used to identify ammunition count for different types of ammunition carried by the user. For example, when each magazine is registered with the weapon, the type of ammunition loaded in the magazine

may also be identified. Thus, when calculating the total count of cartridges available to the user, the hardware processor may separately count magazines that are loaded with different types of ammunition. In some such cases, the weapon may display the total ammunition available to the user and the total ammunition of each type of ammunition available to the user. Thus, for example, the weapon may display the total amount of round nose cartridges and the total number of hollowpoint cartridges available to the user.

In some embodiments, the weapon may count the total number of cartridges available to a user for magazines the user carries for a different weapon. For example, the user may register each magazine the user carries with a weapon that includes the ammunition count hardware. The weapon may then communicate with each of the magazines to determine the total cartridges available to the user regardless of whether the weapon can load each of the magazines. Advantageously, in certain embodiments, the weapon can count the total number of cartridges available for the user's rifle as well as the user's handgun or other weapon and display it on a single display enabling the user to track the total ammunition available to the user. The display may identify separately the cartridge counts for the different weapons of the user.

Example Magazine Registration Process

FIG. 34 presents a flowchart of an embodiment of a magazine registration process 3400. The process 3400 can be implemented by any system that can register a magazine with a weapon. For example, the process 3400, in whole or in part, can be implemented by electronic circuitry included in the magazine 100, such as the electronic circuitry 122, and/or electronic circuitry in the weapon 1300, such as the electronic circuitry 2002. To simplify discussion and not to limit the disclosure, portions of the process 3400 will be described with respect to particular systems, such as the electronic circuitry 122 or 2002. However, it should be understood that operations of the process 3400 may be performed by other systems. For example, one or more operations of the process 3400 may be performed by a computing device that is configured to communicate with a weapon 1300 or one or more magazines.

The process 3400 begins at block 3402 where, for example, a hardware processor included in the electronic circuitry 2002 identifies the existence of an unregistered magazine. The unregistered magazine may include any magazine that is not currently registered with the weapon regardless of whether the magazine was registered with the weapon at some previous point in time. The hardware processor may identify the existence of the unregistered magazine when the unregistered magazine is inserted into the weapon. Upon insertion of the magazine into the weapon, the hardware processor may obtain an identifier for the magazine. The hardware processor may access a list or other data structure stored within a memory of the weapon and may determine whether the identifier for the magazine is included in the list or the data structure to determine whether the magazine is unregistered with the weapon.

Alternatively, or in addition, the existence of an unregistered magazine may be determined when the magazine is brought within a particular distance of the weapon. For example, when the unregistered magazine is brought within radio distance of the weapon, the weapon may access radio frequency identifier (RFID) tag or other type of tag of the magazine to obtain identifier of the magazine. Using the identifier of the magazine, the hardware processor can determine whether the magazine is registered with the weapon. In some embodiments, when a user presses a button

on the magazine or interacts with the user interface of the magazine, the magazine may transmit an identifier within a particular distance. In certain embodiments, this particular distance is generally a short distance (e.g., 10 feet or less) to prevent weapons of other users from receiving the identifier from the magazine. Further, the identifier may be used by the weapon to determine whether the magazine is registered with the weapon. Further, the identifier may be used to facilitate the weapon communicating with the magazine to establish a network identifier for further communication between the magazine and the weapon.

In certain embodiments, the user may deregister a magazine with the weapon by interacting with the user interface to indicate that a magazine inserted into the weapon is to be deregistered. Alternatively, or in addition, the user may interact with the user interface of the magazine to indicate to the weapon that the magazine within RFID communication distance of the weapon is to be deregistered from the weapon.

At block **3404**, the hardware processor receives a magazine identifier for the unregistered magazine. The magazine identifier may be a unique identifier associated with the magazine of the weapon and can be used to distinguish the magazine from other magazines that may be within communication distance of the weapon. In some cases, the magazine identifier is a default identifier associated with any magazine that is not registered to a weapon. Upon registering the magazine with a weapon, the default identifier may be replaced with a unique identifier assigned by the weapon. Upon the registering the magazine with a weapon or upon the magazine no longer being within communication distance with the weapon, the identifier assigned by the weapon may be reset or replaced with the default identifier. In some embodiments, the identifier of the magazine may be or may be similar to a media access control (MAC) address.

At block **3406**, the hardware processor provides a weapon identifier to the unregistered magazine. The weapon identifier may be a unique identifier associated with the weapon that is registering the magazine. In some embodiments, the weapon identifier of the weapon may be or may be similar to a media access control (MAC) address. In certain embodiments, the magazine may combine the weapon identifier with the magazine identifier to create an identifier used for further communication with the weapon. In certain embodiments, instead of providing the weapon identifier to the unregistered magazine, the block **3406** may include providing a unique identifier to the unregistered magazine that is based at least in part on the magazine identifier and the weapon identifier, and which may be used to uniquely identify the magazine. In some embodiments, the block **3406** may be optional or omitted. For example, in some embodiments, the weapon may provide the network identifier created at the block **3408** below to the magazine.

At block **3408**, the hardware processor combines the magazine identifier and the weapon identifier to create a network identifier for the magazine. The network identifier may be a unique identifier that the weapon uses to identify the magazine. In some embodiments the weapon may use the network identifier is a network address to establish communication with the magazine. Advantageously, in certain embodiments, by assigning each magazine a unique network identifier, each magazine within communication range of the weapon can determine whether a communication is intended for the magazine or another magazine registered with the weapon.

At block **3410**, the hardware processor stores the network identifier of the magazine. The network identifier may be stored in a non-volatile memory of the weapon.

At block **3412**, the hardware processor obtains a count of cartridges loaded into the magazine. In certain embodiments, the block **3412** may include performing the process **3100**, or portions thereof. In some cases, the hardware processor may request the count of cartridges from the magazine by establishing a communication connection with the magazine via a secure channel and using the network identifier assigned to the magazine. In some embodiments the block **3412** may include one or more of the embodiments described with respect to the block **3310**.

At block **3414**, the hardware processor adds the count to a total count of cartridges registered with the weapon. The total count of cartridges registered with the weapon may be stored in the nonvolatile memory of the weapon. In some embodiments, performing the block **3414** may include performing the process **3300**, or parts thereof. In some embodiments determining the total count of cartridges registered with the weapon may include aggregating the count of cartridges loaded within each magazine registered with the weapon including a magazine inserted into the weapon. Further, determining the total count of cartridges registered with the weapon may include determining whether the cartridge is within a chamber of the weapon. In some embodiments, the block **3414** may include adding the count of cartridges for the magazine newly registered to the weapon to a running count of cartridges previously determined to be registered with the weapon. In other embodiments, the block **3414** may include performing a new count of cartridges available across all magazines registered with the weapon. In some embodiments determining the total count of cartridges registered with the weapon may include subtracting the count of cartridges loaded into a magazine that is determined to be deregistered or no longer registered with the weapon. The deregistered magazine may include a magazine that the weapon is no longer able to indicate with to you, for example, a distance between the weapon in the magazine.

Example Weapon Display State Machine

FIG. **35** presents a flowchart of an embodiment of a weapon display state machine **3500**. The weapon display state machine **3500** includes a number of states and decision points for a controller of a display of a weapon system and/or a hardware processor included in the weapon system. Each of the states is associated with performing a process as described herein. The process may be performed by a weapon system or a hardware processor of a weapon system. Although particular processes are identified, it should be understood that in certain embodiments, alternative embodiments of the processes described herein may be performed.

When the weapon system is initialized a magazine display state **3510** may be entered causing performance of a magazine display process **3800**. Further, the magazine display process **3800** may be performed continuously or intermittently while a hardware processor and/or display controller is powered or active. Moreover, the weapon system may remain in a magazine display state **3510** when each of the decisions blocks **3502-3506** return a "no" determination. The magazine display process **3800** is described in more detail below with respect to FIG. **38**.

At decision block **3502**, it is determined whether an edge buffer tube magnetic sensor has been triggered. If one or more edge buffer tube magnetic sensors have been triggered, the weapon system enters the chamber count display state **3512** causing performance of a chamber count display

process **3700**. The chamber count display process **3700** is described in more detail below with respect to FIG. **37**.

At decision block **3504**, it is determined whether a central buffer tube magnetic sensor has been triggered. If one or more central buffer tube magnetic sensors have been triggered, the weapon system enters a jam display state **3514** causing performance of the jam display process **3600**. The jam display process **3600** is described in more detail below with respect to FIG. **36**.

At decision block **3506**, it is determined whether a redraw has been triggered. If it is determined that a redraw has been triggered, the weapon system enters a redraw state causing performance of the redraw process **3900**. The redraw process **3900** is described in more detail below with respect to FIG. **39**.

Example Jam Display Process

FIG. **36** presents a flowchart of an embodiment of a jam display process **3600**. The process **3600** can be implemented by any system that can determine whether a weapon is jammed and display a jam state of the weapon. For example, the process **3600**, in whole or in part, can be implemented by electronic circuitry included in the magazine **100**, such as the electronic circuitry **122**, and/or electronic circuitry in the weapon **1300**, such as the electronic circuitry **2002**. To simplify discussion and not to limit the disclosure, portions of the process **3600** will be described with respect to particular systems, such as the electronic circuitry **122** or **2002**. However, it should be understood that operations of the process **3600** may be performed by other systems. For example, one or more operations of the process **3600** may be performed by a computing device that is configured to communicate with a weapon **1300** or one or more magazines.

The process **3600** begins at block **3602** where, for example, a hardware processor included in the electronic circuitry **2002** detects a signal generated by a central buffer tube magnetic sensor. Detecting a signal at the central buffer tube magnetic sensor may include determining that the central buffer tube magnetic sensor generated a signal based on a magnet or magnetic field detected by the magnetic sensor. The signal may be generated by a magnetic sensor within the buffer tube that is between the two ends of the buffer tube in response to a magnetic field produced by a magnet within sensor detection range of the magnetic sensor.

The central buffer tube magnetic sensor may include any of the magnetic sensors that are between the two ends of the buffer tube. For example, with reference to FIG. **20**, some weapons may include five magnetic sensors **2018** within the buffer tube **2012** of the weapon **1300**. The central buffer tube magnetic sensors may include the middle three magnetic sensors **2018** that are between the magnetic sensors at each end of the buffer tube **2012**. In certain embodiments, the magnetic sensors **2018** may generate a signal regardless of detection of a magnet **2016**. In some such embodiments, detecting the signal generated at the central buffer tube magnetic sensor **2018** may include determining that the signal generated by one of the central buffer tube magnetic sensors **2018** is stronger than a signal generated by an edge buffer tube sensor.

At block **3604**, the hardware processor initiates a jam detection timer. The jam detection timer may include a timer that counts the length of time that the signal is received from one or more of the central buffer tube magnetic sensors. Alternatively, or in addition, the jam detection timer may measure the amount of time that the strongest signal is received from one of the central buffer tube magnetic

sensors as supposed to one of the edge buffer tube magnetic sensors. Initiating the jam detection timer may include setting the timer to a value of zero and starting the timer.

At decision block **3606**, the hardware processor determines whether a signal has been generated at an edge buffer tube magnetic sensor. Determining whether a signal has been generated at the edge buffer tube magnetic sensor may comprise determining whether an edge buffer tube magnetic sensor has generated a signal in response to a magnet or magnetic field. In certain embodiments, the decision block **3606** includes determining whether the signal has been generated at a particular edge buffer tube magnetic sensor. In other words, the decision block **3606** includes determining whether the signal has been generated at the magnetic sensor **2018** at one edge of the buffer tube or by the magnetic sensor **2018** at the opposite edge of the buffer tube of FIG. **20**. In other embodiments, the decision block **3606** includes determining whether a signal has been detected as being generated at either of the edge buffer tube magnetic sensors. In some embodiments, the decision block **3606** may include one or more of the embodiments associated with the block **3602**, but with respect to the edge sensors instead of the central buffer tube magnetic sensors.

In some embodiments, the decision block **3606** may include determining whether a signal generated at an edge buffer tube magnetic sensor is stronger than one or more signals that may be generated by one or more other buffer tube magnetic sensors **2018**. In some cases, the magnetic field generated by the magnet **2016** may be sufficiently strong enough to be detected by multiple magnetic sensors **2018**. Alternatively, or in addition, multiple magnetic sensors **2018** may be sensitive enough to detect a magnetic field generated by the magnet **2016**. In some such cases, the determination of whether an edge magnetic sensor **2018** or a central magnetic sensor **2018** within the buffer tube generates a signal responsive to detecting a magnet, or magnetic field, may be based on the strength of the signal generated. In some cases, the decision block **3606** includes determining whether the strength of the signal at the edge buffer tube magnetic sensor exceeds the strength of the signal at the central buffer tube magnetic sensor by a threshold amount.

If a signal is detected as being generated at the edge buffer tube magnetic sensor, or if the signal from the edge buffer tube magnetic sensor is stronger than signals from a central buffer tube magnetic sensor, the hardware processor resets a jam detected flag at the block **3608**. In certain embodiments, the block **3608** may be optional or omitted. For example, if the jam detected flag was not previously set, or does not indicate the existence or potential existence of a jam, the block **3608** may be omitted. Further, the block **3608** may include resetting or stopping a jam detection timer initiated at the block **3604**.

If a signal is not generated at the edge buffer tube magnetic sensor, or if the signal generated at the edge buffer tube magnetic sensor is lower in strength than a signal generated at a central buffer tube magnetic sensor, the hardware processor determines whether the jam detection timer has expired at the decision block **3610**. Determining whether the jam detection timer has expired may include determining whether a passage of time measured by the jam detection timer exceeds a threshold length of time. The threshold length of time may vary based on the type of weapon. In some embodiments, the user may set the threshold length of time. If the jam detection timer has not expired, or has not exceeded the threshold length of time, the process **3600** returns to the decision block **3606** where the hardware

processor continues to monitor whether a signal has been generated at the edge buffer tube magnetic sensor.

If it is determined at the decision block **3610** that the jam detection timer has expired, the hardware processor sets a jam detected flag at the block **3612**. Setting the jam detected flag may include setting a value of the jam detected flag to indicate that a jam has been detected in the weapon. Further, setting the jam detected flag may include storing at a non-volatile memory of the weapon that the jam has been detected. Further, the hardware processor sets a redraw flag at the block **3614**. Setting the redraw flag may trigger performance of the redraw process **3900** associated with the state **3516** as illustrated in the FIG. **35**. The redraw process **3900**, as described below, may be used to output an indication to a user that the weapon is jammed. In certain embodiments, the blocks **3612** and/or **3614** may be optional or omitted. For example, in some embodiments, if the jam detection timer has expired, the hardware processor may cause an output of an indication that the weapon is jammed without setting any internal flags within the weapon.

Example Chamber Count Display Process

FIG. **37** presents a flowchart of an embodiment of a chamber count display process **3700**. The process **3700** can be implemented by any system that can determine whether a cartridge is in a chamber of a weapon. For example, the process **3700**, in whole or in part, can be implemented by electronic circuitry included in the magazine **100**, such as the electronic circuitry **122**, and/or electronic circuitry in the weapon **1300**, such as the electronic circuitry **2002**. To simplify discussion and not to limit the disclosure, portions of the process **3700** will be described with respect to particular systems, such as the electronic circuitry **122** or **2002**. However, it should be understood that operations of the process **3700** may be performed by other systems. For example, one or more operations of the process **3700** may be performed by a computing device that is configured to communicate with a weapon **1300** or one or more magazines.

The process **3700** begins at block **3702** where, for example, a hardware processor included in the electronic circuitry **2002** determines a current weapon state of a weapon. Determining the current weapon state of the weapon may include determining the configuration of a bolt of the weapon, whether a magazine is inserted into the weapon, a count of cartridges within the magazine inserted into the weapon, or any other status information for the weapon the may be used to determine whether a cartridge is loaded into a chamber of the weapon. As indicated by the state graph **3500**, the process **3700** may be initiated when a signal is received from an edge buffer tube magnetic sensor. Accordingly, in certain embodiments, the current weapon state of the weapon is a state other than a jam state.

At decision block **3704**, the hardware processor determines whether a bolt of the weapon is open. Determining whether the bolt of the weapon is open may be based at least in part on the current weapon state of the weapon. Further, determining whether the bolt of the weapon is open may be based at least in part on one or more signals generated by one or more magnetic sensors within the buffer tube. For example, if a signal is generated by the magnetic sensor **2018** that is closest to the butt or back end of the stock or buffer tube, it may be determined that the bolt is open. If instead the signal is generated by the magnetic sensor **2018** that is closest to the chamber or magazine, it may be determined that the bolt is closed. Further, if the signal is generated by a magnetic sensor other than the magnetic sensor closest to the butt of the stock, and may be deter-

mined that the bolt is not open. In some cases, determining which magnetic sensor generated a signal may include determining which signal generated by a plurality of magnetic sensors is strongest.

If it is determined at decision block **3704** that the bolt is open, the hardware processor sets the chamber count to zero at block **3706**. Setting the chamber count to zero indicates that there is not a cartridge within a chamber of the weapon.

If it is determined at the decision block **3704** that the bolt is not open, the hardware processor determines whether the magazine count decremented at the decision block **3708**. Determining whether the magazine count was decremented may include obtaining a count of the number of cartridges within a magazine that is inserted into the weapon. The count of the number of cartridges within the magazine inserted that is into the weapon may be compared to a previously determined count of the number of cartridges within the magazine to determine whether the count of cartridges in the magazine decremented.

If it is determined at the decision block **3708** that the magazine count did not decrement, the hardware processor sets the chamber count to zero at block **3710**. Setting the chamber count to zero indicates that the cartridge is not loaded within the chamber of the weapon.

If it is determined at the decision block **3708** that the magazine count did decrement, the hardware processor sets the chamber count to one at block **3712**. Setting the chamber count to one indicates that a cartridge is loaded within the chamber of the weapon.

After setting the chamber count at one of the blocks **3706**, **3710**, or **3712**, the hardware processor sets the redraw flag at block **3714**. Setting the redraw flag may trigger performance of the redraw process **3900** associated with the state **3516** as illustrated in the FIG. **35**. This redraw process may be used to output to a user an indication of whether a cartridge is loaded into a chamber. Further, the redraw process may be used to output to a user and updated count of cartridges within a magazine that is inserted into the weapon.

Optionally, at block **3716**, the hardware processor resets a jam detection timer. Further, the hardware processor optionally resets the jam detected flag at block **3718**. In certain embodiments, because the hardware processor is able to determine whether a cartridge was loaded into the chamber and was able to determine whether a cartridge count of a magazine inserted into the weapon decremented, the hardware processor is able to confirm that the weapon is not in a jam state or is no longer in a jam state. Accordingly, the blocks **3716** and **3718** may be used to reset a previous identification of a jam state and/or to reset the process used to determine whether the weapon is a jam state. Blocks **3716** and **3718** may be optional or omitted because, for example, a jam was not detected or a jam detection timer was not active.

Example Magazine Display Process

FIG. **38** presents a flowchart of an embodiment of a magazine display process **3800**. The process **3800** can be implemented by any system that can determine and display the status of a magazine with respect to a weapon. For example, the process **3800**, in whole or in part, can be implemented by electronic circuitry included in the magazine **100**, such as the electronic circuitry **122**, and/or electronic circuitry in the weapon **1300**, such as the electronic circuitry **2002**. To simplify discussion and not to limit the disclosure, portions of the process **3800** will be described with respect to particular systems, such as the electronic circuitry **122** or **2002**. However, it should be understood that

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operations of the process **3800** may be performed by other systems. For example, one or more operations of the process **3800** may be performed by a computing device that is configured to communicate with a weapon **1300** or one or more magazines.

The process **3800** begins at decision block **3802** where, for example, a hardware processor included in the electronic circuitry **2002** determines whether a magazine is detected in the weapon. Determining whether the magazine is inserted into the weapon may include communicating with the magazine via an optical connection. Alternatively, or in addition, determining whether the magazine is inserted in the weapon may include determining whether a mechanical switch or other physical indicator has been triggered. Advantageously, in certain embodiments, by including a physical mechanism that is triggered when the magazine is inserted into the weapon enables the weapon to determine that a magazine is inserted even if the magazine is a conventional magazine that does not include the embodiments of the present disclosure.

If it is determined at the decision block **3802** that a magazine is not detected in the weapon, the hardware processor sets the magazine count to zero at the block **3804**. Setting the magazine count to zero indicates to the weapon that there are no cartridges available to the weapon. In some embodiments, setting the magazine count to zero indicates that a weapon that there are no cartridges available to the weapon other than the cartridge that may be within a chamber of the weapon. It should be understood that, in certain embodiments, the operations of the block **3804** are associated with a counter for cartridges currently available for the weapon to fire and may be distinct from the count of cartridges available to the user, for example, in magazines registered to the weapon, the not inserted into the weapon.

At the block **3806**, hardware processor clears the magazine inserted flag or sets the magazine inserted flag to a value indicating that a magazine is not inserted into the weapon. Clearing the magazine inserted flag may include modifying a register or memory of the weapon to indicate that a magazine is not inserted.

If it is determined at the decision block **3802** that a magazine is detected in the weapon, the hardware processor obtains the magazine cartridge count at the block **3808**. Obtaining the magazine cartridge count may include performing the process **3100** or portions thereof. In some embodiments, the block **3808** may be optional or omitted. For example, if a conventional magazine is inserted into the weapon that does not include the embodiments disclosed herein, the hardware processor of the weapon may be unable to determine the cartridge count for the inserted magazine. In certain embodiments, if a conventional magazine is inserted into the weapon, the hardware processor may set a status unknown flag indicating that the weapon is unable to determine the number of cartridges loaded within the magazine.

At the block **3810**, the hardware processor sets the magazine inserted flag indicating that a magazine is inserted into the weapon. Setting the magazine inserted flag may include setting a register or memory address to a value that indicates that a magazine is inserted into the weapon. In some embodiments, the magazine inserted flag is set to a value indicating a count of the cartridges within the magazine as obtained at the block **3808**.

At the decision block **3812**, the hardware processor determines whether a change in the magazine cartridge count or a change in the magazine inserted flag has occurred. If it is determined at the decision block **3812** that a change in the

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magazine cartridge count or in the magazine inserted flag has not occurred, the process **3800** returns to the decision block **3802** where the hardware processor may continuously or intermittently determine whether a magazine is detected in the weapon.

If it is determined at the decision block **3812** that there is a change in the magazine cartridge count. Or in the magazine inserted flag, hardware processor sets the redraw flag at the block **3814**. Setting the redraw flag may trigger performance of the redraw process **3900** associated with the state **3516** as illustrated in the FIG. **35**.

Example Redraw Process

FIG. **39** presents a flowchart of an embodiment of a redraw process **3900**. The process **3900** can be implemented by any system that can output weapon system status information for a weapon. For example, the process **3900**, in whole or in part, can be implemented by electronic circuitry included in the magazine **100**, such as the electronic circuitry **122**, and/or electronic circuitry in the weapon **1300**, such as the electronic circuitry **2002**. To simplify discussion and not to limit the disclosure, portions of the process **3900** will be described with respect to particular systems, such as the electronic circuitry **122** or **2002**. However, it should be understood that operations of the process **3900** may be performed by other systems. For example, one or more operations of the process **3900** may be performed by a computing device that is configured to communicate with a weapon **1300** or one or more magazines.

The process **3900** begins at decision block **3902** where, for example, a hardware processor included in the electronic circuitry **2002** determines whether a redraw flag has been set. Determining whether the redraw flag has been set may include accessing a memory location in a memory of the weapon that stores the redraw flag or a state of the redraw flag. An indication that a redraw flag has been set may be used to cause a display of the weapon, or displaying truncation with the weapon, to display information about a state of the weapon, such as whether the weapon is jammed, and/or information about the number of cartridges available to the user. The number of cartridges available to the user may indicate the number of cartridges in the magazine inserted into the weapon, whether a cartridge is within a chamber of the weapon, and/or the number of cartridges inserted in a set of one or more magazines registered to the weapon whether or not a magazine is inserted into the weapon.

If it is determined at the decision block **3902** that a redraw flag has not been set, the process **3900** returns to the decision block **3902** where the hardware processor continuously or intermittently determines whether the redraw flag has been set. In certain embodiments, the determination that the redraw flag has not been set may indicate that a change in status of the weapon or of cartridges available to the weapon has not changed. Moreover, in certain embodiments, instead of the hardware processor continuously or intermittently determining whether the redraw play has been set, the hardware processor receives an interrupt signal or a push notification alerting the hardware processor that the redraw play has been set.

If it is determined at the decision block **3902** that a redraw flag has been set, the hardware processor determines at the decision block **3904** whether a magazine inserted flag has been set. Determining whether the redraw inserted flag has been set may include accessing a memory location in a memory of the weapon that stores the magazine inserted flag or a state of the magazine inserted flag.

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If it is determined at the decision block **3904** that a magazine inserted flag has been set, the magazine inserted status is displayed on a user interface or display of the weapon system at block **3906**. Displaying the magazine inserted status may include displaying a symbol corresponding to the magazine inserted status. In some embodiments, the display of the weapon system may be within the scope of the weapon or may be projected onto a scope of the weapon. Alternatively, or in addition, the magazine inserted status may be displayed on a user interface or display that is separate from the weapon, such as on a heads up display of a helmet or any other visual system carried or used by the user.

If it is determined at the decision block **3904** that a magazine inserted flag is not set, the magazine not inserted status is displayed on the user interface or the display of the weapon system at the block **3908**. The block **3908** may include one or more of the embodiments described with respect to the block **3906**, but with the output of the display corresponding to the magazine not inserted status.

At decision block **3910**, the hardware processor determines whether a jam detected flag is set. Determining whether the jam detected flag has been set may include accessing a memory location in a memory of the weapon that stores the jam detected flag or a state of the jam detected flag.

If a jam detected flag is set, a jam state notification is displayed on the user interface or the display of the weapon system at the block **3912**. The block **3912** may include one or more of the embodiments described with respect to the block **3906**, but with respect to displaying the jam state of the weapon.

At decision block **3914**, the hardware processor determines whether an unknown state flag is set. Determining whether the unknown state flag has been set may include accessing a memory location in a memory of the weapon that stores the unknown state flag or a state of the unknown state flag. In some embodiments, determining whether the unknown state flag has been set may include accessing or attempting to access the magazine count for the inserted magazine. If the hardware processor is unable to determine the magazine state for the inserted magazine, the hardware processor may determine that the weapon system is in an unknown state.

In some embodiments, a weapon may be in an unknown state when first powered or initialized because, for example, it may be undetermined whether a cartridge is in a chamber. Once a user has cycled the bolt, the weapon can determine its state. If the weapon has a cartridge in the chamber, cycling the bolt may eject the cartridge. Further, if a magazine is inserted into the weapon, cycling the bolt may load a cartridge into the chamber if the magazine is not empty. One or more of the processes disclosed herein can be used to determine the ammunition or cartridge count loaded into a chamber of the weapon and/or a magazine inserted into the weapon. Accordingly, in certain embodiments, once a user has cycled the bolt, a weapon in an unknown state may transition to a known state and the unknown state flag may be reset or otherwise set to a value indicating a known weapon state. In certain embodiments, once the weapon is transitioned from an unknown state to a known state via, for example, the cycling of the bolt, the weapon may remain in a known state by performing the one or more embodiments disclosed herein for determining ammunition count and weapon state. In some embodiments, a display of the weapon may instruct the user to cycle the bolt upon turning on or powering the electronic circuitry of the weapon

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enabling the weapon to determine its state before the user takes further action with the weapon.

In some embodiments, a weapon may be unable to determine a number of cartridges within a magazine because, for example, the magazine may be a conventional magazine that does not include the ammunition count capabilities disclosed herein. In some such embodiments, the weapon may indicate an unknown ammunition count or an unknown state. In other such embodiments, the weapon may indicate a state of the chamber (e.g., loaded or not loaded), but may indicate that a state of a magazine is unknown. In some cases, the weapon may indicate whether or not a magazine is inserted, but may not display a cartridge count due, for example, to the magazine being a conventional magazine.

If it is determined at the decision block **3914** that an unknown state flag is set, an unknown state notification is displayed on the user interface or the display of the weapon system at the block **3916**. The block **3916** may include one or more of the embodiments described with respect to the block **3906**, but with respect to displaying the unknown state.

If it is determined at the decision block **3914** that an unknown state flag is not set, a magazine count and a chamber count is displayed on the user interface or the display of the weapon system at the block **3918**. Determining the magazine count and/or the chamber count for cartridges or ammunition available to the user may include performing one or more of the processes **2800**, **3100**, or **3300**. In some embodiments, determining the magazine count and/or the chamber count may include accessing a memory location in a memory of the weapon that stores the magazine count and/or the chamber count. Further, the block **3918** may include one or more of the embodiments described with respect to the block **3906**, but with respect to displaying the ammunition available to the user or included in magazines registered to the weapon.

Additional Embodiments

A number of additional embodiments are possible based on the disclosure herein. For example, embodiments of the present disclosure can be described in view of the following clauses:

1. A magazine configured to hold ammunition, the magazine comprising:
 - a housing comprising an ammunition chamber, wherein the ammunition chamber is configured to store one or more cartridges of a particular ammunition type;
 - an ammunition counter configured to determine a quantity of cartridges of the particular ammunition type within the ammunition chamber; and
 - an optical transceiver configured to transmit a count of the quantity of cartridges of the particular ammunition type within the ammunition chamber to a weapon system when the magazine is installed in the weapon system.
2. The magazine of clause 1, wherein the ammunition counter comprises:
 - a magnet configured to generate a magnetic field that is at least partially within the ammunition chamber; and
 - a Hall effect sensor positioned within the housing, the Hall effect sensor positioned with respect to the housing and the magnet, wherein the Hall effect sensor generates a signal when the magnet is within a particular distance of the Hall effect sensor.
3. The magazine of clause 2, wherein the ammunition counter further comprises electronic circuitry configured to determine the quantity of cartridges of the particular ammu-

munition type within the ammunition chamber based at least in part on one or more signals generated by the Hall effect sensor.

4. The magazine of any one of the preceding clauses, further comprising a digital to optical signal adapter configured to convert a digital signal corresponding to the count of the quantity of cartridges to an optical signal for transmission by the optical transceiver.

5. A weapon system comprising:

a magazine configured to hold one or more cartridges; and a weapon comprising:

an insertion port configured to accept the magazine, wherein the magazine is configured to be inserted into the insertion port of the weapon and to provide the one or more cartridges to the weapon for firing; and

an optical transceiver configured to receive a count of the one or more cartridges within the magazine when the magazine is installed in the weapon system.

6. The weapon system of clause 5, wherein the magazine comprises:

a first magnet configured to generate a first magnetic field that is at least partially within an ammunition chamber of a housing of the magazine; and

a first set of Hall effect sensors positioned along the housing, at least some of the first set of Hall effect sensors aligned with respect to the first magnet when the ammunition chamber includes a particular number of cartridges.

7. The weapon system of any one of the preceding clauses, wherein the weapon further comprises a plurality of light-emitting diodes controllable to display the count of the one or more cartridges within the magazine.

8. The weapon system of any one of the preceding clauses, wherein the weapon further comprises a plurality of light-emitting diodes controllable to display a jam state of a weapon when the magazine is loaded into the weapon.

9. The weapon system of any one of the preceding clauses, wherein the weapon further comprises a second magnet configured to generate a second magnetic field, the second magnet located within a buffer tube of the weapon.

10. The weapon system of any one of the preceding clauses as modified by clause 9, wherein the weapon further comprises a second set of Hall effect sensors positioned along the buffer tube and configured to detect a position of the bolt based at least in part on the second magnetic field.

11. The weapon system of any one of the preceding clauses as modified by clause 10, wherein the weapon further comprises electronic circuitry configured to determine a count of cartridges within the weapon system.

12. The weapon system of any one of the preceding clauses as modified by clause 11, wherein the electronic circuitry determines the count of cartridges within the weapon system based at least in part on one or more signals generated by the second set of Hall effect sensors and a signal received at the optical transceiver.

13. The weapon system of any one of the preceding clauses, wherein the weapon further comprises an optical to digital signal adapter configured to convert an optical signal received from the magazine by the optical transceiver to a digital signal, the optical signal corresponding to the count of the one or more cartridges within the magazine.

14. A weapon system comprising:

a weapon comprising:

an insertion port configured to accept a first magazine from one or more magazines, wherein the first magazine is configured to be inserted into the insertion

port of the weapon and to provide one or more cartridges to the weapon for firing; and

a transceiver configured to receive one or more magazine status signals corresponding to the one or more magazines, each magazine status signal corresponding to a different magazine from the one or more magazines.

15 The weapon system of clause 14, further comprising the one or more magazines.

16. The weapon system of any one of the preceding clauses, wherein each magazine status signal corresponds to a number of cartridges within the corresponding magazine.

17. The weapon system of any one of the preceding clauses, wherein each magazine status signal corresponds to a use state of the corresponding magazine indicating whether the corresponding magazine has been used with the weapon.

18. The weapon system of any one of the preceding clauses, further comprising a registration system configured to register the one or more magazines with the weapon system.

19. The weapon system of any one of the preceding clauses as modified by clause 18, wherein the registration system comprises an optical scanner configured to scan a machine-readable code associated with the first magazine from the one or more magazines.

20. The weapon system of any one of the preceding clauses as modified by clause 19, wherein the machine-readable code includes a unique identifier associated with the first magazine.

21. The weapon system of any one of the preceding clauses as modified by clause 19, wherein the machine-readable code includes a unique identifier associated with the first magazine.

22. The weapon system of any one of the preceding clauses as modified by clause 18, wherein the weapon further comprises the registration system.

23. The weapon system of any one of the preceding clauses, wherein the weapon further comprises a user interface system configured to display a count of the number of cartridges within the first magazine when the first magazine is inserted into the weapon.

24. The weapon system of any one of the preceding clauses, wherein the weapon further comprises a user interface system configured to display a count of the number of cartridges within the one or more magazines.

25. The weapon system of any one of the preceding clauses, wherein the weapon further comprises a user interface system configured to display a count of the one or more magazines.

26. The weapon system of any one of the preceding clauses as modified by clause 25, wherein the user interface system is further configured to display the count of the one or more magazines that include one or more cartridges.

27. The weapon system of any one of the preceding clauses, wherein the transceiver is further configured to transmit a status of the one or more magazines to an external system that is separate from the weapon system.

28 The weapon system of any one of the preceding clauses as modified by clause 27, wherein the external system comprises a headgear system.

29. The weapon system of any one of the preceding clauses as modified by clause 27, wherein the external system comprises a computing system.

30. The weapon system of any one of the preceding clauses as modified by clause 27, wherein the status is transmitted via a network to a central command center.

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31. The weapon system of any one of the preceding clauses, wherein the transceiver comprises an optical transceiver.

32. The weapon system of any one of the preceding clauses, wherein the transceiver comprises a wireless transceiver.

33. A magazine configured to hold ammunition, the magazine comprising:

a housing comprising an ammunition chamber, wherein the ammunition chamber is configured to store one or more cartridges of a particular ammunition type;

a magnet configured to generate a magnetic field that is at least partially within the ammunition chamber;

a Hall effect sensor positioned within the housing, the Hall effect sensor positioned with respect to the housing and the magnet, wherein the Hall effect sensor generates a signal when the magnet is within a particular distance of the Hall effect sensor; and

electronic circuitry configured to determine a quantity of cartridges of the particular ammunition type within the ammunition chamber based at least in part on one or more signals generated by the Hall effect sensor.

34. The magazine of clause 33, wherein the Hall effect magnet is positioned in the follower.

35. The magazine of any one of the preceding clauses, wherein the one or more signals are generated based on a location of the magnet with respect to the Hall effect sensor.

36. The magazine of any one of the preceding clauses, further comprising a plurality of Hall effect sensors within the housing, the plurality of Hall effect sensors including the Hall effect sensor, the plurality of Hall effect sensors positioned along the housing.

37. The magazine of any one of the preceding clauses, further comprising a plurality of light-emitting diodes controllable to display the quantity of cartridges of the particular ammunition type.

38. The magazine of any one of the preceding clauses as modified by clause 37, wherein the plurality of light-emitting diodes are further controllable to display a jam state of a weapon when the magazine is loaded into the weapon.

39. The magazine of any one of the preceding clauses, further comprising an optical transceiver configured to transmit the quantity of cartridges of the particular ammunition type to a user interface device that is separate from the magazine.

40. The magazine of any one of the preceding clauses as modified by clause 39, wherein the optical transceiver is further configured to receive a jam state of a weapon when the magazine is loaded into the weapon.

41. The magazine of any one of the preceding clauses, further comprising an alignment pin configured to align a circuit board that includes the Hall effect sensor.

42. The magazine of any one of the preceding clauses, further comprising a spring-loaded plunger positioned below the ammunition chamber and configured to control the position of the plurality of cartridges within the ammunition chamber.

43. The magazine of any one of the preceding clauses as modified by clause 42, further comprising a gap between a sealing cap of the magazine and the spring-loaded plunger, the gap configured to house at least some of the electronic circuitry.

44. The magazine of any one of the preceding clauses, further comprising a power source in electrical communication with the electronic circuitry.

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45. A weapon system comprising:

a magazine configured to hold one or more cartridges; and
a weapon comprising an insertion port configured to accept the magazine, a buffer tube, and a bolt, wherein the magazine is configured to be inserted into the insertion port of the weapon and to provide the one or more cartridges to the weapon for firing, the magazine comprising:

a first magnet configured to generate a first magnetic field that is at least partially within an ammunition chamber of a housing of the magazine; and

a first set of Hall effect sensors positioned along the housing, at least some of the first set of Hall effect sensors aligned with respect to the first magnet when the ammunition chamber includes a particular number of cartridges.

46. The weapon system of clause 45, wherein the weapon further comprises a second magnet configured to generate a second magnetic field, the second magnet located within the buffer tube of the weapon.

47. The weapon system of clause 46, wherein the weapon further comprises a second set of Hall effect sensors positioned along the buffer tube and configured to detect a position of the bolt based at least in part on the second magnetic field.

48. The weapon system of any one of the preceding clauses, further comprising electronic circuitry configured to determine an amount of cartridges within the ammunition chamber based at least in part on one or more signals generated by the first set of Hall effect sensors.

49. The weapon system of any one of the preceding clauses as modified by clause 48, wherein the electronic circuitry is further configured to determine a total amount of available cartridges based at least in part on the amount of cartridges within the magazine and a determination of whether a cartridge is in a chamber of the weapon.

50. The weapon system of any one of the preceding clauses as modified by clause 48, wherein the electronic circuitry comprises an application-specific integrated circuit.

51. The weapon system of any one of the preceding clauses as modified by clause 48, wherein the electronic circuitry is further configured to determine a jam state of the weapon.

52. The weapon system of any one of the preceding clauses as modified by clause 48, wherein at least a portion of the electronic circuitry is within the magazine.

53. The weapon system of any one of the preceding clauses as modified by clause 48, wherein at least a portion of the electronic circuitry is within a handle of the weapon.

54. The weapon system of any one of the preceding clauses, further comprising a mountable display mounted on a barrel of the weapon, the mountable display configured to at least display an available count of cartridges within the magazine.

55. The weapon system of any one of the preceding clauses as modified by clause 54, wherein the mountable display is further configured to display a number of loaded magazines available to a user.

56. The weapon system of any one of the preceding clauses as modified by clause 54, wherein the mountable display is further configured to display a number of cartridges fired.

57. The weapon system of any one of the preceding clauses as modified by clause 54, wherein the mountable display is further configured to display a jam state of the weapon.

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58. A method of determining a number of available cartridges, the method comprising:

generating a magnetic field using a magnet located in a magazine;

detecting, using a set of sensors, a location of the magnet within the magazine, wherein each of the set of sensors is configured to generate a voltage based at least in part on the magnetic field, and wherein the location of the magnet is determined based at least in part on one or more voltage values generated by one or more of the sensors from the set of sensors; and determining a number of cartridges of ammunition within a magazine based at least in part on the location of the magnet.

59. The method of clause 58, further comprising displaying a count of the number of cartridges on a display interface included with the magazine.

60. The method of any one of the preceding clauses, further comprising transmitting count data corresponding to the number of cartridges to a separate display interface that is separate from the magazine.

61. The method of any one of the preceding clauses as modified by clause 60, wherein the count data is transmitted using an optical transceiver.

62. The method of any one of the preceding clauses, further comprising: generating a second magnetic field using a second magnet located in a weapon;

detecting, using at least one sensor, a location of the second magnet within the weapon, the at least one sensor separate from the set of sensors; and

determining whether a cartridge is within a chamber of the weapon based at least in part on the location of the second magnet.

63. The method of any one of the preceding clauses as modified by clause 62, further comprising determining whether the weapon is jammed based at least in part on the location of the second magnet within the weapon.

64. The method of any one of the preceding clauses, further comprising:

determining a number of magazines assigned to a user; determining a number of cartridges within each of the number of magazines;

summing the number of cartridges within each of the number of magazines and the number of cartridges of ammunition within the magazine to obtain a total number of available cartridges; and

causing the total number of available cartridges to be displayed to the user.

65. A computer-readable, non-transitory storage medium storing computer executable instructions that, when executed by one or more computing devices, configure the one or more computing devices to perform operations comprising:

generating a magnetic field using a magnet located in a magazine;

detecting, using a set of sensors, a location of the magnet within the magazine, wherein each of the set of sensors is configured to generate a voltage based at least in part on the magnetic field, and wherein the location of the magnet is determined based at least in part on one or more voltage values generated by one or more of the sensors from the set of sensors; and

determining a number of cartridges of ammunition within a magazine based at least in part on the location of the magnet.

66. The computer-readable, non-transitory storage medium of clause 65, wherein the operations further com-

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prise displaying a count of the number of cartridges on a display interface included with the magazine.

67. The computer-readable, non-transitory storage medium of any one of the preceding clauses, wherein the operations further comprise transmitting count data corresponding to the number of cartridges to a separate display interface that is separate from the magazine.

68. The computer-readable, non-transitory storage medium of any one of the preceding clauses as modified by clause 67, wherein the count data is transmitted using an optical transceiver.

69. The computer-readable, non-transitory storage medium of any one of the preceding clauses, wherein the operations further comprise:

generating a second magnetic field using a second magnet located in a weapon;

detecting, using at least one sensor, a location of the second magnet within the weapon, the at least one sensor separate from the set of sensors; and

determining whether a cartridge is within a chamber of the weapon based at least in part on the location of the second magnet.

70. The computer-readable, non-transitory storage medium of any one of the preceding clauses as modified by clause 69 wherein the operations further comprise determining whether the weapon is jammed based at least in part on the location of the second magnet within the weapon.

71. The computer-readable, non-transitory storage medium of any one of the preceding clauses, wherein the operations further comprise:

determining a number of magazines assigned to a user; determining a number of cartridges within each of the number of magazines;

summing the number of cartridges within each of the number of magazines and the number of cartridges of ammunition within the magazine to obtain a total number of available cartridges; and

causing the total number of available cartridges to be displayed to the user.

Terminology

The embodiments described herein are exemplary. Modifications, rearrangements, substitute processes, etc. may be made to these embodiments and still be encompassed within the teachings set forth herein. One or more of the steps, processes, or methods described herein may be carried out by one or more processing and/or digital devices, suitably programmed.

Depending on the embodiment, certain acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the algorithm). Moreover, in certain embodiments, acts or events can be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially.

The various illustrative logical blocks, modules, and algorithm steps described in connection with the embodiments disclosed herein can be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. The described functionality can be implemented in

varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

The various illustrative logical blocks and modules described in connection with the embodiments disclosed herein can be implemented or performed by a machine, such as a processor configured with specific instructions, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A processor can be a microprocessor, but in the alternative, the processor can be a controller, microcontroller, or state machine, combinations of the same, or the like. A processor can also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

The elements of a method, process, or algorithm described in connection with the embodiments disclosed herein can be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module can reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of computer-readable storage medium known in the art. An exemplary storage medium can be coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium can be integral to the processor. The processor and the storage medium can reside in an ASIC. A software module can comprise computer-executable instructions which cause a hardware processor to execute the computer-executable instructions.

Conditional language used herein, such as, among others, “can,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” “involving,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list.

Disjunctive language such as the phrase “at least one of X, Y or Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to present that an item, term, etc., may be either X, Y or Z, or any combination thereof (e.g., X, Y and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y or at least one of Z to each be present.

The terms “about” or “approximate” and the like are synonymous and are used to indicate that the value modified

by the term has an understood range associated with it, where the range can be $\pm 20\%$, $\pm 15\%$, $\pm 10\%$, $\pm 5\%$, or $\pm 1\%$. The term “substantially” is used to indicate that a result (e.g., measurement value) is close to a targeted value, where close can mean, for example, the result is within 80% of the value, within 90% of the value, within 95% of the value, or within 99% of the value.

Unless otherwise explicitly stated, articles such as “a” or “an” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.

While the above detailed description has shown, described, and pointed out novel features as applied to illustrative embodiments, it will be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As will be recognized, certain embodiments described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

What is claimed is:

1. A magazine comprising:

a follower configured to move towards an egress of the magazine as ammunition leaves the magazine and towards a base of the magazine as ammunition is inserted into the magazine;

a round counter comprising:

a linear encoder formed from a magnet positioned on the follower and a plurality of sensors; and

a hardware processor configured to determine an ammunition count of ammunition within the magazine based on a signal generated by the linear encoder, the signal corresponding to a position of the magnet with respect to the plurality of sensors; and an output device in communication with the round counter and configured to output the ammunition count, wherein the round counter is configured to account for ammunition movement that occurs responsive to the magazine being inserted into a magazine well of a firearm and pressure being applied to ammunition within the magazine.

2. The magazine of claim 1, wherein the output device comprises a transceiver configured to receive a data signal from the hardware processor corresponding to the ammunition count and to transmit the data signal to a firearm to display the ammunition count on a display device of the firearm.

3. The magazine of claim 2, wherein the transceiver comprises an optical transceiver configured to output an optical signal through a transmission window of the magazine.

4. The magazine of claim 3, further comprising a collimating lens configured to collimate the optical signal to be transmitted through the transmission window.

5. The magazine of claim 1, wherein the output device comprises a display device configured to display the ammunition count.

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6. The magazine of claim 5, wherein the display device comprises ultra-bright LEDs viewable in daylight.

7. The magazine of claim 6, wherein the ultra-bright LEDs have a luminosity exceeding 100 millicandelas.

8. The magazine of claim 6, wherein the ultra-bright LEDs are dimmable to a degree that enables the display device to be readable with night vision goggles.

9. The magazine of claim 1, wherein the magnet generates a magnetic field that is detectable by at least one sensor of the plurality of sensors and causes the at least one sensor to generate the signal.

10. The magazine of claim 1, wherein the hardware processor is further configured to access an ammunition count table and to determine the ammunition count using the ammunition count table based on the signal.

11. The magazine of claim 10, wherein the ammunition count table comprises a lookup table, and wherein the ammunition count is determined from the lookup table using a value generated from the signal as an index to the lookup table.

12. The magazine of claim 10, wherein the ammunition count table is selected from a plurality of ammunition count tables based at least in part on a determination of whether the magazine is inserted into a firearm.

13. The magazine of claim 10, wherein the ammunition count table comprises a lookup table, and wherein the ammunition count is determined from the lookup table using a value generated from the signal as an index to the lookup table.

14. The magazine of claim 10, wherein the ammunition count table is a magazine-specific table that is calibrated for the magazine.

15. The magazine of claim 10, further comprising non-volatile memory configured to store the ammunition count table.

16. The magazine of claim 1, wherein the signal generated by the linear encoder is generated based at least in part on an interpolation of a first signal generated by a first sensor of the plurality of sensors and a second signal generated by a second sensor of the plurality of sensors.

17. The magazine of claim 1, wherein the plurality of sensors comprises Hall effect sensors.

18. The magazine of claim 1, wherein the plurality of sensors comprises anisotropic magnetoresistive sensors.

19. The magazine of claim 1, wherein each of the plurality of sensors are positioned in series along a side of the

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magazine in a direction defined between the egress of the magazine and the base of the magazine.

20. A magazine comprising:

a follower configured to move towards an egress of the magazine as ammunition leaves the magazine and towards a base of the magazine as ammunition is inserted into the magazine;

a round counter comprising:

a linear encoder formed from a magnet positioned on the follower and a plurality of sensors; and

a hardware processor configured to determine an ammunition count of ammunition within the magazine based on a signal generated by the linear encoder, the signal corresponding to a position of the magnet with respect to the plurality of sensors wherein the hardware processor is further configured to access an ammunition count table and to determine the ammunition count using the ammunition count table based on the signal; and

an output device in communication with the round counter and configured to output the ammunition count.

21. A magazine comprising:

a follower configured to move towards an egress of the magazine as ammunition leaves the magazine and towards a base of the magazine as ammunition is inserted into the magazine;

a round counter comprising:

a linear encoder formed from a magnet positioned on the follower and a plurality of sensors; and

a hardware processor configured to determine an ammunition count of ammunition within the magazine based on a signal generated by the linear encoder, the signal corresponding to a position of the magnet with respect to the plurality of sensors; and

an output device in communication with the round counter and configured to output the ammunition count, wherein the output device comprises a transceiver configured to receive a data signal from the hardware processor corresponding to the ammunition count and to transmit the data signal to a firearm to display the ammunition count on a display device of the firearm, and wherein the transceiver comprises an optical transceiver configured to output an optical signal through a transmission window of the magazine.

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